SPCA106B

MASTER OF COMPUTER APPLICATIONS

FIRST YEAR - FIRST SEMESTER

INTER DISCIPLINARY - I

ACCOUNTING & FINANCIAL MANAGEMENT

VOLUME - II

INSTITUTE OF DISTANCE EDUCATION

UNIVERSITY OF MADRAS

MCA
FIRST YEAR
SEMESTER - I

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With best wishes from mind and heart,

DIRECTOR

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SYLLABUS

Objective of the course: This course introduces the basic concepts of Accounting & Financial Management.

Course Outcomes: After successful completion of this course, the students should be able to understand the basic idea and they can able to work in accounts and finance for managing the business.

Unit-I: Principles of Accounting: Principles of double entry Assets and Liabilities Accounting records and systems Trial balance and preparation of financial statements Trading, Manufacturing, Profit and Loss accounts, Balance Sheet including adjustments(Simple problems only).

Unit-II: Analysis and Interpreting Accounts and Financial Statements: Ratio analysis - Use of ratios in interpreting the final accounts (trading accounts and loss a/c and balance sheet) - final accounts to ratios as well as ratios to final accounts.

Unit-III: Break-even analysis and Marginal Costing: Meaning of variable cost and fixed cost Cost Volume Profit analysis – calculation of breakeven point, Profit planning, sales planning and other decision – making analysis involving break - even analysis - Computer Accounting and algorithm.(differential cost analysis to be omitted).

Unit-IV: Budget/Forecasting: preparation of and Characteristics of functional budgets, Production, sales, Purchases, cash and flexible budgets.

Unit-V: Project Appraisal: Method of capital investment decision making: Payback method, ARR method - Discounted cash flows Net Present values Internal rate of return Sensitivity analysis Cost of capital.

Reference Books:

1) Shukla M.C. & T.S. Grewal, 1991, Advanced Accounts, S.Chand & Co. New Delhi.
2) Gupta R.L. & M. Radhaswamy, 1991, Advanced Accounts Vol. II, Sultan Chand & Sons, New Delhi.
3) Man Mohan & S.N. Goyal, 1987, Principles of Management Accounting, Arya Sahithya Bhawan.
4) Kuchhal, S.C., 1980, Financial Management, Chaitanya, Allahabad.
5) Hingorani, N.L. & Ramanthan, A.R, 1992, Management Accounting, 5th edition, Sultan Chand, New Delhi.

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SCHEME OF LESSONS

LESSON: 11

COSTING AN INTRODUCTION

Cost Accountancy is an essential part of accountancy which has been developed to meet the managerial needs of business and other organisations which incur cost.

Starting off as a branch of financial accounting, cost accountancy has developed so fast in the last few decade that it is difficult to give a suitable definition which fully covers its scope. As per the Terminology published by the Institute of Cost and Management Accountants London Cost Accountancy is "the application of costing and cost accounting principles methods and techniques to the science art and practice of cost control and the ascertainment of profitability. It includes the presentation of information derived there from for the purpose of managerial decision making".

Cost Accountancy is a science and art and practice of a Cost Accountant. It is a science in the sense that it is a body of systematic knowledge having costing principles which a cost accountant should follow for the proper discharge of his duties. It is an art as it requires, the ability and skill on the part of a cost accountant in applying the principles of cost accountancy to various managerial problems. Practice refers to the continuous effort of a cost accountant in the field of Cost Accountancy. The theoretical knowledge alone would not enable a cost accountant to deal with inricacies; he also needs to have a sufficient practical training.

The fields of Cost Accountancy include (1) Costing, (2) Cost Accounting, (3) Cost Control, (4) Budgetary Control and (5) cost Audit.

1. Costing

I.C.M.A. London has defined costing as "the technique and Process of ascertaining cost". As a technique costing is the body of principles and rule for ascertaining the cost. As a process it denotes the procedure for ascertaining costs. The technique and process to be used depend upon the nature of industry, type of product and method of production.

2. Cost Accounting

Cost Accounting has been defined as the process of accounting for cost which begins with the recording of income and expenditure and ends with the preparation of periodical statements and reports for ascertaining and controlling costs.

3. Costing

The techniques and process of ascertaining costs.

According to Wheldon, "Costing is the classifying, recording and appropriate allocation of expenditure for the determination of costs of products or services and for the presentation of suitability arranged data for purposes of control and guidance of management. It includes the ascertainment of every order, job, contract, process service units as may be appropriate. It deals with the cost of production, selling and distribution."

In the words of J.M. Fremgen, "Cost Accounting is the process of recording, classifying, allocating and reporting the various costs included in the operation of an enterprises."

In simple words, costing is a systematic procedure for determining the unit cost of outout oroduced or service rendered. It provides for an analysis of expenditure which enables the management to know not only the total cost but also its constituents. To take an example. It is not sufficient for the management to know that the cost of the shirt is Rs.15. The management is also interested in knowing the amount spent for cloth in the shirt, the amount of labour and other expenses so that it can control and reduce its cost.

Objects of Costing

The main objects of costing are :

1. To ascertain the costs of products and/or services.
2. To determine selling price.
3. To control costs.
4. To provide guidelines for management policy.

1. Ascertainment of Cost

The first object is ascertain the cost of each product process or operation and to ensure that all the expenses have been absorbed in the cost of products. For this, the Cost Accountant has to make certain preliminary investigations and to introduce a system for recording costs. Proper records are to be maintained for materials, labour and overheads.

2. Determination of Selling Price

The cost accounting provides detailed and relevant cost figures for determining the selling price, of products or services. Sometimes the management has to quote prices even below the total cost for the purpose of increasing sales volume.

This can be done only when cost data are analysed into variable and fixed elements. During periods of depression, costing is put to its greatest test because production begins to fall and prices have to be cut scientifically. Only a sound costing system can be of help under such circumstances.

3. Control of Cost

The third important object is the control of cost. This may be achieved with the help of budgets and standards set up for the guidance of management. Actuals are compared with standards and in case of deviations, remedial measures are taken. For setting up a system of control, the Cost Accountant has to study the organisational set-up, allocation of responsibilities and the needs of management. He indicates to the management any inefficient and the extent of wastage in each process of manufacture and how to control such wastage. A check is exercised over the stocks of raw materials, work-in-process, and finished goods so that least capital is locked up in these stocks.

4. Guidelines for Management Policy

Cost accounting has developed primary to serve the needs of management. It assists the management in conducting its business with utmost efficiency. Cost data provide guidelines for various managerial decisions like (a) Introduction of a new product or new line of production, (b) whether to make or buy, (c) utilisation of spare capacity, (d) replacement of existing machines and automation, and (e) future expansion policies and capital outlays.

Advantages of Costing

Costing has many advantages which are discussed belo:

But all of these advantages do not automatically flow from the installation of a costing system. The nature and extent of these advantages depends upon the type, adequacy and efficiency of the costing system and the co-operation it receives from the various levels of management.

A sound system of costing provides the following advantage:

(1) Profitable and unprofitable activities are disclosed. Management can take steps to eliminate or to reduce those activities from which little or not profit is obtained. It can change the method of production in order to render such activities more profitable.

(2) Costing provides such information upon which estimates and tenders may be based. Prices may be adjusted to meet market conditions. In case of big contracts or jobs, it is difficult to give quotations without knowing estimated cost.

(3) It reveals losses or inefficients occurring in any form such as idle time. Excessive spoilage or scrap, under utilization of plant and machinery.

(4) The exact cause of a decrease or increase in profit or loss of business can be located. Management can know whether the profits have come down because of increase in the cost of material or increase in labour rates or both.

(5) Costing guides future production policies. If explains the costs incurred and profits made in various lines of business

and processes and thereby provides data on the basis of which future production can be appropriately planned.

(6) It provides independent and reliable check on the accuracy of financial accounts with the help of the reconciliation of the two at the end of the year.

(7) It provides a perpetual inventory system which helps in the preparations of interim Profit and Loss Account and Balance Sheet without any stock-taking.

(8) It helps in controlling the cost with the application of standard costing and budgetary control. Cost comparison also helps in cost control.

(9) Costing information helps the management in taking vital decisions, such as :

(a) Whether it would be more profitable to purchase a component, or to continue its production.

(b) Whether to accept orders below cost.

(c) Comparative costs of different types of a machine or methods of manufacture and the profitability of the proposed capital expenditure.

(10) Costing system helps the Government, wage boards and trade unions in providing data for price fixation and price control, tariff protection and wage fixation.

COST ACCOUNTING AND FINANCIAL ACCOUNTING

The function of accounting is to provide the financial information for different parties who are interested in the welfare of that enterprise. Both financial accounting and cost accounting are concerned with the accumulation and presentation of information to serve the needs of management and outsiders. The source of the two accounts for recording the transactions is the same. Cost accounting is based on the same principles regarding debit and credit as are applied in financial accounting. Both are in monetary terms. Nevertheless, the two differ in their purpose and scope.

The important points of difference may be discussed as under :

1. Purpose

Cost accounting and financial accounting have different purposes. Financial accounting provides information about the enterprise in a general way. It serve the interests of business and other interested parties by providing suitable information in the financial statements, i.e. Profit and Loss Account and Balance Sheet, Cost Accounting renders information for the guidance of the management for the proper planning; operation, control and decision making.

2. Forms of Account

Financial accounts are required to be kept as per the requirements of the Companies Act and Income Tax Act. The Companies Act lays down the contents of Profit and Loss Account and Balance Sheet, Cost Accounting renders information for the guidance of the management for the proper planning, operation, control and decision making. Cost accounts on the other hand, are voluntarily kept to serve the management in the discharge of its functions. Recently the Companies Act has made obligatory the keeping of costing records in some manufacturing companies.

3. Recording

Financial accounting consists of classification, recording and analysis of transactions in a subjective manner. i.e., according to the nature of expenditure. Cost accounting records expenditure in an objective manner, i.e. according to the purpose for which costs are incurred. In financial accounts, items of cost are expressed in totals whereas in cost accounts, all the expenses are analysed, classified apportioned and allocated so that cost per unit can be calculated.

4. Analysis of Profit

Financial accounts reveal the profits of the business as a whole. Cost accounts show the result of each operation, process and product. Cost accounts disclose the unprofitable products lines so that these may be eliminated and alternative measures taken.

5. Control

Financial accounting lays emphasis on the recording aspect, no consideration is given to control whereas Cost Account provides for a detailed system of control with the help of standard costing and budgetary control.

6. Periodicity of Reporting

Financial accounting involves reporting of business performance and financial state of affairs usually at the end of the accounting year. In cost accounting, there is a continuous flow of cost data and other selected information in the form cost reports to management.

Cost Accounting and Management Accounting

Cost accounting is an effective tool of management in the performance of its functions. The I.Ç.M.A. defines management accounting as "the presentation of accounting information such a way so as to assist management in the creation of policy and in the day-to-day operation of an undertaking." Thus, management accounting aims at reporting to management for planning, decision making, controlling and other managerial functions.

Management accounting makes use of various techniques which include marginal costing, standard costing, budgetary control, break even analysis, cost-volume-profit relationship, ratio analysis, interfirm comparison and uniform costing, internal audit, etc. Most of these techniques are also employed by a cost accountant. Thus the objectives of cost accounting are quite similar to those of management accounting. Infact management accounting is an extension of cost accounting. For instance a cost accountant reports to management variances from standard or budgeted cost and presents a comparative data of various periods. The management accountant, on the other hands, goes a step further to suggest ways and means to improve upon the operations on the basis of available data of variances. In case material cost is more, the management accountant would work out the prospects of controlling the material cost and bring it to the level of standard materials cost.

The management accounting uses the principle and practices not only of cost accounting but also of financial accounting. Cost accounting is a more detailed application of financial accounting and management accounting is still a step further. The chart on next page shows the evolution of management accounting its relationship to cost accounting and financial accounting.

Preparing Income Statement and Balance Sheet	_____	Financial Accounting
Analysing Cost for Control and Maximising Efficiency	_____	Cost Accounting
Assisting Management for Planning Decision Making and Control	_____	Management Accounting

"Cost accounting has become an essential tool of the management"

Cost accounting assists the management in carrying out its day- to-day functions of control and formulation of business policies. They enable the management to know the causes of losses and wastages. It is so much allied to management that it is difficult to indicate where the cost accounting ends and management control begins.

Cost accounting serves the management in the following ways:

1. Classification and subdivision of costs

Costs are accumulated and classified by every possible and division of business. Detailed cost information is provided to the management for control purposes. Profitability of each product sales area, division, etc., are ascertained and it helps management to take steps to eliminate or to reduce those activities from which little or no profit is obtained.

2. Control of Material, Labour and Overheads Costs

It provides a detailed control of materials and supplies labour and overhead costs. Perpetual inventory system is operated to control material and supplies. For labour standard are set up to measure their efficiency and to assist the management in assigning each labour the work for which he is best fitted. Overheads are classified into fixed and variable and controllable and uncontrollable, so that management can concentrate on, a few items of overheads for control.

3. Standards for Measuring Efficiency

It provides the use of standards to assist management in making estimates and plans for the future and to provide the basis of measurements of efficiency. Actuals are compared with the predetermined standards to determine operating efficiency.

4. Budgeting

It provides use of budgets and enables the management to correct inefficiencies before these enter into business operations. The budget system is a co-ordinated plan of action for every executive, salesman and foreman which, by means of the cost system, can be compared with actual results monthly or more frequently.

5. Aids in Price Determination

It provides accurate cost data with help in the fixation of selling prices and for submitting quotation. In periods of depression, it enables the management to determine the extent to which prices can be reduced.

6. Special Factors

It keeps the management informed of special factors like levels of optimum profitability, seasonal fluctuations in volume and costs idle time of labour and machines.

7. Business Policies

It helps the management to take various decisions involving financial consideration like profitability or otherwise of new lines of production, replacement of labour by machines, alternative methods of production and such other matters.

Thus, costing has become an essential tool of management. It serves the management in the execution of policies and in the comparison of actual and estimated results so that the value of each policy may be appraised and changed, if necessary, to meet future conditions.

Limitations of Cost Accounting

Cost Accounting has a few critics. They point out the following limitations of Cost Accounting :

1. Expensive-costing system is expensive to install and run since it involves apportionment of costs and absorption of overheads requiring huge amount of clerical work.

2. Costing is not an independent tool-much of the efficiency of costing system depends upon the philosophy management. If the management is foresighted, the expenses incurred on a costing system may be worth while, otherwise it is waste of the owner's equity.

3. Problem of reconciliation of cost and financial accounts - The analysis provided by the cost accounts and financial accounts may be radically different. It would be quite difficult and costly to reconcile the two types of accounts.

4. Problem of apportioning a blame for inefficiency - Although inefficiency may be pointed out by cost accounting, yet it may become quite difficult to apportion blame on one of the other members of the management team, e.g. where authority is highly centralised it would be impossible to blame an individual. In such a case it is necessary to change the organisation structure of the firm.

Some of the above limitations are genuine but not formidable. They can be easily overcome if a costing system is adopted considering the various principles involved in cost accounting. If a system of costing is well thought out and is backed by expert advice, it is bound to succeed in its objectives.

The steps which should be taken to install Cost Accounting

Cost accounting is a systematic procedure for determining the unit cost of output produced or services rendered. Financial accounting fails to serve the management in the discharge of its functions. Financial accounting is primarily concerned with the recording of transactions affecting the business and the preparation of the profit and Loss Account and Balance Sheet as required by law. Cost accounting has developed primarily because of limitations of financial accounting. Cost accounting serves the purpose of cost ascertainment as well as cost control.

The need for cost accounting arises because of its following benefits:

1. It helps in the ascertainment of costs by products, processes, cost centres departments.

2. Costs are classified according to functions (manufacturing, administration, selling and distribution). elements (material labour and expenses) and variability (fixed, variable and semi- variable). These classifications indicator controllable and uncontrollable portions of costs.

3. It provides detailed control of materials and supplies, labour and overhead costs. perpetual inventory system is operated to control materials and supplies. For labour, standards are set up to measure their efficiency and to assist the management in assigning each labourer the work for which he is best suited. Overheads are classified into fixed and variable controllable, and uncontrollable so that management can concentrate on a few items of overheads for control.

4. Costing provides a system of cost control. Standards are set up and actual costs are compared with standard costs to develop variances for analysis. This will indicate operating efficiency or in-efficiency so that right remedical action may be taken at the right moment.

5. Cost data help the management in taking vital decisions such as (a) introduction of a new product, (b) whether to make or buy, (c) replacement of existing machines and introduction or automation, (d) utilisation of spare capacity, and (e) future expansion policies and capital outlays.

6. Adequate information is available for reports to various interested parties, such as trade associations, customers, shareholders, Government, insurance companies and banks. It helps in providing data for price fixation and price control, tariff protection and wage fixation.

Installation of Cost Accounting

There is no single costing system which can be installed in all types of organisations. Each organisation should choose and design its own system in accordance with requirements of management.

Before installation of costing system it should be considered whether it would be profitable to have a cost accounting department. The benefits from such a department must exceed the amount to be spent on it. This will depend upon many factors like the nature of business and the quality of management. The management must feel business and the need of the costing system and should make full use of the information available from this system in taking various decisions.

Costing system should be so designed as to serve the purpose of a particular organisation. The following steps should be taken at the time of installation to ensure smooth and efficient working of a cost accounting system.

1. The system should be designed after careful analysis of the nature of cooperations involved, the object of costing and the type of cost data required by management. The areas of operation of business wherein management's action will beneficial should be considered for designing the system of costing.

2. The system should be framed in such a way that it suit the general organisation of the business. In case, financial book, can be maintained such a way that they can provide the necessary costing information, separate costing books need not be maintained.

3. The technical aspects of the business should be taken into consideration and an attempt should be made to get the sympathetic assistance and support of the supervisory staff and workers for costing system.

4. Costing system should be simple and easy to operate. The procedure should be easily understood by the operating staff of all categories. Printed forms can be used to record data so as to reduce the clerical work as far as possible. Unnecessary details should be avoided.

5. The system should ensure that cost reports are prepared and presents promptly and regularly. It should serve all the requirements of management. The degree of accuracy desired should be determined. The frequency and regularity of supplying accurate figures very late.

6. The financial accounts should be capable of reconciliation or they should be interlocked in one integral accounting system.

A cost accountant is confronted with some problem while installing a system. Apart from the technical costing problems, there are certain practical difficult which a cost accountant has to face. A list of these difficulties together with suggestions to overcome them are given below :

(i) Lack of support from top management

Many a time, the costing system is introduced at the best of the management in other financial director and without the support of the management in other functional areas like production and sales. They view the system as an interference in their work and do not make use for system.

Before the installation of the system, the cost accountant should ensure that the management is fully committed to the costing system. A sense of cost consciousness is to be created in their minds by explaining them that they system is meant for their benefit. A cost manual may be prepared giving clearly the details and functions of the system.

(ii) Resistance from the existing accounting staff

The existing accounting staff may not welcome the new system. The existing staff may feel that they would lose their importance with the introduction of the costing system.

The cost accountant has to impress upon the staff that the costing system is needed to supplement the financial accounting system. The staff is to be impressed upon that it broaches the job of an accountant and creates new opportunities for the accounting staff.

(iii) Non-cooperation at other levels of organisation

The supervisory staff and workers may not cooperate with the system. They may not provide correct activity data which is very necessary for the system. They may also resist the additional paper work arising as a result of the introduction of the system.

Proper education should be given to staff regarding the advantages of the system and important roles they have to play in it. Meetings should be organised to clear their doubts and difficulties. It is essential to gain their confidence in the system.

(iv) Shortage of trained staff

At the initial stages, there may be a shortage of trained staff. The staff is to be properly trained so that costing department can run efficiently.

After installation also, continuous check over the progress of the system should be kept. It is only through the continuous efforts of a cost accountant that the system will turn out successful and can fulfill its objectives.

The Principal objections against the installation of a costing system in a factory.

(A) It is Unnecessary

It is said that the maintenance of cost records in unnecessary and involves duplication of work. But this argument has got no validity. In the present world of competition, management should know not only the exact cost but also how it has been made up, i.e., what are its various elements. Management is benefited in the following ways from the costing system.

(a) It enables management to ascertain the exact cost as well as its break-up into various elements of cost.

(b) It provided a basis for fixing the most competitive selling price and quotation for tender.

(c) It provides a detailed control of materials and supplies labour and overhead costs. It facilitates the detection and prevention of waste; leakage and inefficiency.

(d) It provides for cost control with the help of standard costing and budgetary control which leads to economy in manufacture and maximisation of profits.

(e) It helps the management to take various decisions involving financial considerations like profitability or otherwise of new lines of production replacement of labour by machines and such other matters.

Taking into consideration all the above benefits, it may be said that costing is not unnecessary.

(B) **It is Expensive**

Secondly, it is pointed out that costing system is quite expensive which only large organisations can afford. In this respect it may be seen that costing system should be treated as an investment. It should be seen that the benefits derived from the system exceed the amount spent on it. It should not prove a burden on the organisation. For this purpose, the system to be adopted should be simple, elastic and should suit the general organisation of business. The existing organisation should be disturbed to the minimum. The details of maintenance of records should be kept to the minimum taking into account the needs and use of each record. Therefore, it cannot be said that a costing system is expensive.

It is essential to devise a costing system suiting the requirements of business and industry and, therefore, there is no one system that will suit all business. There are different methods of costing for different industries. This is because the nature of work carried out by different industries varies. Industries can be grouped into two basic types: (a) industries doing job work, (b) industries engaged in mass production of a single product or identical products. In case of job industries each job needs special treatment and production is strictly according to customer's specifications. There is no standard product and costs have to ascertained for each job separately. Examples of such industries are printing press, construction company, ship-building, manufacturing, of heavy machinery, etc. In case of mass production industries firms manufacture identical products, e.g., chemical, plants processes and cost is accumulated for each process separately. Therefore, it becomes necessary to use different methods of costing at the very nature of production process and methods of production differ in different industries.

The cost method will depend on the type of manufacture and nature of industry. For these two basic types of industries there are two general types of cost accounting system - for job industries Job Costing and for mass production industries Process Costing.

JOB COSTING

This is a method where costs are collected and accumulated for each job separately. This is done because, each job require different work and production is accounting to customer's specifications. Each job has a separate identify and, therefore, it become essential so analyse and segregate costs according to each job separately. Industries where this method of costing is used are: printing presses, ship building, repair shops, manufacture of locomotive engines, etc.

There are some methods which are based on the principle of job costing, but they vary due to special characteristics. Such methods are :

1. Contract or Terminal Costing

This method is applied to ascertain the costs incurred on each contract separately. This is used in industries carrying out the building or construction work. Contract is a big job and therefore, the principles of job order costing are applied here.

2. Batch Costing

Under this method, orders for like products are arranged in convenient batches and each batch is treated as one job and cost is calculated accordingly. Here, in place of a single job, there is a batch or group of identical products. Cost is collected for each batch separately as in the case of job costing. This method is mainly applied in biscuit manufacture, garments manufacture and spare parts and components manufacture.

PROCESS COSTING

It is a method where costs are collected and accumulated according to departments or processes and cost of each department or process is divided by the quantity of production to arrive at cost per unit. The cost is calculated process-wise or departmentwise. This method is used in mass production industries, having continuous production of uniform standard products. Examples of such industries are paper, soap, textiles, chemicals, sugar, food products, etc.

Other methods which are based on the principle of process costing but vary due to special characteristics are :

1. Operation Costing

This is a refinement and more detailed application of process costing. This involves costing by every operation instead of a process. This is used where the manufacturing process consists of a number of distinct operations. This method provides minute analysis of costs and ensures greater accuracy and better control.

2. Single, Output or Unit Costing

The method is applied where production is uniform and consists of only single product or two or three types of similar products with variations only in size, shape, quality, etc. It is a simple method of costing in which the total cost is divided by the number of units produced to determine the cost per unit. This method is applied in industries like mines, quarries, oil drilling etc.

3. Operating Costing

This method is applicable to undertaking that render services. This is used to determine the costs of rendering services by airways railways, road transport, hospital, power house, etc. For example, transport company is interested in knowing the cost of carrying one tonne of goods per kilometer of the cost of carrying one passenger per kilometer. Here the cost unit is to be determined and it varies from undertaking to undertaking. It is similar to unit costing, the only difference being that in place of goods, the cost of services is determined in operating costing.

4. Multiple or Composite Costing

This involves the application of more than one method of costing in respect of the sample product. For example, process costing may be applied upto a certain stage and then job costing may be used. In the case of bicycle manufacturing, the cost of each part of bicycle may be ascertained by using batch costing and for cost of assembling the parts, single or output costing method may be applied. Examples of industries in which this method is used are cycle manufacture, motor car manufacture, aeroplanes, television, etc.

UNIFORM COSTING

When a number of firms in an industry agree to use the same costing principles and/or practices, it is known as uniform costing. It is not a separate method of costing like job of or process casting. It only introduces uniformity regarding costing principles among different undertakings engaged in the same industry. It attempts to establish uniform costing methods so that comparison of performances in various undertakings can be made to the common advantage of all the participating units.

Cost Unit:

A cost unit is a unit of product, service or time in relation to which costs may be ascertained or expressed. It is a unit of quantity in terms of which costs may be computed. For example, the cost of steel is ascertained in terms of per tonne, cost of carrying a passenger in terms of per kilometer.

The terms of measurement used in cost units are number, length, area, volume, weight, time and value. Cost unit should be selected very carefully. It should be natural to the nature of business operations.

A few examples of cost units in different industries are given below :

Industry	Cost Unit
Steel	Per Tonne
Cement	Per Tonne
Sugar	Per Quintal
Cable	Per Metre or Kilometre
Textile (cloth)	Per Metre
Colliery	Per Tonne
Paper	Per Kg. or Per Tonne
Chemical	Kg. Litre or Tonne
Power	Kilowatt hour (KWH)
Transport	Passenger Kilometre, or Tonne Kilometre
Automobile	Per automobile i.e. number

COST CENTRE

Cost centre is defined as a location, person or items of equipment (or group of these) for which costs may be ascertained and used for purposes of cost control. In simple words, it is a part of an enterprise to which costs can be charged.

A cost centre can be: (1) a location, i.e., an area such as a department, storeyard or sales area; (2) an item of equipment, e.g., lathe, machine, delivery vehicle; or (3) a person, e.g., salesman, foreman.

Thus, cost centres may be the following types.

1. The Impersonal Cost Centre. It is a cost centre which consists of a location or items of equipments or group of these.

2. The Personal Cost Centre. It is a cost centre which consists of a person or group or persons.

3. The Operation Cost Centre. It is a cost centre which consists of machines and or persons carrying out similar operations.

4. The Process Cost Centre. It is a cost centre which consists of a specific process or a continuous sequence of operations.

The determination of a suitable cost centre is very important for ascertainment and control of cost. The manager in charge of a cost centre is held responsible for control of cost of his cost centre. Further, cost centres enable the accumulation of all such costs at one place for which a common base of recovery may be used.

The size and number of cost centres vary from concern and are dependent upon the amount of expenditure and requirements of management for cost control. The selection of suitable cost centres depends upon the following factors:

1. Layout and organisation of factory;
2. Availability of information; and
3. Management policy regarding selection of cost centres.

CLASSIFICATION OF COST AND COST SHEET

Classification is the process of grouping costs according to their common characteristics. It is the systematic placement of like items together according to their common features.

There are various methods of classifying costs. The method selected is decided by the purpose to be achieved. The same cost figures are classified according to the different methods depending upon requirements. The important basis of classification are :

1. by nature or element
2. by function
3. as direct or indirect
4. by variability
5. by controllability
6. as capital and revenue, and
7. by normally
8. by time
9. for making management decisions.

1. Cost Classified by Nature of Element

One of the basis of classification is nature or element i.e., what they are. On this basis, it is classified in three categories: Material, Labour and Expenses. There can be further sub-classification of these; for example, material into war material, components, maintenance material etc.

This classification is important as it serves to find out the various elements of the cost of a product and what is the relative importance of each element in the total cost of a product. This classification also helps in valuation of the work- in-process.

2. Cost Classified by Function

Here costs are classified according to functions to which they relate. There are three major functional divisions in an organisation viz. Production Administration, and Selling & Distribution. In the cost sheet or cost statement, the total cost is split up according to the function and element.

3. Direct and Indirect

Direct means that which can be conveniently identified with and allocated to a particular unit of cost, i.e., job, product or process. On the otherhand, the term indirect means that which is of general character and that cannot be identified with a particular unit of cost. It has to be distributed or shared.

Costs which can be directly identified with cost centres, processes or production units are direct costs. Examples are: the cost of cloth in the ready-made shirt, wages payable to a worker who is directly involved in production, etc. Costs which cannot be identified with cost centres or cost units and are to be distributed on some equitable basis, are indirect costs. Examples are: the cost of consumable stores, salary of a foreman or supervisor rent of the factory. etc.

There are certain costs which can be identified with a particular unit of cost but the process of identifying them is so costly and cumbersome that it is not worth while to do so and therefore, they are treated as indirect costs For example, in the case of thread used in stitching a shirt, though it is possible to find out the amount of thread used in a particular shirt, but it would not be of much use, therefore it is treated as indirect.

Such a classification is essential for cost ascertainment and cost control. While preparing the cost sheet, first of all the direct cost like direct material, direct wages and direct expenses are added to arrive at the prime cost. To this, all the direct expenses are to arrive at the total cost. It helps in cost control through the tools of standard costing and budgeting.

4. Costs classified according to variability

Variability of cost is reckoned in relation to the volume or production. Some costs vary in sympathy with production while some remain constant. On this basis costs are classified into three groups, viz., fixed variable and semi-variable.

(a) Fixed Cost :

Fixed cost is that cost which in aggregate tends to be unaffected by variations in volume of output. The amount of fixed cost tends to remain constant for all volumes of productions within the installed capacity of the plant. For example rent of office and factory manager's salary shall remain the same even if productions goes up or comes down.

(b) Variable Cost:

This is a cost which in aggregate tends to vary directly with variations in volume of output. Such cost increases as the production goes up, it correspondingly decreases when productions falls. However, variations may not always be in the same proportion. Practically, all the direct elements of costs are variable.

(c) Semi-fixed or Semi-variable Cost:

This is the cost which is partly variable. It has the characteristics of both the fixed and variable. For example, maintenance of building and plant.

5. Costs classified according to controllability

Under this basis, costs classified on the consideration whether they are capable of control or not. The concept of control of cost is related to a specified level of managerial authority and with the reference to specified level, costs are classified as controllable and non-controllable.

6. Capital and Revenue

Capital cost is that cost which is incurred in purchasing some asset for purpose of increasing the earning capacity of business. In case of steel plant, the cost of the rolling machine is capital cost. Revenue cost is one which is incurred to maintain revenue earning capacity of the business, e.g., cost of maintaining assets, cost of running business.

7. Normality

There are two types of cost which display the normality characteristic. They are :

(a) Normal Cost

It is a cost which is normally incurred at a given level of output in the condition in which that level of output is normally attained.

(b) Abnormal Cost

It is a cost which is not normally incurred at a given level of output in the condition in which that level of output is normally attained.

Normal cost is part of the cost of production, whereas abnormal cost is excluded from the cost of production.

8. Classification by Time

On the basis of time, cost can be classified into (a) Historical Cost and (b) Pre-determined Cost.

(a) Historical Cost

It means the ascertainment of costs after they have been incurred. Historical costs are available only after the completion, of production. Such cost figures have only historical value.

(b) Pre-determined Cost

Pre-determined costs are estimated costs. These costs are computed in advance of production on the basis of actual cost and the factors affecting the cost. Pre-determined costs made on a more or less scientific basis result in a standard cost. Standard costs are compared with actual costs so as to find out the differences or variances. These variances are analysed so as to enable the management to take remedical action, if necessary.

9. Classification for Managerial Decision

On this basis costs may be classified into the following types :

(a) Marginal Cost

Marginal cost is the variable cost comprising prime cost and variable overheads. It has been defined as "the amount at any given volume of output by which aggregate costs are changed if the volume of output is increased or decreased by one unit." With the increase in output by one unit, the total cost increase the resultant increase in the total cost is known as marginal cost.

(b) Out of Pocket Cost

It is a cost which, with respect to given decision of managements, gives rise to cash expenditure. Such costs are relevant for the purpose of price fixation during trade recession or for taking a 'make or buy' decision.

(c) Differential Costs

These are the increases or decreases in total cost, or the changes in specific elements of cost, that result from any variation in operations. These are the costs which arise as a result of change in the level, pattern, method of production. These may be incremental or decremental, depending upon whether they result in increase in cost or decrease in cost.

(d) Imputed Costs

Imputed Costs are costs that do not involve at any time actual cash outlay and which do not, as a consequence, appear in the financial records. Nevertheless, such costs involve a forgoing on the part of the person or persons for whom costs are being calculated. They are hypothetical costs which are computed only for the purpose of decision-making. For example, interest on capital not payable, rent on own building, etc.

(e) Opportunity Cost

It is the measurable advantage forgone as a result of the rejection of alternative use of resources, whether of materials, labour or facilities. For example, if an unowned building is proposed to be used for a project, the likely rent of the building is the opportunity cost which should be taken into account while evaluating the profitability of the project.

(f) Replacement Cost

This is the cost in the present market. Replacement cost of an asset is the cost which would be incurred if the asset is to be purchased at the current market price and not the cost at which it was purchased earlier.

(g) Avoidable and Unavoidable Cost

Avoidable Costs are those costs which, under given conditions of perform efficiency, should have not been incurred. Unavoidable costs are those which are essential to be incurred and are inescapable.

(h) Sunk Cost

This is that cost which has already been incurred and which is, therefore, not relevant to decision making process. This is historical and is sunk in the past. For example, when the management has to decide about the replacement of plant and machinery the depreciated book value of the old plant is not relevant in this decision-making process as the amount is a sunk cost and will be written off at the time of replacement.

(i) *Conversion Cost*

This is the total of direct labour cost and manufacturing overheads. In otherwords, conversion costs is the production costs minus costs of direct materials. It is a special purpose cost which may be used for a particular purpose.

Distinction between Cost Estimation and Cost Ascertainment Cost Estimation

Cost estimation is the process of pre-determining the cost of products or services. These costs are prepare in advance of production and precade the operations. Estimated costs are definitely future costs and are based on the average of the part actual figures adjusted for anticipated changes in future.

Cost estimates are often required for bidding for contracts or for offering quotation of prices in respect of a job to be undertaken. Extreme care has to be taken in cost estimation, as a high quotation may result in a loss of business and a low quotation may lead to lower profits or sometimes losses. Thus, in order to safeguard against errors in cost estimation the estimates should be reviewed and checked by someone other than the person responsible for its preparation.

In addition to the use for bidding and quotations, the cost estimates are applied for various purposes, some of which are :

(a) Budgeting

(b) Measuring efficiency of performance

(c) Make or buy, or lease or buy decisions

(d) Preparation of projected financial statements

When cost estimates are prepared with extreme care to serve as targets to be achieved, and for the purpose of controlling costs, these are termed as standard costs.

Cost Ascertainment

Cost ascertainment refers to the methods used and process of finding costs. It is concerned with computation of actual costs after cost has been incurred. In different types of industries, different methods are employed for cost ascertainment, such as job costing, contract costing, batch costing, operating costing, process costing etc. The basic principles underlying all these methods are the same, but these methods have been designed to suit the needs of individual conditions. The use of ascertaining actual costs is very limited because of the following reasons :

1. Ascertainment of actual costs has to no practical utility for control purposes.

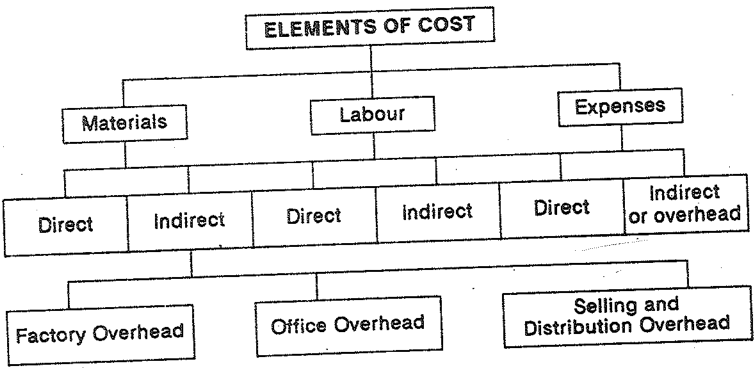
2. Actual costs cannot be used for the purpose of price quotations.

3. Actual costs are ineffective as means of measuring performance efficiency.

In spite of the above limitations, ascertainment of actual costs proves very useful in certain cases. For instance, unprofitable activities are revealed, losses or inefficiencies occurring in the form of idle time, excessive scrap, etc., come into light.

Thus, both cost estimation and cost ascertainment have proved useful in their respective fields.

Management is not satisfied merely with total cost data. For control purposes and managerial decisions, it is necessary to have a proper classification and analysis of total cost. For this purpose the total cost is analysed by the 'elements' of cost, i.e., material, labour and overheads. The various elements of cost may be illustrated as below :



By grouping elements of cost, the following divisions of cost are obtained :

1. Prime Cost = Direct Material + Direct Labour + Direct Expenses

2. Works or Factory Cost = Prime Cost + Works or Factory Overhead

3. Cost of Production = Works Cost + Office & Administration Overhead.

4. Cost of Sales or Total Cost = Cost of Producing + Selling & Distribution Overhead

The difference between the total cost of Sales and the total sales represent profit or loss.

All these terms are examined in detail.

1. Direct Material

Direct Material is all that material which can be conveniently measured and directly charged to the product. For example number in chair or table, cloth in shirt. However, in some cases though material forms a part of finished product yet it is a not treated as direct material, e.g., thread used in shirt stitching, nails and in shoes. This is because value of such materials in small and it is quite difficult and futile to measure them. Such materials are treated as indirect materials.

The following are examples of direct materials: (a) all raw materials like pig iron in foundry, fruits in canning industry; (b) materials specially purchased for a particular job; order or process; (c) partly manufactured materials passing from one process to another or from one operation to another; (d) basic or essential packing materials like cortons, boxes, papers, etc.

2. Direct Labour

Direct Labour is that labour which can be conveniently identified or attributed wholly to a particular job, product or process. It is the labour which is directly engaged on productive jobs or which can be directly identified with jobs. In other words it is all labour expanded in converting raw materials into manufactured articles or in altering the construction, composition or condition of the product.

Direct Labour includes payment made to the following groups of labour; (a) labour engaged in the actual production or in carrying out of an operation or process; (b) inspectors, analysis etc., specially required for such production; and (c) labour engaged in aiding the manufacture by way of supervision, maintenance, tool setting, transportation of materials, etc.

The wages of foremen, inspectors, etc., are not direct in this sense but as they are specifically connected with production and can be identified for costing purposes, they are treated as direct labour.

3. Direct Expenses

All expenses which are specially connected with production and can be identified for costing purposes, are treated as direct labour. Directly charged expenses are: (a) cost of special patterns, drawings, tools, etc. made for a specific product or process (b) hire charges of special tools or equipment for a particular job (c) maintenance costs of such special costs of such special tools or equipments, and (d) costs of any special process for manufacture.

Overhead

All indirect costs are termed as overhead. These are classified as follows :

1. *Factory Overhead:*

It refers to those expenses which are incurred in the factory and are concerned with the running of the factory. It includes indirect material, indirect labour and indirect expenses in producing an article. Examples of factory overheads are: rent of factory building, repairs, depreciation, wages of indirect workers, store-keeping expenses etc.

2. *Office of Administrative Overhead:*

This is the indirect expenditure incurred in formulating policies, planning and controlling the functions, directing and motivating the personnel of an organisation in the attainment of its objectives. In simple words, it includes all the expenses like expenses incurred by the administrative office, examples are all office expenses like rent, staff salaries, postage, stationery, expenses of general management, i.e. Board of Director etc.

3. *Selling Overhead:*

This is the cost of promoting sales and retaining customers, examples are: advertisement, salaries or salesmen, commission on sales, market research, etc.

4. *Distribution Overhead:*

In includes all the expenditure incurred from the time the product is completed until it reaches destination. It refers to all the expenses incurred in executing orders. It includes: (a) cost of warehousing the finished product, (b) cost of packing, (c) despatch and transportation cost; and (d) cost of reconditioning empty containers returned by customers for re-use.

Cost Sheet

The various expenses incidental to production, as recorded in the financial accounts, are extracted there from, analysed under various elements of expense heads, and put down in the form of a statement. Such as tabulated statement is called a 'cost sheet'. Bigg defines a cost sheet thus: "The expenditure which has been incurred upon production for a period is extracted from the financial books and the stores records, and set out in a memorandum or a statement. If this statements is confined to the disclosure of the cost of the units produced during the period, it is termed a cost sheet". The Terminology has defined a cost sheet as, "a document which provides for the assembly of the detailed sheet of a cost centre of cost units". In the light of this definition, it may be said that a cost sheet is a detailed statement of the elements of cost arranged in a logical order under different heads such as materials, labour and overheads. The period covered by the cost sheet may be a week, a fortnight or a month. Since the tabulated statement is essentially for the internal reporting of cost information to the management which may consist of non-technical men, it does not follow the financial double entry accounting style of debits and credits. The statement should be simple and easy to comprehend but not over simple so as to conceal the truth. It should endeavour but not over simple so as to conceal the truth. It should endeavour to give only so much information as is necessary to enable the management to asses the operational efficiency by directing their attention to any possible source of weakness. The statement should have a suitable descriptive heading indicating the period for which it is prepared. It should present cost information in total as well as per unit of quantity. If quantities are also presented, suitable headings should be given for each column. It would be better for the purpose of a comparison, identical information is also given side by side, relating to the cost date of the previous period.

A cost sheet serves the following purposes :

(a) It gives the break-up of the total cost by elements and divisions.

(b) It discloses the total as well as the cost per unit of production.

(c) It facilitates comparison.

(d) It helps the preparation of cost estimates for the submission of tenders.

(e) It helps the fixation of selling price.

(g) Within its . own limited sphere. It facilitates cost control by disclosing operational inefficiency, if any

As pointed out above, the cost sheet shown the cost per unit of each item of expenditure, the total of each elements, and sometimes, the proportion which each item bars to the total cost per unit, expressed as a percentage. If it is desired to include sales and profit also, the information may be presented in the form of a Production Account or Manufacturing Account or Working Account. Although the term 'account' is used in this context, it is not necessary that the production account should being the conventional T, form. Even with same title, the form may in the form of a statement. A production. account is, thus an extended form of a cost sheet. If the cost sheet should show profit also, it can be prepared in two parts. The first part is a cost sheet or a statement of cost as such, showing only the cost of production, while the second part, which may be called the statement of profit is made beginning and at the end of the period, the selling and distribution expenses, and sales. If both these statements should be put the 'T' form the account prepared would consist of three parts the first part to show the cost components and the cost of production, and second on part to show the cost components and the cost of production, and second on part to show stocks of finished goods and the cost of production to show stocks of finished goods and the cost of production to show the goods sold, and the third part selling and distribution expense, sales and profit for the period. Accordingly, there is nothing sacred about the heading or title, as long as the production account prepared in form either an account or a statement, shows, besides the cost of production; profit or loss also.

Unit or output Costing

This is a method of costing by the units of production. This is adopted where production is continuous and the units produced are identical. It is a simple method in which the total cost is divided by the number of units produced to determine the unit cost.

This system of costing is generally adopted where an undertaking is engaged in producing only one type of product or a few grades of the same product. Units of output are homogeneous or they can be measured by a common unit. The industries where this method of costing is adopted are collieries, breweries, sugar mills, cement works etc. The unit of cost is the unit in which the ultimate production is measured, e.g., per tonne of coal, per 100 bricks, per lb of paper etc.

Collection of Costs

1. *Materials*

Materials are purchased from time to time. If there is one product, these can be directly charged to production. Cost of raw materials consumed is ascertained in the following way. Opening stock of raw material + purchases - closing stock of raw materials. Normal loss of material is adjusted by inflating the issue price of materials.

Labour

A distribution is to be made between direct labour and indirect labour. Indirect labour form part of the overheads. The direct labour costs are collected periodically through pay rolls which are prepared separately for each section of the work.

2. *Overheads*

Overheads are classified into three broad categories factory overheads, office and administration overheads, and selling and distribution overheads. Overheads are usually charged at a pre- determined rate.

Treatment of Stock:

Stock may be raw materials, of finished goods, or of work-in- progress.

Stock of Raw Materials

In calculating the value of raw materials consumed during the period, the value of stock of raw material in the beginning of the period is added to the purchases and the value of stock of raw materials at the end of the period is substracted from the purchases.

PRODUCTION STATEMENT FOR THE PERIOD ENDING

Previous Year's Figures	Particulars	Current Year's Figures	
		Total Cost	Cost Per unit
	Raw Materials		
	Direct Wages		
	Chargeable Expenses		
	Prime Cost		
	Works Overheads		
	Works Cost		
	Office and Administration		
	Overheads Cost of Production		
	ADD: Opening Stock of Finished Goods		
	LESS: Closing Stock of Finished Goods		
	Cost of Goods Sold		
	Selling and Distribution		
	Overheads		
	Cost of Sales		
	Profit or Loss		
	Selling Price		

Stock of Finished Goods:

An adjustment for stock of finished goods is made after calculating cost of production. The opening stock of finished goods is added to and closing stock of finished goods is substracted from the cost of production. The balance figure will be the cost of goods sold.

Stock of Work-in-progress:

Work-in-progress means partly finished goods, i.e., goods which are in the manufacturing process. The cost of work-in-progress consists of cost of materials consumed.

The above information can also be presented in the form of an account in which case it will be known as Production Account.

Illustration: 1

ABC Cool Air Ltd., are manufacturing room coolers and the following details are given in respect of its factory operations for the year ended 31st December, 1980

	Rs.	Rs.
Work in Progress, 1st January, 1980:		
At Prime Cost	25,500	
Manufacturing Expenses	7,500	3,300
Work in Progress 31st December, 1980:		
At Prime Cost	22,500	
Manufacturing Expenses	4,500	27,000
Stock of Raw Materials 1st Jan. 1980		1,12,500
Purchase of Raw Materials		2,38,500
Direct Labour		85,500
Manufacturing Expenses		42,000
Stock of Raw Materials on 31st Dec. 1980		1,02,000

On the basis of the above data, prepare a statement showing the cost of production. Also indicate separately the amount of manufacturing expenses which enter into cost of production.

Solution:

ABC COOL AIR LTD.
(Cost Sheet)

For the month ending 30th September, 1975

	Rs.	Rs.
Opening Stock of Raw Materials 1.1.1980	1,12,500	
ADD: Purchase of Raw Materials	2,38,500	
	3,51,000	
LESS: Closing Stock of Raw Materials 31.12.1980	1,02,000	
Cost of Material Consumed		2,49,000
Direct Labour		85,500
		3,34,500
ADD: Work in Process 1.1.1980 (at prime cost)		25,500
		3,60,000
LESS: Work in Process 13.12.1980 (at prime cost)		22,500
Prime Cost		3,37,500
ADD: Manufacturing Expenses	42,000	
Manufacturing Expenses included in W.I.P. on 1.1.1980	7,500	
	49,500	
LESS: Manufacturing included in W.I.P. on 31.12.1980	4,500	45,000
Works Cost		2,92,500

Illustration: 2

The following details have been obtained from the cost records of Comet Paints Limited :

	Rs.
Stock of Raw Materials on 1st Sept., 1975	75,000
Stock of Raw Materials on 30th Sept., 1975	91,500
Direct Wages	52,500
Indirect Wages	2,750
Sales	2,11,000
Work-in-Progress on 1st Sept., 1975	28,000
Work-in-Progress on 30th Sept., 1975	35,000
Purchases of Raw Materials	66,000
Factory Rent, Rates and Power	15,000
Depreciation of Plant & Machinery	3,500
Expenses on Purchases	1,500
Carriage Outward	2,500
Advertising	3,500
Office Rent and Taxes	2,500
Traveller's Wages and Commission	6,500
Stock of Finished Goods on 1st Sept., 1975	54,000
Stock of Finished Goods on 30th Sept., 1975	31;000

Prepare a Production Statement giving the maximum possible break- up of cost and profit.

COMET PAINT LIMITED
(Production Statement)

For the month ending 30th September, 1975

	Rs.	Rs.
Stock of Raw Materials (1st Sept 1975)	75,000	
ADD: Purchases	66,000	
Expenses of Purchases	1,500	
	1,42,500	
LESS: Stock of Raw Materials (30th Sept. 1975)	91,500	
Materials Consumed		51,000
Direct Wages		52,000
Prime Cost		1,03,500
ADD: Work-in-Progress (1st Sept. 1975)		28,000
Factory Overheads :		
Indirect Wages	2,750	
Factory Rent, Rates & Power	15,000	
Depreciation of Plant & Machinery	3,500	
		21,250
		1,52,750
LESS: Work-in-Progress (30th Sept. 1975)		35,000
Works Cost		1,17,750
Office and Administration Overhead:		
Office Rent and Taxes		2,500
Cost of Production		1,20,250
ADD: Stock of Finished Goods (1st Sept. 1975)		54,000
		1,74,250
LESS: Stock of Finished Goods (30th Sept. 1975)		31,000
Cost of Goods Sold		1,43,250

Selling and Distribution Overhead		
Carriage Outward	2,500	
Advertising	3,500	
Travellers Wages & Commission	6,500	
		12,500
Cost of Sales		1,55,750
Profit		55,250
Sales		2,11,000

Illustration: 3

M/s.A B Shoes Co. manufactures two types of shoes. A and B production costs for the year ended 31st March, 1978 were:

Direct Material	Rs.15,00,000
Direct Wages	8,40,000
Production Overhead	3,60,000
	Rs.27,00,000

There was no work-in-progress at the beginning or at the end of the year. It is ascertained that.

(a) Direct material in type. A shoes consists twice as much as that in type B shoes.

(b) The direct wages for type A shoes were 60% of those of type A shoes.

(c) Production overhead was the same per pair of A and B type.

(d) Administration overhead for each type was 150% of direct wages.

(e) Selling cost was Rs.1.50 per pair.

(f) Production during the year were: Type A 40,000 pairs of which 36,000 were sold: Type B 1,20,000 pairs of which 1,00,000 were sold.

(g) Selling price was Rs.44 for Type A and Rs.28 for Type B per pair.

Prepare a statement showing cost and profit.

Solution:

STATEMENT OF COST AND PROFIT

For the year ending 31st March, 1978

Particulars	Type A		Type B	
	Total Rs.	Per Pair Rs..	Total Rs.	Per Pair Rs.
Direct Materials	6,00,000	15.00	9,00,000	7.50
Direct Wages	3,00,000	7.50	5,40,000	4.50
Prime Cost	9,00,000	22.00	14,40,000	12.00
Production Overhead	90,000	2.25	2,70,000	2.25
Production Cost	9,90,000	24.75	17,10,000	14.25
Administration Expenses	4,50,000	11.25	8,10,000	6.75
Cost of Production	14,40,000	36.00	25,20,000	21.00
LESS: Closing Stock	1,44,000	--	4,20,000	--
Cost of Goods Sold	12,96,000	36.00	21,00,000	21.00
Selling Expenses	54,000	1.50	1,50,000	1.50
Cost of Sales	13,50,000	37.50	22,50,000	22.50
Profit	2,34,000	6.50	5,50,000	5.50
Sales	15,84,000	44.00	28,00,000	28.00

NOTE :

1. Material Cost has been allocated as follows :

Suppose Type B's material cost is	=	X
Then Type A's material cost would be	:	2X
Therefore (1,20,000x + 80,000x)	=	Rs.15,00,000
2,00,000x	=	Rs.15,00,000
x	=	Rs.7.50 per pair
Therefore Type B Material Cost	=	Rs.7.50 per pair
Type A - material cost	=	Rs.15.00 per pair

2. Wages :

Let labour charge for Type A be	=	X
Let labour charge for Type B be	=	$\frac{60X}{100}$
Therefore		
40,000x + 1,20,000 x $\frac{60X}{100}$	=	Rs.40,000x + 70,000
	=	Rs.8,40,000
x	=	Rs.7.50 per pair
Therefore Type A's labour charges	=	Rs.7.50 per pair
Therefore Type B's labour charges	=	Rs.7.50 x $\frac{60X}{100}$
	=	Rs.4.50 per pair

Production overhead has been distributed in the ratio of production as charge for each pair is same.

Illustration: 4

A company makes two distinct types of vehicles, A and B. The total expenses during a period as shown by the books for the assembly of 600 of the Type A and, 800 of the Type B vehicles are as under :

	Rs.
Material	1,98,000
Direct Wages	12,000
Stores Overheads	19,800
Running Expenses of Machine	4,400
Depreciation	2,200
Labour Amenities	1,500
Works General	30,000
Administration and Selling	26.800

The other data available to you is : A:B

Material cost ratio per unit	1:2
Direct Labour ratio per unit	2:3
Machine utilisation ratio per unit	1:2

Calculate the cost of each vehicle per unit giving reasons for the basis of apportionment adopted by you.

Solution:

Statement of Cost

Particulars	Vehicle A Rs.	Vehicle B Rs.	Total Rs.
Material	54,000	1,44,000	1,98,000
Direct Wages	4,000	8,000	12,000
Stores Overhead	5,400	14,400	19,800
Running Expenses of Machines	1,200	3,200	4,400
Depreciation	600	1,600	2,200
Labour Amenities	500	1,000	1,500
Works General	10,000	20,000	30,000
Works Cost	75,700	1,92,200	2,67,900

Administration and Selling	7,573	19,237	26,800
Total Cost	83,273	2,11,427	2,94,700
Cost per unit of each vehicle	13,895	2,84,275	--

Notes :

The material cost has been apportioned on the basis of the cost ratio per unit multiplied by the units produced.

A		B
1		2
600 x 1	:	800 x 2
600:		1600
3		8

Direct Wages have been apportioned in the same manner

600 x 2	:	800 x 3
1.200		2.400

The stores overhead has been apportioned as a percentage of the materials cost.

$$\text{i.e.,} \quad \frac{19{,}800 \times 100}{1{,}98{,}000} = 10\%$$

$$A : \frac{54{,}000 \times 10}{100} = \text{Rs. } 5{,}400$$

$$B : \frac{1{,}44{,}000 \times 10}{100} = \text{Rs. } 14{,}400$$

Running expenses have been apportioned on the basis of the machine utilisation ratio per unit.

A		B
1		2
600 x 1	:	800 x 2
600	:	1600

Depreciation has to be apportioned on the same basis as the running expenses. Labour amenities have been apportioned on the basis of the direct labour ratio.

A	B
2 x 600	300 x 800 i.e., 1:2
$1 \times \frac{1500}{3}$	$2 \times \frac{1500}{3}$
= Rs.500	= Rs. 1,000

The works general expenses have been apportioned as a percentage of wages.

$$\text{i.e.} \quad \frac{30{,}000 \times 100}{12{,}000} = 250\%$$

$$A \quad \frac{4{,}000 \times 250}{100} = Rs.\ 10{,}000$$

$$B \quad \frac{8{,}000 \times 250}{100} = Rs.\ 20{,}000$$

Administration and selling expenses have been apportioned as percentage of the works cost.

$$\text{i.e.} \quad \frac{26{,}800 \times 100}{2{,}67{,}900} = 10.00\%$$

Tenders and Product Pricing

Frequently a manufacturer of capital goods and consumer durable goods is required to quote the price at which he can supply a particular article. Moreover, demand for certain articles like refrigerators, coolers and air conditioners increases during summer season. The manufacturer has to fix a competitive price keeping in view the likely impact of inflationary trends on the input. Before submitting a tender of fixing price a manufacturer must have a detailed information regarding cost of raw materials, based on these information he can prepare a cost sheet incorporating the likely increase or decrease in price levels of various components of production and submits the tenders or fixes the price of his products. It must, however, be noted here that the amount of overhead in these cases is to be estimated on certain basis. The following illustrations will be useful in preparing a tender or fixing price of consumer durable goods.

Illustration: 5

From the information you are required to prepare a cost sheet for the period ended on 31st December, 1978:

Opening Stock of Raw Materials Rs.20,000; Furchases Rs.1,70,000; Closing Stock of Raw Materials Rs.8,000; Direct Wages Rs.40,000; Other Direct Expenses Rs.20,000; Factory Overheads 100% of Direct Labour; Office Overheads 10% of work cost; Selling and Distribution Expenses Rs.4 per unit sold; Unit of finished product at the beginning 1000 units (value Rs.32,000) produced during the year 10,000 units at hand at the end of the period 2,000.

Also, find out the selling price per unit on the basis that profit mark-up is uniformly made to yield a profit of 20% of the selling price. There was no work-in-progress either at the beginning or at the end of the period.

COST SHEET FOR THE PERIOD ENDED 31ST DECEMBER, 1978

	Rs.	Rs.
Output 10,000 units		
Opening Stock of Material	20,000	
ADD: Purchases	1,70,000	
	1,90,000	
LESS: Closing Stock of Raw Material	8,000	

Material Used		1,82,000
Direct Wages		40,000
Other Direct Expenses		20,000
Prime Cost		2,42,000
ADD: Factory Overhead 10% of direct wages		40,000
Works Cost		2,82,000
ADD: Office Overheads 10% of works cost		28,200
Cost of Production		3,10,200

Cost of Production per unit $= \frac{3,10,200}{10,000} = 31.02$

Statement of selling price (Sales 9,000 units)

	Rs.
Total cost of production	3,10,200
ADD: Opening stock of finished products	32,000
	3,42,200
LESS: Closing stock of finished products (2000 units @ Rs.31.20)	62,040
Cost of goods sold	2,80,160
ADD: Selling and distribution overhead @ Rs.4 per unit sold	36,000
Cost of Sales	3,16,160
Profit (20% on selling price or 25% on cost price)	79,040
Sales	3,95,200

Selling Prince per unit $= \frac{3,95,200}{9,000} = Rs.43.92$

Notes:

1. Number of Units sold have been calculated as follows: Opening Stock of Finished Products + Produced during the year - Closing Stock of Finished Products.

= 1,000 units + 10,000 units - 2,000 units = 9,000 units.

2. Price per unit of the closing stock has been on the basis of cost of production per unit.

Illustration: 6

The undermentioned figures have been extracted from the books of the Metal Co. Ltd. whose accounting year close on 31st December, 1978.

	Rs.
Stock of Raw Materials on 1st January, 1978	35,000
Stock of Raw Materials on 31st January, 1978	4,900
Purchase of Materials	52,500
Factory Wages	95,000
Factory Expenses	17,500
Establishment Expenses	10,000
Completed Stock in Hand on 1st January, 1978	Nil
Completed Stock in Hand on 31st January, 1978	35,000
Sales	1,89,000

The company manufactured 4,000 electric stoves during 1978. The company was required to quote for the supply of 1,000 Electric Stoves during 1979. The Stoves to be quoted for, are of uniform quality and make, and similar to those manufactured during the period ended 31st December, 1978. However, as from 1st January, 1979, cost of material has increased by 15% and cost of factory labour by 10%.

Prepare a statement showing the price to be quoted, to give the same percentage of net profit on sales as was realised during 1978.

Solution:

COST AND PROFIT STATEMENT FOR THE YEAR, 1978

		Output 4,000 stoves	Amount per unit
	Rs.	Rs.	Rs.
Opening Stock of Materials	35,000		
Purchase of Materials	52,500		
	87,500		
LESS: Closing stock of material	4,900		
Value of materials used		82,600	20.65
Direct Wages		95,000	25.75
Prime Cost		1,77,600	44.40
ADD: Factory Expenses		17,500	4.37
ADD: Establishment Expenses		10,000	2.50
Cost of Production		2,00,100	51.27
ADD: Opening completed stock		Nil	--
		2,05,100	
LESS: Closing completed stock		35,000	
Cost of Sales		1,70,100	
Profit		18,900	
		1,89,000	

Percentage of profit on sales during the year 1978 is 10%

Statement showing Quotation Price for 1,000 stoves

	Rs.	Rs.
Materials required		
Material used + 15% increase	20,650	
Factory wages + 10% increase	3,098	23,748
	23,750	
	2,375	26,125
Prime Cost		49,873
Factory Expenses		4,375
		54,208
Establishment Expenses		2,500
Total Cost		56,748
ADD: Profit (10% on Selling Price) or 1/9 of cost of (56748 x 1/9 Production)		6,805
Selling Price		63,053

Illustration: 7

The Metal Products Company Ltd. manufactured and sold 1,000 Ceiling Fans during the year ended 31st December, 1978. The summarised accounts of the company are given below :

Manufacturing, Trading and Profit & Loss Account for the year ended 31st December, 1978.

	Rs.		Rs.
To Materials used	80,000	By Sales	4,00,000
Direct Wages	1,20,000		
Manufacturing Costs	50,000		
To Gross Profit	1,50,000		
	4,00,000		4,00,000

	Rs.		Rs.
To Salaries	60,000	By Gross Profit	1,50,000
To Rent, Rates & Insurance	10,000		
To General Expenses	20,000		
To Selling Expenses	30,000		
To Net Profit	30,000		
	1,50,000		1,50,000

For the year ending 31st December, 1979 it is proposed that :

(i) Output and Sales will be 1,200 Fans; (ii) Price of materials will rise by 20% on the previous year level; (iii) Wages per unit will rise by 5%; (iv) Manufacturing cost will be in proportion to the combined cost of materials and wages; (v) Selling expenses per unit will remain unchanged; and (vi) Other expenses will remain unaffected by the rise in output.

Prepare a statement of cost, showing the price at which the fans should be marketed so as to earn a profit of 10% on the selling price.

Solution:

Statement of Cost and Profit for the manufacture of 1,000 Fans for the year ended 31st December, 1978.

	Total cost	Cost Per Fan
Materials used	80,000	80.00
Direct Labour	1,20,000	120.00
Prime Cost	2,00,000	2,00.00
ADD: Factory Overheads	50,000	50,00

Balance B/F		12,50,000	250.00
ADD: Office Overheads			
Salaries	60,000		
Rent, Insurance	10,000		
General Expenses	20,000	90,000	90.00
Cost of Production		3,40,000	340.00
ADD: Selling Overheads		30,000	30.00
		3,70,000	370.00
Profit		30,000	30.00
Selling Price		4,00,000	400.00

ESTIMATE FOR THE MANUFACTURE OF
1200 FANS DURING 1979

	Rs.	Cost Per Fan Rs.	Total Cost Rs.
Materials	80		
ADD: 20% increase $\left[\frac{20 \times 80}{100}\right]$	16	96	1,15,200
Direct Labour	120		
ADD: 5% increase $\left[\frac{5 \times 120}{100}\right]$	6	126	1,51,200
Prime Cost		222	2,66,400
ADD: Manufacturing Overheads 25% on Prime Cost		55.50	66,600
		277.50	3,33,000
ADD: Office Overheads Salaries Rent		75.00	90,000
General Expenses		--	--
(unaffected by increase in output)		--	--
Cost of Production		352.50	4,23,000

ADD: Selling Overheads (Remain uncharged)	30.00	36,000
Cost of Sales	382.50	4,59,000
ADD: Profit (10% on Sales or 1/9 of Cost)	42.50	51,000
Selling Price	425.00	5,10,000

Note :

1. Profit is 10% on Selling Price

i.e.: $= \dfrac{100 \times 10}{90} = \%$ on cost

$= \dfrac{100}{9} \%$

Profit $= \dfrac{100}{9} \times \%$ of 4,59,000

$=$ Rs.51,000.

Illustration: 8

The following figures have been extracted from the Trial Balance of a Scooter Manufacturing Company for the year 1978.

Materials Used	16,00,000
Direct Wages	17,40,000
Factory Overheads Expenses	2,82,000
Establishment and General Expenses	2,78,000

For the year ending 31st December, 1979 it is estimated that :

(i) Each scooter will require materials worth Rs.2,900 and expenditure in direct wages Rs.1,200.

(ii) During 1979 the factory overhead expenses will bear the same ratio to direct wages as in 1978.

(iii) The percentage of establishment and general expenses on factory will be the same in 1979 as in 1978.

(iv) The company has decided to earn a profit of percent on the selling price.

You are required to prepare a statement showing the price at which the company should sell its scooter during 1979.

Solution:

Statement showing the price of Scooter

	Rs.
Materials	2,900
Direct Wages	1,200
Prime Cost	4,100
ADD: Factory overhead expenses (16.2% of Direct Wages)	194
Factory Cost	4,294
ADD: Establishment and General Expenses (7.7% of Works Cost)	331
	4,625
ADD: Profit (10% on Selling Price or 1/9 of Cost)	514
Selling Price	5,139

Notes:

1. Percentage of factory overhead to direct wages.

$$= \frac{2,82,600 \times 100}{1,74,000} \times = 16.2\ \%$$

2. Percentage of establishment and general expenses, to works cc

$$= \frac{2,78,000}{36,22,000} \times 100 = 7.7\ \%$$

TEST YOURSELF

1. The following information is given to you for a factory in 1981:

Materials	1,00,000.00
Direct Wages	75,000.00
Factory Expenses	60,000.00
Office and Administrative Expenses	44,000.00

On the basis of the above particulars ascertain the cost of a job to be done in 1982 for which materials required will be Rs.2,000 and wages will amount to Rs.4,000. What will be the quotation if a profit of 25% on Sales is to be earned.

Hint: Factory overheads on direct wages and office expenses on works cost.

2. Following information is extracted from a manufacturing concern:

Stock of finished goods 1.1.1972	25,500.00
Stock of raw materials 1.1.1972	10,000.00
Productive Wages	1,95,000.00
Sales of finished goods	6,06,000.00

Purchase of raw materials	2,90,000.00
Stock of raw materials 31.12.1972	12,500.00
Stock of finished goods 31.12.1972	24,000.00
Works overhead charges	43,000.00
Office and general expenses	36,000.00

Prepare a statement showing Total Cost in all its expenses.

LESSON : 12

MARGINAL COSTING

Marginal Costing, also known as Direct costing of variable costing is a special technique which may be used in conjunction with other techniques like standard costing and budgetary control. Its importance lies in the assistance it may give in solving managerial problems. This lesson discusses some of the important aspects of the technique.

Definition :

The Institute of Costs and Management Accountants, England defines marginal cost as "the amount at any given volume of output by which aggregate costs are changed if the volume of output is increased or decreased by one unit". And marginal costing is defined as "the ascertainment by differentiating between fixed costs and variable costs and of the effection of marginal profit of changes in volume of type of output". As referred to here, a unit may mean a single article a batch of articles, an order, a stage of production, a process or a department ie. it relates to the change in output in the capacity particular circumstance under consideration.

To the economist, marginal cost is an incremental cost he considers the addition to total cost which results from the production of one more unit of output. The Institute of Chartered Accountants, England defines marginal cost as "every expense (whether of production, selling or distribution) incurred by the taking of a particular decision". Marginal cost will be regarded here as the prime cost plus all overheads which vary with volume.

From the above definitions it is clear that under marginal costing, costs are classified as fixed and variable, and only variable costs are charged to products while fixed costs are charged to Profit & Loss Account fixed costs otherwise known as period costs, such as rent rates insurance etc. are incurred on the basis of period and therefore closed to Profit & Loss Account without carrying them to next period through closing stocks. On the other hand, marginal costs vary with the volume of output and therefore, are charged to production. Marginal costing is not a system of costing such as process or job costing but a technique which presents management with information enabling it to measure the profitability of an undertaking by considering the behaviour of costs. All companies use marginal costing to aid management decisions bsorption Costing vs. Marginal Costing.

Absorption Costing Vs. Marginal Costing :

Absorption Costing generally refers to the practice of charging all costs, both variable and fixed, to operations, processes, or products. Under this type of costing system, all costs, whether fixed or variable are treated as product costs and as such the product is made to bear the burden of full costs even though they have no relevance to current operations. But under management costing technique only variable costs are regarded as product and the fixed costs are transferred to the Profit & Loss Account.

Absorption Costing also differs from marginal costing with regard to inventory valuation. While under marginal costing, closing stock are valued at the marginal cost, they are valued at full cost under absorption costing. Such a method of valuation has the effect of carrying over fixed costs from one period to another. Such a procedure would normally affect the trading results and vitiate cost comparison also.

In absorption costing, fixed overheads can never be absorbed exactly because of the difficulty in forecasting costs and volume of output. But the exclusion of fixed costs from product costs in the case of marginal costing does not give rise to the problem of over-or under-absorption of fixed overhead.

In the case of absorption costing, profit is the difference between sales revenue and total cost. But under the marginal costing technique, the excess of sales revenue over marginal cost of sales is known as contribution. It is the contribution that is the guiding principle of decision making under the marginal costing technique. In order to ascertain net profits fixed costs are absorbed from total contribution. The following example would illustrate the various points explained above.

Example :

SPN Ltd supplies the following particulars for the preparation of Income Statement for the year ended 31.12.1980.

Sales	=	2000 units at Rs.50 per unit
Manufacturing Cost :		
Fixed	=	Rs.19,800
Variable	=	2,200 units at Rs.31 per unit
Closing Stock	=	200 units
Selling and Administrative Expenses :		Rs.
Fixed	=	5,600
Variable	=	4,400

Solution:

INCOME STATEMENT OF SPN LTD
(for the year ended 31.12.1980)

(i) Under Absorption Costing Method	Rs.	Rs.
Sales (2,000 units at Rs.50/- unit)		1,00,000
Less: Cost of Sales Manufacturing		
Cost-Fixed	19,800	
Variable	68,200	
	88,000	
Less: Closing Stock		
(200 units at Rs.40/- unit)	8,000	80,000
Gross Profit		20,000
Less: Selling and Administrative Expenses		
Fixed	5,600	
Variable	4,400	10,000
Net Profit		10,000

Workings:

Total Cost for 2,200 units 88,000

Cost of Closing Stock of 200 units $\frac{88,000 \times 200}{2,200} =$ Rs.8,000

(ii) Under Marginal Costing Method :	Rs.	Rs.
Sales (2,000 units at Rs.50/- each)		1,00,000
Less: Variable cost of sales		
Variable manufacturing cost	68,200	
Less: Closing Stock	6,200	62,000
Variable Gross Margin		38,000
Less: Variable Selling and Administrative Expenses		4,400
Contribution Margin		33,600
Less: Fixed Cost		
Manufacturing Cost	19,800	
Selling and Administration	5,600	25,400
Net Profit		8,200

Workings :

Closing Stock is valued on the basis of the variable manufacturing cost

$$\frac{\text{Rs. } 68{,}200 \times 200}{2{,}200} = \text{Rs. } 6{,}200$$

The difference in profits is due to the difference in the valuation of closing stock.

A proforma Marginal Cost Statement is given below which may be used by a concern producing and selling multiple products.

	Marginal Cost Statement Products						
	A		B		C		
	Per Unit Rs.	Total Rs.	Per Unit Rs.	Total Rs.	Per Unit Rs.	Total Rs.	Rs.
Sales	x	x	x	x	x	x	
Less: Marginal Cost Prime Cost	x	x	x	x	x	x	
+ Variable Overheads	x	x	x	x	x	x	
Contribution	x	x	x	x	x	x	xx
Less: Specific Fixed Costs		x		x		x	xx
Net Contribution		x		x		x	xx
Less: General Fixed Costs		–		–		—	xx
Net Profit							xx

The presentation is very helpful in order to decide what each product contributes towards fixed costs and profit. The format shown above should be used for working out problems under marginal cost method. From the marginal cost statement the following equations may be derived :

Sales - Marginal Cost = Contribution

Fixed Cost + Profit = Contribution

Combining these two equations, we get the fundamental marginal cost equation :

Sales - Marginal Cost = Fixed Cost + Profit.

Symbolically :

S - M = F + P = C (Contribution)

S - C = M

M + C = S

C - F = P

C - P = F

F - C = L (Loss)

F - L = C

C + L = F

These equation may be used for solving problems of different types involving cost volume profit relationship.

Role of Contribution :

Contribution is the difference between sales and marginal costs and it contributes towards fixed costs and profit. When a manufactures more than one product, the computation of profit realised on individual products may be difficult due to the problem of apportionment of fixed costs to different products. The rationale of contribution lies in the fact that fixed costs are done away with under marginal costing. The concept of contribution helps to determine the break-even point, profitability of products, departments, etc., to select product mix for profit maximisation, and to fix selling prices under different circumstances such as trade depression, export sales price discrimination etc. Contribution is the definite test whether a product or process is worthwhile to continue among different products or processes.

Example :

The following information has been obtained from the accounts of a departmental store having three departments :

Particulars	Departments		
	A Rs.	B Rs.	C Rs.
Sales	50,000	75,000	1,25,000
Marginal Costs	45,000	50,000	75,000
Fixed Costs (apportioned on the basis of sales)	10,000	15,000	25,000
Total Cost	55,000	65,000	1,00,000
Profit (Loss)	(5,000)	10,000	25,000

On the basis of the above information management wants to discard Department-A immediately as it shows a loss. Also give your opinion on the comparative profitability of different departments if specific. Fixed costs are ascertained to be Rs.2,500 for Department-A, Rs.27,500 for Department-B and Rs.15,000 for Department-C, the remaining Rs.5,000 being general fixed cost.

Prepare appropriate statements and also give your comments explaining the position presented in the statement.

Solution:

(a) When all fixed costs are regard as general fixed costs.

MARGINAL COST STATEMENT

Particulars	Department			Total
	A Rs.	B Rs.	C Rs.	Rs.
Sales	50,000	75,000	1,25,000	2,50,000
Less: Marginal Cost	45,000	50,000	75,000	1,70,000
Contribution	5,000	25,000	50,000	80,000
Less: Fixed Cost	–	–	–	50,000
Profit				30,000

Comment :

The loss of Department-A, Rs.5,000 is due to the arbitrary basis of the apportionment of fixed cost to different departments. If the Department-A is closed the total contribution as well as profit will be reduced by Rs.5,000. Hence it is not admissible to close Department-A.

(b) When there are specific as well as generalized costs :

MARGINAL COST STATEMENT

Particulars	Department A Rs.	Department B Rs.	Department C Rs.	Total Rs.
Sales	50,000	75,000	1,25,000	2,50,000
Less: Marginal Cost	45,000	5,000	75,000	1,70,000
Contribution	5,000	70,000	50,000	80,000
Net Contribution	2,500	(2,500)	35,000	35,000
Less: General Fixed Costs	–	–	–	50,000
Cost Profit				30,000

Comment :

According to the above statement it is the Department-B which is to be closed since it shows a loss of Rs.2,500 after apportioning the specific fixed costs. If the Department-B is closed, the total contribution as well as profit would increase by Rs.25,000. Hence it is always better to close Department-B if either its working cannot be improved or the price of its products cannot be increased. The position of closure of Department-B can be shown as below :

STATEMENT OF COMPARATIVE PROFITABILITY

Particulars	Department		Total
	A Rs.	C Rs.	Rs.
Sales	50,000	1,25,000	1,75,000
Less: Marginal Cost	45,000	75,000	1,20,000
Contribution	5,000	50,000	55,000
Less: Specific Fixed Costs	2,500	15,000	17,500
Net Contribution	2,500	35,000	37,500
Less: General Fixed Costs	–	–	50,000
Profit			32,500

It is clear from the above statement that the profit has increased on the closure of Department-B by Rs.2,500 which exactly equal to the loss incurred by that Department before closure.

Separation of Semi-Variable Expenses into Fixed and Variable :

Determination of marginal cost is the main task to be carried out in the application of marginal costing. In this task the challenging problem is the segregation of semi-variable overheads into fixed and variable components.

The following methods may be used for this purpose :

Comparison by period or level of activity method :

Under the method or level of output at two levels is compared with Corresponding level of expenses are obtained by ratio :

$$\frac{\text{Change in amount of expense}}{\text{Change in activity or quantity}}$$

This and each of the other methods would be explained in detail with the help of the following example.

Example :

The information extracted from the books of a manufacturing company relates to semi-variable expense (maintenance cost). Find out the fixed and variable elements and also the semi-variable expense for the month of July when the volume of output would be 8,000 units.

Month	Production Units	Maintenance Cost (Rs. in '000's)
January	4,000	220
February	2,000	180
March	5,000	260
April	10,000	380
May	7,000	300
June	8,000	340

Solution :

Taking the levels of activity of any two months say, March and May the variable element may be calculated as follows :

$$\text{Variable element} = \frac{\text{Change in the amount of expense}}{\text{Change in quantity}}$$

$$= \frac{\text{Rs. } 3,00,000 - \text{Rs. } 2,60,000}{(7,000 - 5,000) \text{ units}} = \frac{\text{Rs. } 40,000}{2,000 \text{ units}}$$

$$= \text{Rs.}20 \text{ per unit.}$$

Month	Production Units Rs.	Semi Variable Expense Rs.	Variable Rs.	Fixed Rs.
May	7,000	3,00,000	1,40,000	1,60,000
March	5,000	2,60,000	1,00,000	1,60,000
	2,000	40.000		

Total semi-variable expense for the month of
July = (8,000 units x Rs.20) + Rs.1,60,000)
= 3,20,000.

This method is very simple though not scientific and accurate. It is important that in selecting the period or level of activity and costs incurred in such periods, care is taken to see that the figure are not distorted by any abnormal factors.

(b) High and Low Points Method :

This method is similar to the previous one except that the data relating to high and low business activity out of various levels are considered.

Considering the high and low points from the previous examples.

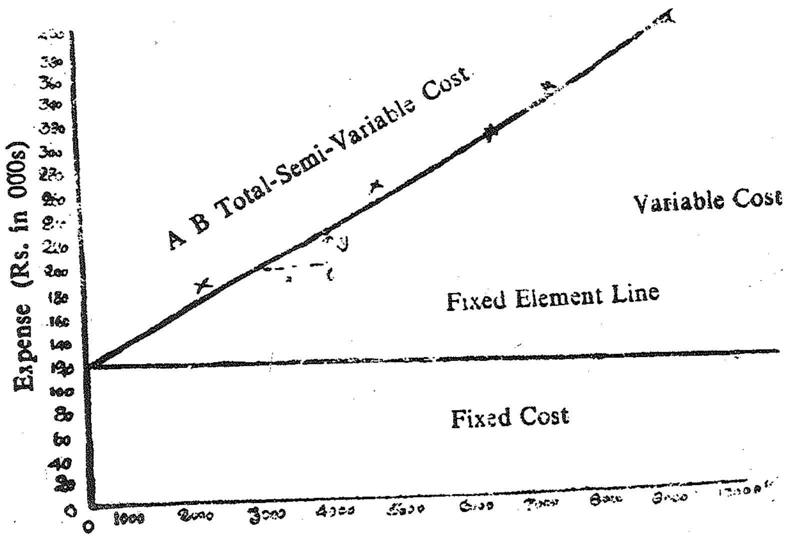
Month	Production Units Rs.	Expense Rs.	Variable Rs.	Fixed Rs.
April	10,000	3,80,000	2,50,000	1,30,000
Feb.	2,000	1,80,000	50,000	1,30,000
	8,000	2,00,000		

Variable element = 8,000 units = Rs.25/-

For July, the total
semi-variable expense = (Rs.25 x 8,000) + Rs.1,30,000 = (Rs.3,30,000)

(c) Scattergraph Method :

According to this method costs, are plotted on along Y axis and production unit on X axis of a graph. Then the various points are plotted on the graph taking the amount of expense corresponding to each volume of production. A 'line of best fit' is to be drawn by inspection through these points extending it to intersect the vertical axis. This is total cost line and the point of intersection would give the amount of fixed element. A line drawn parallel to the horizontal axis passing through this point of intersection would be the fixed element line. The difference between fixed cost and total cost lines would be the variable element. The resulting graph for the data given in the previous example is shown below:



(Units) Volume of Production
Scattergraph of Semi-variable Expense

From the above graph it is observed that the fixed cost element is Rs.1,60,000. To obtain variable cost element any two points A and B on total cost line may be taken and the change of amount corresponding to change in volume be observed.

Thus, variable element = Slope of the curve

$$= \frac{BC}{AC} = \frac{\text{Rs. } 26{,}000}{1000 \text{ units}}$$

= Rs.26/- per unit

Substituting the values to the straight line equation

$$Y = a + bX, \text{ we may obtain}$$

$$Y = 1,60,000 + 26\ X \text{ where.}$$

For July, putting $X = 8,000$ units

$$Y = 1,60,000 + 8,000 \times 26$$

$$= \underline{\text{Rs.}3,68,000}$$

(b) **Method of Least Squares :**

This method is based on the basic regression equation Y = a + bX where 'a' represents fixed element and 'b' the degree of variability. Assuming a linear equation Y = a + bX and putting various values we may observe :

$$Y_1 = a + bX_1$$

$$Y_2 = a + bX_2$$

$$\underline{Y_3 = a + bX_3}$$

$$\Sigma Y = Na + \Sigma bX \qquad \ldots\ \ldots\ (1)$$

Again multiplying both sides of the linear equation Y = a + b x by X we may obtain for a number of observations the following values :

$$X_1 Y_1 = aX_1 + bX^2{}_1$$

$$X_2 Y_2 = aX_2 + bX^2{}_2$$

$$\underline{X_3 Y_3 = aX_3 + bX^2{}_3}$$

$$\Sigma Xy = a\Sigma X + b\Sigma X^2 \qquad \ldots\ \ldots\ (2)$$

By solving the equations (1) and (2) through simultaneous equation method we may obtain the values of 'a' and 'b'. This is explained below considering the same example used for the previous methods.

Solution :

Month	Production Units (in '000's) (X)	Semi-variable Expense (Rs. in '000's) (Y)	X^2	Xy
January	$x_1 = 4$	$y_1 = 220$	16	880
February	$x_2 = 2$	$y_2 = 180$	4	360
March	$x_3 = 5$	$y_3 = 260$	25	1,300
April	$x_4 = 10$	$y_4 = 380$	100	3,800
May	$x_5 = 7$	$y_5 = 300$	79	2,100
June	$x_6 = 8$	$y_6 = 340$	64	2,720
	$\Sigma = 36$ $N = 6$	$\Sigma xy = 1{,}680$	$\Sigma x^2 = 258$	$\Sigma xy =$ 11,160

Substituting the values to the equations (1) and (2).

$$\Sigma y = Na + b\Sigma x \qquad \ldots \ldots (1)$$

$$\Sigma xy = \Sigma x + b\Sigma x \qquad \ldots \ldots (2)$$

We get,

$$1{,}680 = 6a + 36b \qquad \dots \dots (3)$$

$$11{,}160 = 36a + 258b \qquad \dots \dots (4)$$

Solving equations (3) and (4) we may obtain

$$b = \text{Rs.}25\ 5/7 = \text{(Variable element)}$$

$$a = \text{Rs.}1{,}23{,}333\ 1/3 = \text{(Fixed element)}$$

[Note : In order to reduce the figure work where large figures are given, the scales may be charged by taking deviations from arithmetic mean instead of considering the actual figures.]

For July, the semi-variable cost applying the equation

$$Y = \text{Rs.}1{,}23{,}333\ 1/3 + 8{,}000 \times 25\ 5/7$$

$$= 1{,}23{,}333 + 2{,}05{,}714$$

$$= 3{,}29{,}047/-$$

Managerial uses of Marginal Costing :

Before discussing the managerial uses of marginal costing it would be necessary to mention a few advantages of this technique:

Advantages :

(a) Marginal costing is simple to understand and can be combined with standard costing and budgeting control.

(b) Elimination of fixed overheads from the cost of production avoids the effect of varying charges per unit and also prevents the carry forward of a portion of the current period's fixed overhead to the subsequent period. As such costs and profits are not vitiated cost comparisons become meaningful.

(c) There is no problem of over-or-under-absorbed overheads.

(d) Practical cost control is greatly facilitated through flexible budgeting and by a clear-cut division of costs into fixed and variable elements.

(e) It helps in profit planning by break-even charts and profit graph. Comparative profitability can easily be assessed and brought to the notice of management for decision-making.

(f) It is an effective tool for determining efficient sales of production policies or for taking pricing and tendering decisions particularly when the business is at a low ebb.

From the advantages stated above, the following may be listed as specific managerial uses :

(a) Cost Ascertainment:

Marginal costing technique facilities not only the recording of costs but their reporting also. The classification of costs into fixed and variables components makes the job of costs ascertainment more easy. The main problem in this regard is only the segregation of semi-variables cost which may be overcome by adopting any of the methods already explained for the purpose

(b) **Cost Control:**

Marginal Cost statement can be understood more easily by the management than those presented under the absorption costing. Bifurcation of costs into fixed and variable enables to exercise control over production cost and thereby effect efficiency. In fact, while variable cost are controlled at the lower levels of management, fixed cost can be controlled at the top level. Under technique management could study the behaviour of costs at varying conditions of output and sales and thereby exercise better control over costs.

(c) **Decision-making:**

Modern managements are faced with a number of decision making problems everyday. Profitability is the criterion for selecting the best course of action. Marginal costing through 'contribution' assets management in solving problems. Some of the decision-making problems that can be solved marginal costing are : (a) profit planning (b) pricing of products (c) make or buy decisions (d) suitable product mix etc.

Limitations of Marginal Costing:

Despite its superiority over absorption costing, the marginal costing technique has its own limitations.

(a) Segregation of all costs into fixed and variable costs is very difficult. In practice, a considerable technical difficulty arises in drawing a sharp line of demarcation between fixed and variable costs. The distinction between them holds good only in the short run and in the long run, however, all costs are variable.

(b) In marginal costing greater importance is attached to sales function thereby delegating the production function comparatively to secondary position. But the real efficiency of a business is to be assessed only by considering the selling and production function together.

(c) The elimination of fixed costs from the valuation of inventories is illegal since fixed costs are also incurred in the manufacture of goods. Further it results in the understatement of the value of stocks which is neither the cost nor the market price.

(d) Pricing decisions cannot be based on contribution alone. Sometimes the contribution will be unrealistic when increased production and sales are effected either through the extensive use of existing machineries or by replacing manual labour by machines. Another possibility is that there is the danger of too many sales being effected at marginal cost.

(e) Although the problem of over-or under-absorption of fixed over-heads could be overcome to a certain extent, the same problem may still persist with regard to variable overheads.

(f) The application of this technique is limited in the case of industries in which, according to the nature of business, large stocks have to be carried by way of work-in-progress (eg: contracts)

PRACTICAL APPLICATION OF MARGINAL COSTING

(a) Profit Planning :

A business concern exists with the objective of making profits and profits are the yardstick of its success. Profit planning is therefore necessarily a part of operations planning. It is the basis of planning cash-capital expenditure and pricing. If growth and survival of a business are to be·ensured. Profit Planning becomes absolute necessity. Marginal costing assist profit planning through the calculation of contribution ratio. It enables the planning of future operations in such a way as to either maximise profits or maintain specified level of profit. Basic marginal costing equations which are useful in profit planning are as follows :

Profit/volume ratio (P/V ratio). It is the ratio of contribution to sales and symbolically.

$$\text{C/S ratio or P/V ratio} = \frac{\text{Contribution (C)}}{\text{Sales (S)}} \qquad (1)$$

$$\text{Contribution} = \text{Sales} \times \text{P/V ratio} \qquad (2)$$

$$\text{Sales} = \frac{\text{Contribution}}{\text{P/V ratio}} \qquad (3)$$

Break-Even Point (BEP): It is that volume of sales or production where there is no profit or no loss. It may be derived from the equation (3) we may get,

$$\text{BEP (in Rs.)} = \frac{\text{Contribution at BEP}}{\text{P/V ratio}}$$

At BEP, the contribution will be equal to fixed cost and therefore, the formula may be restructured as follows :

$$BEP = \frac{\text{Fixed cost}}{\text{P/V ratio}}$$

$$BEP\ (\text{in units}) = \frac{\text{Fixed cost (F)}}{\text{Contribution per unit}}$$

Margin of Safety (MS):

It represents the difference between the sales or production at the selected activity and the break-even sales or production.

MS Sales at the Selected activity – BEP.

$$\text{Sales at the selected activity} = \frac{C}{\text{P/V ratio}}$$

$$BEP = \frac{F}{\text{P/V ratio}}$$

$$\text{Therefore, MS} = \frac{C}{\text{P/V ratio}} - \frac{F}{\text{P/V ratio}} = \frac{\text{Profit (P)}}{\text{P/V ratio}}$$

Where $C - F = P$

Margin of safety is also presented in percentages as follows :

$$\frac{\text{MS [Sales]} \times 100}{\text{Sales at selected activity}}$$

Illustration: 1

From the following information, calculate BEP and determine the net profit if sales are 25% above BEP.

	Rs.
Selling price per unit	50
Direct material cost per unit	20
Direct wages per unit	10
Variable overheads per unit	7.5
Fixed overheads (total)	50,000

MARGINAL COST STATEMENT

Solution:

	Rs.	Rs.
Selling price per unit		50.00
Less: Marginal cost per unit		
Materials	20.00	
Wages	10.00	
Variable overheads	7.50	37.50
Contribution		12.50

$$P/V \text{ ratio } \frac{C}{S} \times 100 \ \frac{12.50}{50} \times 100 = 25\ \%$$

$$BEP = \frac{F}{P/V \text{ ratio}} = \frac{Rs.\ 50,000}{25} \times 100 = Rs.\ 2,00,000$$

BEP	=	Rs.2,00,000
25% BEP	=	Rs. 50,000
Total Sales	=	Rs.2,50,000

Contribution = Sales x P/V ratio

Contribution at Rs.2,50,000. = Sales Rs.2,50,000 x 25%

Contribution	=	Rs.62,500
Less: Fixed Cost	=	Rs.60,000
Net Profit	=	Rs.12,500

Illustration: 2

A toy manufacturer earns an average net profit of Rs.6 per unit in a selling price of Rs.30 by producing and selling 30,000 units at 60% of the potential capacity. Composition of his cost of sales is :

	Rs.
Direct Material	8.00
Direct Wages	2.00
Works Overhead	12.00 (10% fixed)
Sales Overhead	2.00 (25% variable)

During the current year, he intends to produce the same number but anticipates that :

(a) his fixed charges will go up by 10%

(b) rates of direct labour will increase by 20%

(c) rates of material will increase by 5%

(d) sell in a price cannot be increased

Under these circumstances, he obtains an order for a further 20% of his capacity. What minimum price will you recommend for accepting the order to give the manufacturer an overall profit of Rs.1,80,000.

Solution:

Marginal Cost Statement for current year :

	Rs.	Number of units Capacity per unit Rs.	30,000 60% Total Rs.
Sales		30.00	9,00,000
Less: Marginal Cost			
Material	8.40		
Wages	2.40		
Variable works overhead	6.00		
Variables sales overhead	0.50	17.30	
			5,19,000
Contribution		12.70	3,81,000
Less: Fixed Cost			
Works overhead	1,80,000		
Add: 10% increase	18,000	1,98,000	
Sales overheads	45,000		
Add: 10% increase	4,000	49,500	2,47,500
Net Profit			1,33,500

Calculation of minimum price for special order (10,000 units at 20% capacity)

	Rs.
Planned Profit	1,80,000
Actual Profit	1,33,500
Increase in Profit to be earned from the order	46,500

Minimum Price :

Marginal cost to be recovered 10,000 x 17.3	=	1,73,000
Increase in profit to be earned	=	46,500
Minimum selling price for 10,000 units	=	2,19,500
Minimum selling price per unit	=	Rs.21.95

Measurement of Profitability:

Marginal costing is useful to measure the profitability of a product or process. Profitability is measured with reference to contribution. Another important factor essential for the determination of profitability is the "Key Factor" or "Limiting Factor" or "Principal Budget Factor" "Scarce Factor". A key factor may be defined as the factor which over a period, will limit the volume of output or which puts a limit on the efforts of the management to produce as many units of the limiting factor, but sometimes materials labour, capital, plant capacity may be limiting factors. When contribution and key factor are given one can assess the relative profitability using the following formula:

$$\text{Profitability} = \frac{\text{Contribution}}{\text{Key Factor}}$$

For example, when rupee sales is the key factor, the profitability is measured by P/V ratio, when labour is in short supply, the profitability is determined by dividing the contribution by labour hours and so on.

Illustration: 3

The following particulars are taken from records at a company engaged in manufacturing two products X and Y from a certain raw material :

	Product X (per unit) Rs.	Product Y (per unit) Rs.
Sale	25.00	50.00
Material Cost (Rs.2.5 per kg)	5.00	12.50
Direct Labour (Rs.1.5 per hour)	7.50	15.00
Variable Overhead	2.50	5.00
Total Fixed Overhead	5,000	

Comment on the profitability each product when :

(i) total sales in value is limited

(ii) raw material is in short supply

(iii) production capacity is the key factor and

(iv) when total availability of raw material is 4,000 kg. and maximum sales potential of each product is 1,000 units find the product mix to yield maximum profit. Determine the maximum profit.

Solution:

Statement showing comparative profitability :

	Product X (per unit)		Product Y (per unit)	
	Rs.	Rs.	Rs.	Rs.
Sales	–	25.00	–	50.00
Less: Marginal Cost:				
Materials	5.00		12.50	
Direct Wages	7.50		15.00	
Variable Overhead	2.50	15.00	5.00	32.50
Contribution		10.00		17.50

(i) When total sales in value is limited the profitability is measured by

$$\text{P/V ratio} = \text{Rs. } \frac{10 \times 100}{25} = 40\,\%\,\text{X}\,; \frac{17 - 50}{50} \times 100 = 35\,\%$$

Since the P/V ratio of X is higher, it is more profitable.

(ii) $\text{Contribution per kg.} = \dfrac{\text{Rs. } 10}{2 \text{ kg}} = \text{Rs.5/per kg.}$

$$\frac{\text{Rs. } 17.5}{5 \text{ kg}} = \text{Rs.3.5 per kg.}$$

When raw material is in short supply, contribution per kg. of raw material for product X is more and therefore more profitable.

(iii) Contribution per hour = $\frac{\text{Rs. } 10}{5 \text{ Hrs.}}$ = Rs.2 per hour.

$\frac{\text{Rs. } 17.50}{10 \text{ hrs.}}$ = Rs.1.75 per hour.

When the production capacity is limited, contribution per hour of product X is more and therefore, more profitable.

(iv) It is case of two key factors. When raw material is in short supply, product X is more profitable. So maximum ie., 1,000 units of X could be produced consuming 2,000 kg. of raw material, and with the balance quantity i.e.

4000 kg. - 2000 kg = 2000 kg., 400 units (2000 kg. % 5 kg. of product X will be produced. Hence for maximum profit, the product mix would be :

Product X 1000 units	=	2000 kg
Product Y 400 units	=	2000 kg
Total quantity of materials available	=	4000 kg

Maximum contribution from		Rs.
1000 units of product X	=	10,000
400 units of product Y	=	7,000
Total contribution	=	17,000
Less: Fixed overhead	=	5,000
Maximum profit	=	12,000

(c) Level of Activity Planning:

Business concern may have plans either to expand or contract level of activities depending upon the conditions prevailing in the market. Such a planning is to be considered before the events over-take the business. Marginal costing is very useful for taking such decisions by enabling management to compare the contribution at different level of activities.

Illustration: 4

FOLLOWING IS THE COST STRUCTURE OF AB LIMITED

	Level of Activity		
	60%	70%	80%
Output (in units)	2,400	2,600	32,000
	Rs.	Rs.	Rs.
Materials	48,000	56,000	64,000
Wages	14,400	16,800	19,200
Factory Overheads	25,600	27,200	28,800
Factory Cost	88,000	1,00,000	1,12,000

The factory is considering an increase of production to 90% level of activity. It is not expected that there will be any increase in fixed overheads at this level. The management requires a statement showing all details of factory costs at 90% level of activity.

Solution:

MARGINAL COST STATEMENT

	Level of activities output Total Cost Rs.	90% 3,600 units per unit Rs.
Materials	72,000	20.00
Wages	21,600	6,00
Variable Overheads	14,400	4.00
Marginal Cost	1,08,000	30.00
Fixed Overhead	16,000	
Total Factory Cost	1,24,000	

Note : Factory overheads increase by Rs.1,600 each level of activity Therefore variable overheads must be $\frac{\text{Rs. } 1000}{400 \text{ units}}$ = Rs. 4 per unit. At 80% level of activity factory overheads are Rs.28,800 of which variable cost are Rs.12,800 (Rs.4 x 3,200) resulting in fixed overheads of Rs.16,000 (Rs.28,800 - Rs.12,800).

Profitable Mix of Sales:

Business which have a variety of product lines, shall employ marginal costing in order to determine the most profitable sales mix from a number of selected alternatives.

Illustration: 5

The directors of AB Ltd. are considering the sales budget for the next budget period. The following information has been made available from the cost records :

	Product Z (per unit)	Product Y (per unit)
Direct Materials	Rs. 40	Rs. 50
Selling Price	Rs. 120	Rs. 200
Direct Wages @ Rs.2 per hour	10 hours	15 hours

Variable Overhead : 100% of Direct Wages.

Fixed Overheads : Rs.2,000/- p.a

You are required to present to the Management a statement showing the marginal cost of each product and to recommend which of the following sales mixes should be adopted:

(a) 450 units of Z and 300 units of Y

(b) 900 units of Z only

(c) 600 units of Y only

(d) 600 units of Z and 200 units of Y

MARGINAL COST STATEMENT

	Product Z (per unit)		Product Y (per unit)	
	Rs.	Rs.	Rs.	Rs.
Selling Prize		123		200
Less: Marginal Cost :				
Direct Materials	40		50	
Direct Wages	20		30	
Variable Overhead	20	80	30	110
Contribution		30		90

SELECTION OF SALES ALTERNATIVES

	Products		Total
	Z Rs.	Y Rs.	Rs.
(a) 450 units of Z and 300 units of Y			
Contribution	18,000	27,000	45,000
Less: Fixed overheads			20,000
Profit			25,000
(b) 900 units of Z only			
Contribution	36,000	...	36,000
Less: Fixed Cost			20,000
			16,000

	X Rs.	Y Rs.	Total Rs.
(c) 600 units of Y only			
Contribution	...	54,000	54,000
Less: Fixed Cost			20,000
Profit			34,000
(d) 600 units of Z and 200 units of Y			
Contribution	24,000	18,000	42,000
Less: Fixed Cost			20,000
Profit			22,000

Thus alternative (c) is the one recommended.

Marginal Costing and Pricing:

Determining the prices of products manufactured by a company is often considered to be a difficult problem. However, the basic problem involved in pricing is the matching of demand and supply. Marginal costing is sometimes used to determine prices, a simple and familiar example being the railway ticket. The normal fare will usually be more than the charge collected for excursion fare (concessional fare) for, the normal fare is calculated to cover all the railway costs including fixed which are a considerable item; whereas the excursion fare will probably cover only the marginal cost (which is relatively small) and some contribution towards profit. The marginal costing technique can help management in fixing prices in such special circumstance as;

(a) A trade depression in the industry,

(b) Spare capacity in the factory,

(c) A seasonal fluctuation in demand,

(d) When it is desired to obtain a special contract

Illustration: 6

MI Ltd, manufactures and sells light engineering goods. Due to competition, the company proposes to reduce its selling price. If the present level of profit is to be maintained, indicate the number of units to be sold if the proposed reduction in selling prices is 5%, 10% and 15%.

The following additional information is available :

	Rs.	Rs.
Present Sales (60,000 units)	–	1,50,000
Variable Cost (60,000 units)	90,000	
Fixed Cost	35,000	1,25,000
	Net Profit	25,000

Solution:

MARGINAL COST STATEMENT

			No. of units: 60,000	
	Present Price (Rs.)	Price 5% (Rs.)	Reductions	
			10% (Rs.)	15% (Rs.)
Sales	1,50,000	1,42,500	1,35,000	1,27,500

Less: Marginal Cost	90,000	90,000	90,000	90,000
Contribution	60,000	52,500	45,000	37,500
Less: Fixed Cost	35,000	35,000	35,000	35,000
	25,000	17,500	10,000	2,500
Contribution per unit	1.00	Rs.0.875	Rs.0.75	Rs.0.625

Profit to be maintained = Rs.25,000

Contribution to be earned = Profit to be earned + Fixed cost

= Rs.25,000 + Rs.35,000

= Rs.60,000

The number of units required to be sold at different levels

$$\text{of price reduction} = \frac{\text{Total contribution be earned}}{\text{Contribution per unit}}$$

Hence,

$$\text{At 5\% reduction} = \text{Rs. } \frac{60{,}000}{0.875} = 68{,}572 \text{ units (app)}$$

$$\text{At 10\% reduction} = \text{Rs. } \frac{60{,}000}{0.75} = 80{,}000 \text{ units}$$

$$\text{At 15\% reduction} = \text{Rs. } \frac{60{,}000}{0.625} = 96{,}000 \text{ units}$$

Illustration: 7

Calculate from the following information the Break-Even Point and the net profit if the sales volume is Rs.8,00,000.

P/V Ratio is 40% and Margin of safety is 25%.

Solution:

Contribution - Sales x P/V Ratio

Break-Even Point = Actual Sales - Margin of Safety

Given P/V Ratio = 40% and MS = 25%

Contribution at a sales volume of Rs.8,00,000

= Rs.8,00,000 x 40% = Rs.3,20,000

Calculation of Fixed cost and BEP

$$\text{MS (as a percentage)} = \frac{MS}{AS} \times 100 = 25\%$$

(i.e.) MS is 25 then AS would be 100 and
BEP is 75 (see the second equation above)
BEP = 75% of AS

$$BEP = Rs.\ 8,00,000 \times \frac{75}{100} = Rs.\ 6,00,000$$

Contribution at BEP (which is always equal to fixed costs)

= BEP x P/V Ratio

Therefore,

$$\text{Fixed cost} = Rs.\ 6,00,000 \times \frac{40}{100} = Rs.\ 2,40,000$$

Calculation of profit at a sales volume of Rs.8,00,000

Contribution	=	Rs.3,20,000
Less: Fixed Costs	=	Rs.2,40,000
Profit	=	Rs. 80,000

Illustration: 8

The budgeted (Sales and Variable Costs) results of AB Ltd. are as follows:

Products	Sales	Variable Costs as % of Sales Value
	Rs.	Rs.
O	50,000	60
P	80,000	65
Q	40,000	50
R	60,000	75
[illegible]	30,000	80

Fixed costs for the period are Rs.90,000 (a) show the amount of expected loss and (b) suggest a change in sales volume of each product which will eliminate the expected loss assuming that sale of only one product can be increased at a time.

Solution:

P/V Ratio = 1 - Marginal Cost Ratio

P/V Ratio for Product O $= 1 - \frac{60}{100} = 40\,\%$

(a) STATEMENT SHOWING AMOUNT OF EXPECTED LOSS

Product	Sales x	P/V Ratio =	Contribution
	Rs.	Rs.	Rs.
O	50,000	40	20,000
P	80,000	35	28,000
Q	40,000	50	20,000
R	60,000	25	15,000
S	30,000	20	6,000
	Total		89,000
	Less: Fixed Cost		80,000
	Loss/under-recovery of fixed cost		1,000

(b) Additional volume of sales required under - recovery of fixed costs

$$\text{Additional sales volume} = \frac{(\text{Loss})}{\text{P/V Ratio}}$$

Product	Addition Sales Rs.
O $\frac{\text{Rs. } 1000}{40} \times 100$	= 2,500
P $\frac{\text{Rs. } 1000 \times 100}{35} \times 100$	= 2,857 (approx)
Q $\frac{\text{Rs. } 1000}{50} \times 100$	= 2,000

R. $\frac{\text{Rs. } 1000}{28} \times 100$	$= 4,000$
S $\frac{\text{Rs. } 1000}{20} \times 100$	$= 5,000$
	$= 16,357$ (approx)

Illustration: 9

Following information has been obtained from the books of Cor. Ltd. for the year 1982 :

	Rs.	Rs.
Sales (75% of licensed capacity)		2,40,000
Direct Materials	72,000	
Direct Wages	48,000	
Variable Overheads	18,000	
Fixed Overheads	72,000	2,10,000
Profit		30,000

The key factor is sales demand. In order to match the full capacity level the management proposes to reduce the selling price of the product by 5% in 1983.

You are required to prepare a forecast showing the effects of the proposed reduction is selling price after taking into account the following changes in cost :

Sales forecast (at reduced prices) Rs.3,04,000.

Direct wages and variable overhead are expected to rise by 5%. Direct Materials prices are expected to increase by 5%. Fixed overheads are estimated to increase by Rs.3,000/-.

Solution:

Workings

Sales for 1983

		Rs.
Sales (100% capacity) $\frac{\text{Rs. } 2,40,000}{75} \times 100$	=	3,20,000
Less: Proposed reduction in selling price	=	16,000
Estimated Sales	=	3,04,000

Marginal Cost Statement (capacity)			100%
Total Sales		=	Rs.3,04,000
Less: Marginal Cost	Rs.	Rs.	Rs.
Direct Materials	96,000		
5% increase in prices on Rs.96,000	4,800	1,00,800	
Direct Wages $\frac{\text{Rs. } 48,000 \times 4}{3}$ =	64,000		
+ 5% increase on Rs.64,000	3,200	67,200	
Variable Overheads 16,000 x 4/3	24,000		
+ 5% increase on Rs.24,000	1,200	25,200	1,93,200
	Contribution		1,10,800
Less: Fixed Overhead	72,000		
+ increase	3,000		75,000
		Profit	35,800

Illustration: 10

The income statement of C Ltd. is summarised as follows:

		Rs.
Net Revenue		2,00,000
Less:	Expenses, including Rs.1,00,000 of Fixed Expenses	2,20,000
Net Loss		·20,000

The Sales Manager believes that an increase of Rs.50,000 in advertising outlays will increase sales substantially. His plan was approved by the Chairman of the Board.

Show as at what sales volume will the company break-event?

(b) What sales volume will result in a net profit of Rs.1,000.

		Rs.
Sales		2,00,000
Less: Variable Expenses		
Total Expenses	2,20,000	
Less: Fixed Expenses	1,00,000	1,20,000
	Contribution	80,000
Less: Fixed Expenses		1,00,000
	Net Loss	20,000

$$\text{P/V Ratio} = \frac{80{,}000 \times 100}{2{,}00{,}000} = 40\%$$

When the proposed advertising expenditure is incurred and treated as fixed expenses, the total fixed expenses would be Rs.1,50,000.

(i.e. Rs.1,00,000 + Rs.50,000)

$$\text{Break-Even Point} = \frac{F}{\text{P/V Ratio}} = \frac{\text{Rs. 1,50,000}}{40\,\%}$$

$$= \frac{\text{Rs. 1,50,000}}{40} \times 100 = \text{Rs. 3,75,000}$$

$$\text{Required Sales} = \frac{\text{Fixed Expenses} + \text{Required Profit}}{\text{P/V Ratio}}$$

$$= \frac{\text{Rs. 1,50,000} + \text{Rs. 10,000}}{40\,\%}$$

$$= \frac{\text{Rs. 1,60,000} + 100}{40} = \text{Rs. 4,00,000}$$

The break-even analysis is the most widely known form of the CVP analysis. The study of CVP relationship is frequently referred to as break-even analysis. However, some state that upto the point of activity where total revenue is equal to total expenses the study can be called as break-even analysis and beyond that point, it is the application of CVP relationship.

Thus the narrower interpretation of the break-even analysis refers to a system of determining that level of activities where total revenue equals total cost i.e. the point of zero loss. The broader interpretation denotes a system, of analysis that can be used to determine the probable profit at any level of activities.

Assumptions of Break-Even Analysis :

Break-Even analysis can be carried out in two ways (a) Algebraic Method and (b) Graphical Method. Both the methods are based upon certain assumption which are rarely found in practice.

Some of the assumptions are as follows :

(a) All costs can be classified into fixed and variable elements.

(b) While variable costs vary proportionately with volume, the fixed, costs remain constant.

(c) Selling price remains constant despite volume changes.

(d) In the case of multiple products, sales mix also remains constant.

(e) Productivity per worker, and efficiency of plant etc. remain mostly unchanged.

Any changed in any one of the above factors will affect the break- even point and the profits will be affected by factors other than volume. Hence the results of the break-even analysis should be interpreted subject to the limitations of the above assumptions.

Mathematical Representation of Break-Even Analysis:

Many terms and techniques have been developed which now form part of break-even analysis. The most important terms and the formulae for computations are explained.

Calculation of Break-Even Point:

Break-Even Point is a no loss, no profit point where total sales is equal to total costs (fixed as well as marginal costs) or total contribution is equal to total fixed costs.

$$\text{Break-Even Point (in Rs.)} = \frac{\text{Fixed Costs}}{\text{P/V Ratio}}$$

$$\text{Or} \quad \frac{\text{Fixed Costs}}{1 - \dfrac{\text{Marginal Cost per unit}}{\text{Selling Price per unit}}}$$

$$\text{Break-Event Point (in units)} = \frac{\text{Fixed Costs}}{\text{Contribution per unit}}$$

$$\text{Where P/V ratio is applied} \quad \text{BEP} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

$$\text{Contribution} = \text{Sales - Marginal Cost.}$$

Example:

The fixed costs for the year are Rs.40,000. Variable costs per unit for the single product being made is Rs.6. Estimated sales for the period are valued at Rs.1,60,000. The number of units involved coincides with the expected volume of output. Each unit sells at Rs.10 each. Calculate the Break-Even Point.

Solution:

$$\text{Break–Even Point} = \frac{\text{Fixed Cost}}{1 - \dfrac{\text{Marginal Cost}}{\text{Selling Price}}}$$

$$= \frac{\text{Rs. } 40{,}000}{1 - 6/10}$$

$$= \text{Rs. } 40{,}000 \times 10 = \text{Rs.}1{,}00{,}000$$

Alternatively :

$$\text{Break-Even Point (BEP)} = \frac{\text{Fixed Cost}}{\text{P/V ratio}}$$

$$= \frac{\text{Rs. } 40{,}000 \times 100}{40} = \text{Rs. } 1{,}00{,}000$$

$$\text{(BEP) (in units)} = \frac{\text{Fixed Costs}}{\text{Contribution per unit}}$$

$$= \frac{\text{Rs. } 40{,}000}{4} = 10{,}000 \text{ units}$$

Calculation of Profit Volume Ratio:

This ratio shows the relationship between the value of sales and contribution. A most appropriate term might be the contribution / Sales Ratio. This is often expressed as a percentage.

$$\text{Profit/Volume Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

When total sales and total costs (without break up for fixed and variable components) are given to two periods of activity, the following formula may be used to calculate P/V Ratio.

$$\text{P/V Ratio} = \frac{\text{Change in profits}}{\text{Change in Sales}} \times 100$$

Where, Sales - Costs = Profits.

A few uses of P/V Ratio are as follows :

(a) Determination of marginal costs for any volume of Sales. Marginal cost percentage can be arrived at by deducting P/V Ratio from 100%. For example, if P/V ratio is 25%, then the marginal cost percentage would be 75% (100-25).

(b) Calculations of the desired volume of output, profit or other essential facts.

(c) Comparisons can be made by calculating the P/V Ratio for each factor to be compared; viz.

(i) Line of product

(ii) Sales area

(iii) Method of sale; eg., through whole-salers of retailers.

(iv) Individual factories

(v) Separate companies

Example:

The following products are manufactured and sold by NPSV Ltd Variable costs and prices are also given:

Product	Price Rs.	Variable Cost Rs.
A	100	50
B	200	150
C	400	250
Show the P/V ratio for each line of product.		

Solution:

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

Particulars	Marginal Cost Sheet Products		
	A Rs.	B Rs.	C Rs.
Sales	100	200	400
Less: Marginal Cost	50	150	250
Contribution	50	50	150

$$\text{P/V Ratio} = \frac{\text{Rs. } 50 \times 10}{100} = 50\,\% \quad \frac{50 \times 100}{200} = 25\,\%$$

$$= 150 \times 100400 = 37.5\,\%$$

Example:

ssuming that the cost structure and selling prices remain the same in periods I and II find out P/V Ratio.

Periods	Sales Rs.	Total Cost Rs.
I	1,20,000	1,08,000
II	1,40,000	1,24,000

Solution:

Periods	Sales Rs.	Cost Rs.	Profit Rs.
I	1,20,000	1,08,000	12,000
II	1,40,000	1,24,000	16,000

$$\text{P/V Ratio} = \frac{\text{Change in Profits}}{\text{Change in Sales}} \times 100$$

$$\frac{\text{Rs. } 16{,}000 - \text{Rs. } 12{,}000}{\text{Rs. } 1{,}40{,}000 - \text{Rs. } 1{,}20{,}000} \times 100$$

$$= \frac{\text{Rs. } 4{,}000}{\text{Rs. } 20{,}000} \times 100 = 20\,\%$$

Margin of Safety (MS):

The margin of safety is the difference between the total sales and the break-even sales. It may be expressed in monetary terms or as a percentage i.e., the margin of safety is an extremely valuable guide to the strength of a business.

This is calculated as follows :

(i) Margin of safety = Actual Sales - Break-Even Sales.

(ii) $\text{MS (in Rs.)} = \dfrac{\text{Profit}}{\text{P/V Ratio}}$

(iii) $\text{MS (in units)} = \dfrac{\text{Profit}}{\text{Contribution per unit}}$

Example:

Current sales are 20,000 units p.a.

Selling price is Rs.6 per unit.

Prime costs are Rs.3 per unit.

Variable overheads are Re.1 per unit.

Fixed costs are Rs.30,000.

Calculate (i) P/V Ratio, (ii) Break-Even Points and (ii) Margin of Safety.

Solution:

$$\text{P/V Ratio} = \frac{C}{S} = 100 = \frac{\text{Rs. } 40{,}000}{\text{Rs. } 1{,}20{,}000} \times 100 = 33\frac{1}{3}\%$$

$$\text{Or} \quad = \frac{\text{Rs. } 2}{6} \times 100 = 33\frac{1}{3}\%$$

$$\text{BEP} = \frac{F}{\text{P/V Ratio}} = \frac{\text{Rs. } 30{,}000}{33\ 1/3} \times 100$$

$$\text{BEP} = \text{Rs.}90{,}000$$

$$\text{BEP} = \frac{F}{\text{C per unit}} = \frac{\text{Rs. } 30{,}000}{\text{Rs. } 2} = 15{,}000 \text{ units (in units}$$

$$\text{MS} = \text{Actual Sales - BEP}$$
$$= \text{Rs.}1{,}20{,}000 - \text{Rs.}90{,}000 = \text{Rs.}30{,}000$$

$$\text{Or MS} = 20{,}000 \text{ units} - 15{,}000 \text{ units} = 5{,}000 \text{ units}$$

$$\text{Or MS} = \frac{\text{Profit}}{\text{P/V Ratio}} = \frac{\text{Rs. } 10{,}000}{33\ 1/3} \times 100 = \text{Rs. } 30{,}000$$

$$\text{MS} = \frac{\text{Profit}}{\text{C per unit}} = \frac{\text{Rs. } 10{,}000}{\text{Rs. } 2} = 5{,}000 \text{ units}$$

BREAK-EVEN CHART

The break-even point can also be computed graphically. A break- even chart portrays a pictorial view of the relationships between costs, volume and profit. The break-even point indicated in the chart will be one at which the total cost line and the total sales line intersect. It is an important aid to profit planning.

Construction of Break-Even Chart:

(a) Select scales for sale or production in units on the horizontal axis.

(b) Select a scale for costs and revenue on the vertical axis.

(c) Draw the fixed cost line parallel to horizontal axis.

(d) Draw the variable cost line starting from the point on the vertical axis which represents fixed costs.

(e) Draw the sales line, starting from the point of origin (zero) and finishing at the point of maximum sales.

Example:

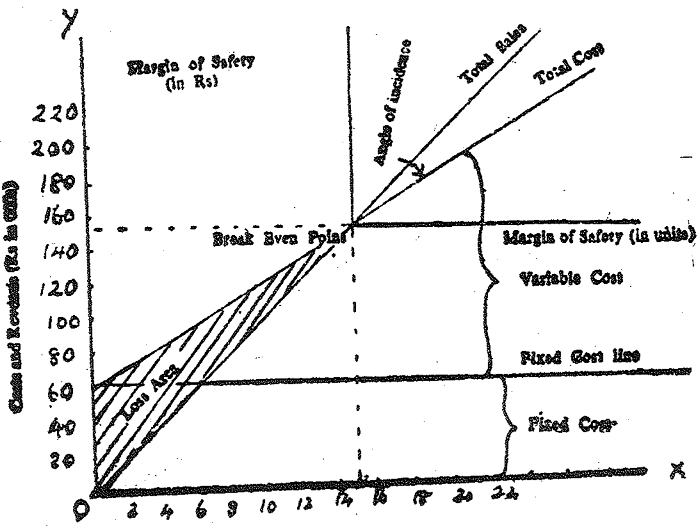
From the following data compute the Break-Even Point by means of a break-even chart:

Particulars	Volume of Sales (Units)			
Sales at Rs.10	5,000	10,000	15,000	20,000
Fixed Cost (Rs.)	60,000	60,000	60,000	60,000
Variable Costs Rs.6 per unit Current Sales 20,000 units				

Solution:

Output units	Variable Cost Rs.	Fixed Cost Rs.	Total Cost Rs.	Sales
5,000	30,000	60,000	90,000	50,000
10,000	60,000	60,000	1,20,000	1,00,000
15,000	90,000	60,000	1,50,000	1,50,000
20,000	1,20,000	60,000	1,80,000	2,00,000

The data can now be shown graphically as follows :



Sales and output (units in '000's)

Break-Event Chart

From the Figure-I it can be noticed that the break-even point occurs when sales are 15,000 units valued at Rs.1,50,000.

This can be verified algebraically.

$$BEP = \frac{F}{C \text{ per unit}} = Rs. \frac{60,000}{4} = 15,000 \text{ units}$$

Angle of incidence.

This is an angle formed between the total sales line and total cost line (Refer to figure No.1). A large angle of incidence indicates a high rate of profit while a narrow angle would show a relatively low rate of profits. This angle is important in boom periods when sales are expanding. Once the break-even point is reached, additional sales show a good profit return.

Practical utility of Break-Even Analysis.

Break-Even Analysis can be used to show the effect of a change in any of the following profit factors :

(1) Change in selling price

(2) Change in volume of sales

(3) Change in variable costs

(4) Change in fixed costs

These changes will now be illustrated using the same figures of the above example.

1. Effect of 20% increases in selling price

Particulars	Marginal Cost Sheet	
	Output Per Unit Rs.	20,000 Units Total Rs.
Sales	12	2,40,000
Less: Variable Cost	6	1,20,000
Contribution	6	1,20,000
Less: Fixed Cost		60,000
Profit		60,000

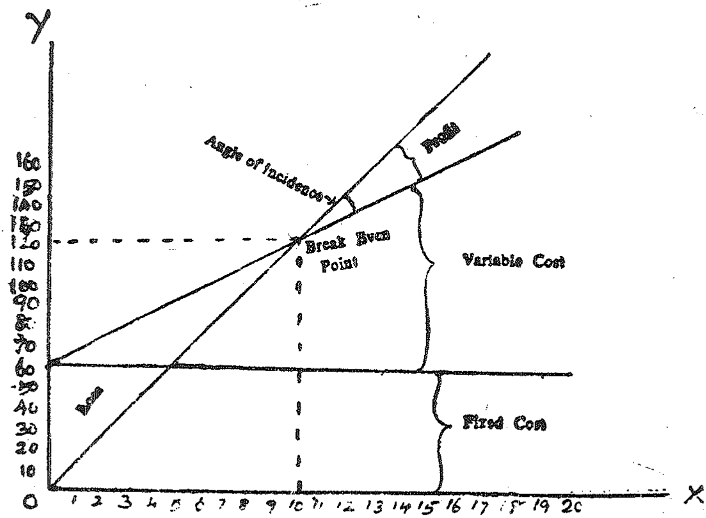
$$\text{Break-Event Point Rs.} = \frac{60,000}{\text{Rs. } 6} = 10,000 \text{ units}$$

Margin of Safety = 20,000 – 10,000 = 10,000 units

MS (on a percentage on total sales)

$$\text{Rs.} = \frac{1,20,000}{2,40,000} \times 100 = 50\,\%$$

It can be observed that due to 20% increase in selling prices the profit of N PSV Ltd. has increased by Rs.40,000 and BEP has decreased to 10,000 units of Rs.1,20,000. The same effects may be illustrated through Break-Even Chart.



Sales and output (units in '000's)
(Showing 20% increase in selling price)

2. Effect of 10% decrease in sales volume :

Particulars	Marginal Cost Sheet	
	Output per unit Rs.	18,000 units Total Rs.
Sales	10	1,80,000
Less: Variable Cost	6	1,08,000
	4	72,000
Less: Fixed Cost		60,000
Profit		12,000

$$\text{P/V Ratio} = \text{Rs. } \frac{4}{10} \times 100 = 40\,\%$$

$$\text{BEP} = \text{Rs. } \frac{60{,}000}{4} = 15{,}000 \text{ units}$$

$$\text{MS} = \frac{\text{Profit}}{\text{C per unit}} = \text{Rs. } \frac{12{,}000}{4} = 3{,}000 \text{ units}$$

It may be observed that 10% reduction in sales volume has effected only the Margin of Safety. There is no change in BEP since the P/V Ratio and Fixed Costs are the same. For graphical representation the student may refer Figure No.1.

3. Effect of 20% decrease in variable costs:

Particulars	Marginal Cost Sheet	
	Output Per Unit Rs.	20,000 Units Total Rs.
Sales	10.00	2,00,000
Less: Variable Cost (Rs.6 Less 20%)	4.80	96,000
Contribution	5.20	1,04,000
Less: Fixed Costs		60,000
Profit		44,000

P/V Ratio $= \frac{C}{S} \times 100$

$\therefore$ P/V Ratio $=$ Rs. $\frac{5.20}{10} \times 100 = 52\%$

BEP $= \frac{FC}{C}$

$\therefore$ BEP $= \frac{\text{Rs. } 60{,}000}{5.2} =$ 11.539 units

(Rs. 10 x 11,539 $=$ Rs. 1,15,390

MS $=$ 20,000 - 11,539 $=$ 8,461 units (Approx.)

Reduction in variable cost will improve the P/V Ratio and reduce BEP ultimately the MS also will increase.

4. Effect of 10% increase in fixed costs

Particulars	Output Per Unit Rs.	20,000 Units Total Rs.
Sales	10	2,00,000
Less: Variable Costs	6	1,20,000
Contribution	4	80,000
Less: Fixed Costs (Rs.60,000 - 10%)		54,000
Profit		14,000

P/V Ratio $=$ Rs. $\frac{4}{10} \times 100 = 40\%$

BEP $= \frac{\text{Rs. } 66{,}000}{40} \times 100 =$ Rs.1,65,000

Or

$$= \frac{\text{Rs. } 1,65,000}{\text{Rs. } 10} = 16,500 \text{ units BEP} = \frac{P}{P/V \text{ Ratio}}$$

$$MS = \frac{\text{Rs. } 14,000}{40} \times 100 = \text{Rs. } 35,000 \; MS = \frac{P}{P/V \text{ Ratio}}$$

$$MS = (\text{Rs.}2,00,000 - \text{Rs.}1,65,000) = \text{Rs.}35,000$$

Increase in fixed cost will increase BEP and thereby wil reduce MS.

Limitation of Break-Even Analysis:

1. When Break-Event Analysis is based on accounting data, as it usually happens, it may suffer from various limitations of such data eg. neglect of imputed cost, arbitrary depreciation estimates and inappropriate allocation of overhead costs.

2. Break-Even Analysis is static in character. Costs and prices are subject to constant change from unit as alone from period to period.

3. Cost in a particular period may not be caused entirely by the output in that period.

4. Selling costs are specially difficult to handle in break-even analysis.

5. The straight line total revenue curve presumes that any quantity might be sold at that one price. In real world, perfect competition is rare.

6. A basic assumption in break-even analysis is that the cost revenue volume relationship is linear. This is realistic only over narrow ranges of output.

7. Break-Even Analysis is not an effective tool for long range use and its use should be restricted to the short run only.

8. The area included in the break-even analysis should be limited. If too many products, too many departments or too many plants are lumped together and graphed on a single break-even chart, both good and bad performances can easily be buried in the total picture of the group.

9. Break-Even Analysis assume that profits are a function of output ignoring the patent fact that they are also caused by other factors such as technological change, improved management, changes in the scale of fixed factors of production and so on.

Additional illustrations.

Test Yourself:

1. What benefits are to be gained from marginal costing? Are there pitfalls in the application of marginal costs?

2. Give a brief account of a practical application of marginal costing which you consider sound from a policy point of view.

3. A.E. Ltd. produces for components, the cost particulars of which are given below:

Elements of Cost :	Components			
	A Rs.	B Rs.	C Rs.	D Rs.
Material	120	150	150	150
Wages	30	40	40	45
Variable Overhead	15	15	20	45
Fixed	20	35	30	30
	185	240	240	270
Output per machine hour (units)	6	3	4	4

The limiting factor is shortage of machine capacity.

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LESSON : 13

STANDARD COSTING

For a business to be efficient all functions which enter into production have to be examined and made to operate in the best and cheapest manner, in order to obtain maximum return for minimum cost. Costs may be the predetermined and used as yardsticks against which future performance can be measured. For any system to achieve these objectives there is, impliedly, a degree of management planning and control which rarely exists in any historical costing system. With actual costs, because of their historical nature and possible inclusion of inefficiencies, no useful comparisons, are possible. The use of carefully predetermined costs for specific conditions and a stated volume of output, overcomes these difficulties. Personal judgements, which may be based, are replaced by scientific methods, which are used to determine what costs should be in any set of circumstances. This lesson would deal with the system of standard costing and material and labour cost variances. Other variances will be dealt with in the next lesson.

Definition of Standard Costing:

Standard Costing is a system of cost accounting which makes use of predetermined standard costs relating to each element of cost, labour, material and overhead-for each line of product manufactured or service supplied. Actual costs incurred are compared with Standard Cost, as the work proceeds, the difference between the two being known as "Variances" these are analysed by "reasons" so that inefficiencies may be quickly brought to the notice of the persons responsible for them and appropriate action maybe taken. Standard Costing, therefore, implies planned and coordinated arrangement of all matters relating to costing. A body of procedure built in haphazard lines, not being chosen as the most suitable for the business concerned, cannot be regarded as a standard costing system, even-though it purports to use Standard Costs. Unless due care is taken, a business may be burdened with a system which is both inappropriate and too costly.

Features of Standard Costing System:

Though it is not possible to have a precise form of Standard Cost System applicable for all businesses, yet the system can be made clear with some principles and features to be found in it.

1. Once Standard Costs have been set, they are entered on a Standard Cost Card. Separate card is used for each component or product made. It shows the breakdown of total unit cost into direct materials, direct labour and factory overhead. It also indicates the physical units to be completed and hours to be worked. Great care is to be taken in preparing these cards.

2. Forms and documents, designed to cover standard procedures, standard quantities and/or Standard Costs, are used. For performances which are "exceptions to the rule" separate cards with different colour and shape are used.

3. Standard records are maintained in addition to the regular books of accounts. This is necessary to analyse the costs at later stages.

4. Variances are isolated and evaluated in terms of rupees.

5. Variance reports are to be prepared for all levels of management.

6. Books of accounts are kept to record all transactions.

The main object of the system is to enable management, when necessary, to take the appropriate action and thus correct any adverse tendencies while maintaining as far as possible anticipated efficiency.

Advantages of Standard Costs:

The weaknesses of historical costing system can be overcome if the standard costs are complied carefully. The following are the advantages to be gained from the use of standard Costs.

1. Standard costs are **a yardstick** against which Actual Costs can be compared.

2. The analysis of variances in costs will help management to have regular checks upon expenditure incurred. Deviations from the pre-determined standards of performances can quickly be localised and taking advantage of the "principle of exception" management can concentrate upon matters which are not proceeding according to plan.

3. The clerical work associated with costing is usually reduced and yet much more useful information is made available to management.

4. Interpretation of management reports is made easier and there is a reduction in the time taken to study these reports as they contain only important matters.

5. Control over costs is greatly facilitated. Cost Control and Cost reduction are probably the most important aims of any costing system and Standard Costing gives due recognition to this fact.

6. Production and Price policies can be formulated with certainty before production takes place.

7. Once the Standard costs have been compiled they can be used as a basis for other gains, such as provision of an incentive scheme of payment for employees.

Use of Standard Costing :

Standard Costing may be regarded as a technique which is complementary to the ordinary costing system, which will be some form of process or job costing. Though standard costs can be of value in all types of commercial and industrial undertaking the highest degree of efficiency in their use is to be obtained where a product or its components can be standardised. They are of maximum value if used in conjunction with a process costing System; In job costing, the products are non-standard nature and hence difficult to set the standard costs. But parts which are common to a number of heterogeneous finished products can be standardised and Standard Costs developed. In this way the advantages outline can accrue to some extent, even to enterprises using job costing. Standard costing can be effectively adopted in any organisation using either "Operating Costing", "Multiple Costing" or "Batch Costing" provided the operations involved, whether for a product or service. are repeated time and again, so that standardisation of product and methods becomes possible.

Limitations of Standard Costing:

The major drawback of standard costing is the difficulty in fixing up reliable and workable standards. Unless standards are correctly fixed, the control and study of the variances on which whole system is built up will not be effective. Unless standards are current, they are as useless as incorrect standards. Reporting on variances on which no action is possible, just because the standards themselves are open to doubt, is detrimental to the system and is worse than having no standards at all. Therefore, in organising a standard cost system, the management must be alive to these limitations and try to overcome them to the extent possible.

Standard Costing and Budgetary Control:

To be able to set and use standard costs, some form of budgeting will be essential. There is a need to forecast the level of output and the conditions which will operate in the period the standard costs are to be used. Both have to be determined before the standard costs are set, to changes in output or conditions will inevitably, tend to increase or reduce standard costs. Budgetary control in its complete form involves a predetermined plan in financial terms, to cover all phases of business activities and the operation of that in such a way that anticipated profit is, as near as possible, achieved. In a way, both the systems are, 'forward looking; in their essence and nature. However, there is a fundamental difference between the two-in as much as budgetary control is concerned with comparison of estimated and actual results of a department or company while standard costing has reference to the comparison of estimated and actual results of manufacturing a product or rendering a service. From this angle, both are valuable aids to management in controlling costs.

It is often emphasized that budgetary control and standard costing cannot function independently as both methods involve estimating cost for a future period. Strictly speaking, this is not true though it is a fact that each function better in conjunction with each other. It is relatively easy to compute budgets for production costs and sales when the standard costs have been determined. Where as in determining standard costs it is essential to ascertain the level of output for the period and this is much easier when budgeted level has been formulated.

Therefore, budgeting must precede the accurate and effective determination of price standard for materials labour and overhead. But conversely budgetary control is more successfully installed in concerns having standardised production processes and operating under standard cost accounting systems. Most of the companies in U.K. and U.S.A which have "forward looking" approach to cost accounts are using both the systems.

Standards Vs Standard Costs:

It is possible and useful to distinguish between standards and standard costs. Standards may be thought of as the physical requirements of the product, such as so many Kgs. of material or hours of labour and processing time. Standard costs on the other hand are the physical requirements of the products expressed in monetary terms. It is usual to refer to standards as standards per unit of physical standards. These physical standards are then expressed in monetary terms and referred to as standard costs of rupee standards. Thus "Standards" should be considered "physical standards" and "standards costs" should be considered "doller standards in units also of making it difficult to segregate the effects of price changes and production efficiency.

Type of Standards:

Two types of standards have been found acceptable for cost control purposes: (1) the attainable standard and (2) the perfection standard. The former can be used for cost control and may supplement the use of attainable standards. The standard costs set as per the later type are called ideal standard costs. Ideal conditions are assumed while setting these standards. But in practice ideal conditions rarely operate and hence if these are used, large unfavourable variances inevitably emerge. But in an automated factory, having as efficient system of production control, including preventive maintenance and also enjoying the trust and confidence of all personnel, perfection or ideal standards may be feasible.

Attainable standards may be defined as that cost which should be attained under efficient operation of existing facilities using specified materials. The expected efficiency may not be what is desired, but rather what can be obtained with the existing plant, facilities and personnel. In each element of cost to be incurred an allowance for contingencies is included. The extent of size of the additional allowances will be largely determined by the plant and facilities to be actually used. Thus, for example, an efficient machine of a particular type will be capable of a definite performance. If however, that machine is now nearing the end of its useful life, or it is operated by trainee operators then maximum efficiency cannot be expected. Accordingly, the standard times will be set having full regard to these shortcomings. A major failing of this standard is the necessity for constant revision.

However. some plants change standards less frequently than each accounting period, with some using the same standards for as long as 10 years. These sometimes, referred to as 'basic or bogey standards" are not revised unless major changes are made in production facilities or product specifications. They are attainable and usually, with important in methods, should be exceeded after they are in use for some time. The major objection to this type of standard is that a variance is usually expected, and for control purposes, variances must be measured against th expected variances rather than the actual cost against the attainable standard cost.

Still other plants change their standards during an accounting period or all changes in production or product specifications. This constant use of current standards provides an excellent measurement of production efficiency but makes it difficult to compare operations at various times during the period. Also cost reductions effected during the period are immediately cancelled out in the reports by the change in standards. This tends to dampen the enthusiasm of the organisation, because the members do not have the results of their efforts reflected during the remainder of the period as gain against the standards.

The disadvantages of basic and current attainable standards become more pronounced when they are converted into standard costs. The middle ground between "basic standards" and " current attainable standards" is standards and standard costs prepared and revised on an annual basic. Of course, if there are significant changes in technology and price, the affected standards and standard costs would be revised at the earliest convenient opportunity.

Setting up of Attainable Standards:

Direct Material Standards: Setting up of standards in material is based on kind and quantity of material which should be used to produce a finished product of specified quality in a manufacturing process, the chemists establish specifications for the raw materials to be used. They set the standard formulate, showing the quantities of each raw materials to be used in a standard lot or batch. Allowance is made for normal losses in initial processing and considering this the yield in quantity of finished or semi-finished product is determined. Standard procedure for manufacturing are also set. The product's density, strength etc. will also be set.

When setting standard quantities, the following aspects are to be considered.

(a) Due allowance must be made for cutting and cther normal waste,

(b) Where the waste from one product can be used on another product it will be necessary, if of significant value, to credit the product causing waste with the value of such waste. The allowance for waste is to be estimated properly. If the allowance is too high many inefficiencies in material wage will not appear in the Material Usage Variance : if low, unfavourable variances many not represent inefficiencies.

Material Price Standards:

Raw-material unit price standards are usually set by the purchasing department. In some companies these standards are approved by an economist or other person outside the purchasing department who is familiar with market prices. The prices are set at the average expected for the period during which the standards will be used. Files of the rates are maintained in the purchasing and accounting departments. From these data as to standard quantities and prices, the accountant builds the standard cost of direct material.

If the product is one which is assembled from purchased parts or fabricated from processed materials, such as steel plate, rods or shapes, the specification would be set by design engineers instead of chemists. However, the procedure for building a standard cost would parallel that described for a product manufactured from basic raw materials.

Direct Labour Standards:

The workers directly engaged upon the manufacture of a product as known as direct labour. A thorough analysis of the labour operations involved in the manufacture of a product will be an essential preliminary. Close attention must be paid to the grading labour, for the time taken from an operation by a particular grade of worker may not be the same if a different grade is used. The possibility and desirability of substituting machines for hand labour and of improving the plant layout should be considered before the standards are set.

Labour "Quantities" Standards:

The labour "Quantity" to be embodied into a product is usually indicated by the number of minutes or hours that the appropriate grade of worker will take to perform the total number of operations necessary to manufacture that product. Generally, two methods are used to determine standard times. They are

(1) based on past performance and

(2) based on test runs of operations by the use of time and motion studies.

Labour Rate Standard:

Labour rates paid in the past-last month or last year-are very often a poor guide to what rates will be paid in the future. The supply and demand conditions relating to labour are far from static, so accordingly labour rates may change quite often. A very careful analysis of all factors likely to effect wage rates in the safest approach. Only by making due allowance for the future trend of wages can be a useful standard cost be calculated.

When a piece rate method of payment is in operation, the standard cost will be the fixed rate per piece. If a bonus scheme is in operation the cost accountant will have to decide whether to include anticipated extra payments in the standard rates or in the manufacturing overhead. In the former case the amount to be added to the basic rate to arrive at the standard may be obtained by reference to past records. A similar estimate, if the alternative method is adopted, could be used to augment manufacturing overheads.

When wages are paid on a straight rate per-hour basis, standard production per-hour is set on the basis of time and motion studies or on the basis of past experience. In small plants where technical staff are not available the accountant may set the time standards on past experience modified by the judgement of the foreman as to expected changes.

Factory Overhead Standards:

Factory overhead standards are much more difficult to develop than are standards for direct material and direct labour. First divide the overhead, costs into fixed. variable and semi-variable overhead costs expected to be incurred for each producing cost center and each service cost center. Normally budgets are used to control factory overheads. These budgets deal with the individual factory overhead costs in each cost centre. In the case of standard cost accounting, the varying degrees of cost variability and fixidity can and should be averaged into one factory overhead rate for variable factory overhead and another factory overhead rate for a fixed factory overhead. If such an averaging process is found objectionable, it can be offset considerably by the use of series of standard overhead rates for various groupings of variable and fixed costs.

Some overhead costs will be incurred by one centre alone, and a direct change to that centre will then be necessary. When a cost is to be incurred on behalf of a number of costs centres it will be necessary to carry out an apportionment. The object should be to trace responsibility to be incurred or benefit to be taken by each cost centre and apportion the cost on the basis of the findings. The possible methods considered for each cost centre are floor space, cubic capacity values, direct wages, number of employees and kilowatt hours. The principle involved is to find a factor which is related to the cost being considered and then apportion on a prorata basis. Thus, for example, heating costs may be regarded as being related to space occupied by a cost centre and apportioned on that basis. The same basis could be adopted for rent and rates and depreciation of buildings. When considering supervision costs, either direct wages or the number of employees may be the appropriate factor.

The next step is the transfer of service cost-centre costs to production. This may be done by the adoption of one of the methods shown below:-

(a) Calculate a standard overhead rate for each service cost centre and then change to producing cost centres according to anticipated usage.

(b) Apportion, on a predetermined basis, the total cost for service cost centres to each producing cost centre.

(c) Use of combination of (a) and 9b).

Under the first method a standard rate per hour or other unit is calculated and then in the budget each production cost centre is charged with the appropriate cost found by multiplying standard hours or standard units by the standard rate. Typical standard units are letters for water, kilowatt hours for power. cubic feet for gas etc.

The formulas to arrive at the standard rate for each cost centre is

$$\frac{\text{Total Standard cost for service cost centre}}{\text{Total Standard service (or hours) for centre}}$$

The rates for all service cost centres, as indicated, are then used, apply the total service costs to producing cost centres. Method (b) is similar to (a), but, instead of using a rate, each service cost centre's total is apportioned on an equitable basis, to the producing cost centres. A combination of the two methods can often be used to advantage. Having transferred all service centre costs to producing cost centres, a standard overhead recovery rate can be calculated. Normally one of the three methods, the machine hour rate the direct-labour hour rate or the direct wages percentage rate is used,. the formulas for the three methods of calculating standard rates are as follows.

(a) Direct Labour hour rate:

$$\frac{\text{Total Standard overhead cost for producing cost centres}}{\text{Standard directed labour hours for producing cost centres}}$$

(b) Machine hour rate:

$$\frac{\text{Total Standard overhead cost for machine}}{\text{Standard machine hours}}$$

(c) Direct Wages Percentage rate:

$$\frac{\text{Total Standard overhead cost for producing cost centre}}{\text{Standard direct wages for producing cost centre}} \times 100$$

Both (a) and (b) give a rate per hour. If a particular job takes one hour, then the over head applied will be equivalent to the appropriate hourly rate. Method (e) gives a percentage wages cost.

The choice of method of recovery will depend upon the circumstances. If the cost centre is highly mechanised with few labourers, then machine hour rate is used. Whereas if the cost centre has no machinery and the work being performed manually, then the direct wages percentage rate may be adopted.

Materials and Labour Cost Variance:

The important feature of a standard cost accounting system is the use of variance accounts to record the differences between actual and standard costs of production. Debit balances in these accounts reflect costs above standard and credit balances reflect costs below standard. The variances show the extent to which costs were controlled and provide for the analysis and interpretation of operating report. The difference between the standard and the actual cost is called the variance. Actual costs can be above or below the standard costs and thus they are called unfavourable or favourable variance respectively.

The study of variances is done according to the conventional division of cost into its elements viz., materials, labour and overhead. However, there are some 'general' variances which can apply to all elements of cost

General Variances:

Many of the following are not different variances from those which are calculated for material, labour and overhead. Rather they are descriptions of the variances appearing under the elements of cost; they indicate particular characteristics of the other variances. Thus; for example, it is possible to have controllable variance in respect of materials; this is nothing more than a material Variance which is the responsibility of a particular person; it is not a different variance. These remarks also apply to uncontrollable variances, favourable variances; and unfavourable variances they describe the principal variances.

Total Cost Variance:

It is the variance which explains the difference between Total Standard Cost and the Total Actual Cost. It is the net variance and can be obtained by setting of all variances, favourable and unfavourable against each other.

Controllable and Uncontrollable Cost Variances:

If a variance can be regarded as the responsibility of a particular person, with the result that his degree of efficiency will be reflected in its size, then it is said to be a CONTROLLABLE VARIANCE.

If a variance is due to extraneous causes, such as a national increase in the labour rates of workers belonging to a particular trade, then it is said to be UNCONTROLLABLE. Here the responsibility for the variance cannot be assigned to a particular department or person. These variances are due to some causes which are outside the control of individuals employed in the business concerned.

This division into controllable and uncontrollable variances is extremely important. The fact that a variance may be controllable in some circumstances but uncontrollable in others is observed earlier. This observation is important and should be kept in mind when analysing variances. The reason for the variance is the factor which determines the category into which it or a part of it falls. In some cases the variance may be due to several reasons. In such cases wherever possible the reasons are to be determined and its financial cost determined. The controllable variances are to be attended and corrective actions are to be taken to minimise such variances.

Favourable and Unfavourable Variances:

When a variance is a reflection of efficiency-Actual Cost lower than Standard Cost-It is said to be a "favourable" one. On the other hand when Actual Cost exceeds Standard Cost, the difference is an "unfavourable" variance. Unfavourable variances are also called "debit" or "adverse" variances Favourable variances are also known as "credit" variances. The assumption that a favourable variance is a sign of efficiency, whereas an unfavourable variance inefficiency, can be justified only if standard cost have been correctly set. If standard are wrong, variances become meaningless.

All variances should be stated in monetary terms; quantities without costs are of limited use and value. This is not to say that physical quantities should never be used; as reports to foreman level may usually show hours lost or quantities of materials wasted; however these are not true variances.

Sub-Variances:

The principal variances are those which related, in total, to quantity or price deviations from standard. Taken together, the two variances equal the appropriate cost variance. Thus for example, for Material Price variance and Material Usage Variance may be regarded as principal variances which, jointly form the Material Cost Variance. If a principal variance is analysed into its constituent parts, separate variances are known as sub- variances. The usual sub-variance are Mixture or Mix Variance, Yield Variance, and Calendar Variance. These form part of a principal variance. For example the total of Yield Variance and Mixture Variance is equal to Usage Variance. These variances are usually applied with materials although labour sub-variance can be calculated.

Material Variances:

Material Cost Variances MCV may be regarded as a " total" variance sub-divided into:

(a) Material price Variance which is the difference between Actual and Standard Prices of the material used, multiplied by Actual Quantity.

(b) Material Usage Variance, which is the difference between the Actual and Standard Quantities of the material used, valued at standard price.

These are the principal material variance, but in addition, there are others which may be regarded as sub-variances and which are contained in the Material Usage Variance.

Reasons for Material Price Variances:

The most usual reasons for a material price variance can be summarised as follows:

(a) Changes in the basic prices of materials.

(b) Failure to purchase the quantities anticipated when standards were set, resulting in higher prices directly, or indirectly, though not obtaining Quantity discounts.

(c) Not taking discounts, when standards were set in anticipation of receiving all such discount.

(d) Changes in the charges incurred for transport inwards purchasing and store keeping. When these are debited to the material cost, thus increasing the issue price per unit of material.

(e) Failure to purchase the standard quality of material, this resulting in a different price being paid.

Reasons for Material Usage Variance:

The following may cause material usage variance:

(a) Defective or sub-standard material.

(b) Wastage in material due to unskilled employees or insufficient production methods.

(c) Pilferage of materials.

(d) Carelessness in the use of materials.

(e) Use of different quality of material other than the one specified as standard.

(f) Use of non-standard material mixture.

(g) Failing to inspect materials.

These reasons for material usage variance may be analysed with the help of Standard Material Requisition. Any extra materials will have to be authorised on a special form called an Excess Material Requisition. The Standard From will be the authority for standard quantities only. For the additional materials a distinctive form-say of the particular colour-can be used. All such forms can readily be sorted into the principal reasons for the variances .

A summary statement can be complied and submitted to the departments concerned. In particular, the aim should be to show the types of materials, the operators and foreman involved and the cause of the deviations from efficient perf~ mances.

The material Usage Variance is of the utmost importance. Accordingly much benefit can be obtained by the prompt presentation of a statement of the reasons for, and details of, the variances. Incipient troubles and inefficiencies can be traced and corrected before large losses have occurred.

Material Yield Variances:

If the actual yield differs from the standard yield it is possible to calculate a yield variance. This being the difference between actual and standard yields calculated by reference to standard output price. In process industries it is often essential to be able to control the input and consequent yield of each operation. To obtain maximum efficiency each stage should be systematically "Watched" the input and yield will have to be weighed or measured and recorded. To achieve maximum control, the checking and recording of input and yield and the calculation of yield variances should be done for each operation.

An unfavourable yield variance shows that, in total, more than the standard quantity of materials has been used. A favourable yield variance indicates that less than the total standard quantity was required. A sudden increase or decrease in yield variance will usually indicate that an investigation is required. This variance is an index of efficiency in operating.

Materials-Mix Variance:

The Material Mixture of Mix Variance considers that the proportions of materials or labour for a specified output. If, for example, the mixture is varied so that a larger than standard proportion of more expensive materials is used, then there will be an unfavourable mix variance. When a larger proportion of cheaper materials is included in the mixture, then a favourable Material Mix variance will result.

Failure to use the standard mixture may be due to one or more of a number of seasons. There may be a shortage of one of the specified materials, an alternative having to be used. Often the Mix Variance in such a case, will be uncontrollable; this will be so when fire, food, labour strike, or similar occurrence cuts off the main supply of a material thus resulting in a grave shortage. On the other hand, failure to record material in good time may be the fault of official in charge making purchase requisitions. Another reason may be that stores have issued substitute material due to, say, the correct material not being unpacked, or can be issued except with great difficulty or cannot be traced due to inefficient store keeping. These and other reasons will be brought to light by analysis and investigation of the Mix variances. Many can be eliminated by the issue of instructions regarding procedures.

Calculation of Material Variances:

One of the most important things to remember in connection with the calculation of variances is that the total variance for each element of cost is divided into other variances and therefore, when the variances have been calculated, the accuracy of the calculation can be checked by "balancing" the total variances. With net total of other variances. Thus material Cost Variance can be calculated and then the result checked against the net total of the Material price Variance and material Usage Variance. It is essential to "net" the price and quantity (usage) variances, for one may be favourable and the other unfavourable.

Material Cost Variance:

This is concerned with the actual and standard material total cost The difference between these two is the variance. The formula is

$$M.C.V. = (Aq \times Ap) - (Sq \times Sp)$$

The standard quantity must relate to the production actually achieved, eg. units or tons of output.

Material Price Variance:

Material Price Variance (MPV) is simply the difference between the standard price per unit and actual price per unit multiplied by the quantity purchased.

The formula is

$$M.P.V. = (AP - SP)\ Aq$$

For example the standard price for Material A is Rs.2-50 per Kg. and the actual price at which the material was bought is Rs.2-45 per Kg. The price variance per Kilo is Re.0-25 and the total price variance if 10000 Kgs. of material is used, is

$$10000 \times 0.05 = \text{Rs.}500\ (F).$$

Material Usage Variance or Quantity Variances:

The Material Usage Variance (MUV) is the difference between quantities actually used and quantities which should have been used This difference between Actual and Standard quantities is multiplied by the standard price per unit in order to determine the amount of material usage variance.

The formula is

$$M.U.V. = (Sq - Aq)\ SP$$

If the actual quantity (Aq) used is more than the standard quantity (Sq), then MUV is unfavourable and vice versa for favourable

Consider the above example where 10000 Kgs. of material were used and the standard usage should have been 9500 Kgs. The usage variance is (9500 - 10000) or 500 kgs. which when multiplied by the Standard Price (SP) Rs.2-50, gives unfavourable (UF) MUV of Rs.1,250 (UF).

Example:

The standard cost card shows the details relating to the materials required to produce component XX 845:

Quantity of material = 200 Kgs.

Price of material = Rs.2 per Kg.

Actual production and related material data are as ollows:

Actual price of material used = Rs.1-75 per Kg.

Actual Quality of material used = 240 Kgs.

Calculate: (a) Material Cost Variance

(b) Material Price Variance

(c) Material Usage Variance.

Solution

(a) Material Cost Variance (MCV)

= (AqxAp) - (SqxSp) or
= Actual Cost - Standard Cost.

Standard Price = 2Rs.per kg.

Standard Quantity = 200 Kgs.

∴ Standard Cost = (SQxSp) = 200 x 2 = Rs.400

Actual price = Rs 1-75 Per Kg.

Actual Quantity = 240 kgs.

Actual Cost = (Aq xAp)
= (240 xRs.1.75) = Rs.420

MCV = (AC - SC)
= Rs.420 - Rs.400 = Rs.20
(Adverse) or (A) or (UF)

(b) Material Price Variance (MPV)

$= Aq\ (AP - Sp)$

$= 240\ (1.75 - 2)$

$=$ Rs.60 (Favourable) or (F)

(c) Material Usage Variance (MUV)

$= SP\ (AQ - SQ)$

$=$ Rs.2 (240 − 200)

$=$ (2 x 4 0 = Rs.80) (Adverse) or (A)

Proof:

MCV = MPV + MUV

Rs.20 (A) = Rs.60 (F) + Rs.80 (A)

Material Yield Variance:

Material Yield Variance (MYV) is the difference between the standard yield specified and the actual yield obtained. In process industries it is possible to set a standard process yield. The standard yield is the output expected from the standard output. The actual yield, however, differs from the standard yield and thereby gives rise to yield variance.

Material Yield Variance = Standard Cost of Standard Output − Standard Cost of Actual Output

= Standard Cost per unit of output (Standard output − Actual output)

= SC per unit (SO − AO)

Material Mix Variance:

When the actual composition of materials & used differs from the standard composition of materials, the materials mix variance (MMV) can be obtained. Mix variance is applicable only when direct materials are physically mixed.

MMV = (Actual proportion of Material – Revised Standard Proportion of Material) x Standard Price.

= (Aq – RSq) SP

Example :

From the following data compute:

(a) Material Cost Variance

(b) Material price Variance

(c) Material Mix Variance

(d) Material Yield Variance.

	Standard			Actual		
	Qnty Kgs	Price Rs.	Value Rs.	Qnty Kgs.	Price Rs.	Value Rs.
Material X	160	2	320	140	2.10	294
Material Y	80	5	400	100	4.50	450
	240		720	240		744
Less:	24		--	36		--
	216		720	204		744

Solution:

(a) Material Cost Variance

MCV = AC - SC

$$= 744 - \left(\frac{720}{216} \times 204 \right) = \text{Rs. } 64 \ (A)$$

(b) Material Price Variance

MPV = Aq(Ap = Sp)

MPV for Material x = 140 (2.10-2) = Rs.14 (A)

MPV for Materially = 100 (4.50-5) = Rs.50 (F)

Net MPV = 36 (F)

(c) Material Mix Variance

MMV (AQ – RSQ) SP

$$RSQ = \frac{\text{Sq of each type}}{\text{Total Std. Mix}} \times \text{Total Actual Mix}$$

$$\text{RSQ of Material x} = \frac{160 \times 240}{240} = 160$$

$$\text{RSQ of Material y} = \frac{80 \times 240}{240} = 80$$

MMV of Material x = (140 – 160) 2 = 40 (F)

MMV of Material y = (100 – 80) 5 = 100 (Ad)

(d) Material Yield Variance

MYV = SC per unit (SQ – AQ)

$$= \frac{720}{216} (216 - 204)$$

Proof:

MCV	=	MPV + MMV + MYV
64(A)	=	36 (F) + 60 (Ad) + 40 (Ad).

Labour Variances:

Labour Cost Variance is the total variance which may be subdivided into.

(a) Labour Rate Variance or Wage Rate Variance which is the different between Actual and Standard rates of the workers employed, multiplied by the Actual hours.

(b) Labour Efficiency Variance, which is the difference between the Actual and Standard labour hours, valued at the Standard Rate.

(c) Idle time variance, is the Standard cost of actua hours any employee remains idle due to abnormal circumstances lock out, power-failure, etc.

(d) Labour Mix Variance is caused by the actual grades of labour being different from the standard. Such a situation may arise due to the shortage of a particular grade of labour.

Labour Rate Variance:

The following are the usual causes for this variance:

(a) Changes in basic wage rates

(b) Employing a man of grade different from the one laid down as the standard grade for the job.

(c) Excessive amount of overtime than the amount set.

The responsibility for the variance is attached to production Departments. Sometimes this may be an uncontrollable variance, for rates of wages are usually determined by the supply and demand conditions of the labour market, backed by the negotiating strength of the particular trade union. If a foreman is responsible for the variance, he should be informed of the details so that necessary action can be taken.

Labour Efficiency Variance:

The reason for this variance may be due to

1. use of substandard employees with Insufficient training of Incorrect instructions or workers dissatisfaction

2. failure to obtain best results from employees due to poor working conditions, efficient or organisation, use of defective machinery, and Incompetent supervision.

The actual causes of the variance when it arises should be carefully listed and where practicable sub-variances calculated. It is normally the best guide to labour efficiency, so prompt action can result in large savings.

Formulae for Calculating Labour Variances:

1. Labour Cost Variance (LCV) = Standard Costs − Actual Cost
   = SC − AC.

2. Labour rate Variance (LRV) = Actual Hours (Actual Rate − Standard Rate)
   = AH (AR -SR)

3. Labour Efficiency Variance (LEV) = Standard Rate (Actual Hours − Standard Hours)
   = SR (AH -SH)

Idle-Time Variance (ITV) = Idle Hours x Standard Rate

Labour Mix Variance (LMV) = Standard Cost of Standard Mix − Standard Cost of Actual Mix

= SC of SM − SC of AM.

Example:

From the following data relating to a component xx 503, calculate.

(a) Labour Cost Variance

(b) Labour Rate Variance

(c) Labour Labour Efficiency Variance

<u>Standards-per Component :</u>

Labour Rate = Rs.5 per hour

Hours = 2 per unit

<u>Actual Production Data :</u>

1000 units of component are produced

Labour rate = Rs.5.50 per hour

Hours-Worked = 1950.

Solution :

(a) Labour Cost Variance = SC -AC

Standard labour rate = Rs.5 per hour

Standard number of hours for actual production = (1000 units x 2) = 2000 hours

Standard Cost = Rs.2000 x 5 = Rs.10,000

Actual Cost = 1950 x Rs.5.50 = Rs.10,725

∴ LCV = Rs.10,000 = Rs.10,725 = Rs.725 (A)

(b) Labour Rate Variance (LRV) = (AR -SR) AH
= (5.50-500) 1950
= 0.50 x 1950 = Rs.975 (A)

(c) Labour Efficiency Variance = SR(SH AH) (LEV)
= 5(2000 -1950)
= 5 x 50 = Rs.250 (F)

(Standard hours = Actual production x
Standard hours per unit = 1000 x 2 = 2000 hours)

Proof :

LCV = LRV + LEV

Rs. 725(A) = Rs.975 (A) + Rs. 250 (F)

Examples :

	Standard		Actual			Standard hours in Actual	
	Hours	Rates	Wages	Hours	Rates	Wages	Rates
Skilled	400	2.00	800	600	2.20	1320	450
Unskilled	200	1.00	200	200	0.90	180	150
	600		1000	800		1500	600

Compute Different Labour Variances:

(a) Labour Cost Variance (LCV) = AC - SC
= 1500-1000 = Rs.500(A)

(b) Labour Rate Variance (LRV) = AH (AR-SR)
= 600 (2.20 – 200) +
200 (0.90 – 100)
= (600 x 0.20) + (200x0.10)
120 (A) + 20(F) = Rs.100 (A)

(c) Labour Efficiency Variance (LEV) = SR (AH-SH)
= 2(600-450) + 1(200-150)
= 300 (A) + 50(A) = Rs.350(A)

(d) Labour Mix Variance = SCSM – SCAM
= 1000-(450x2) + (150x1)
= 1000-1050 = Rs. 50(A)

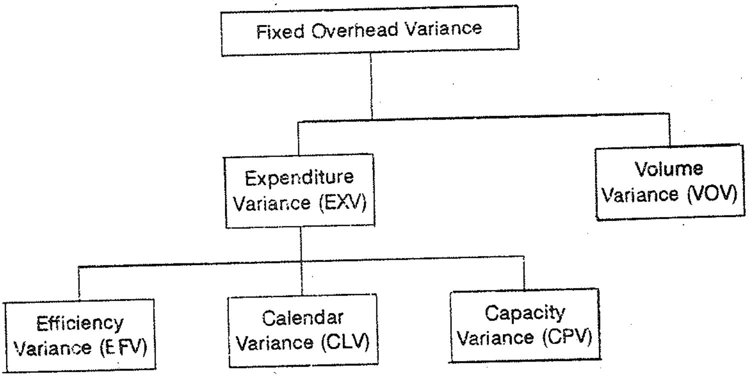
Proof :

LCV = LRV + LEV + LMV

500(A) = 100(A) + 330(A) + 50(A)

Overhead Cost Variances:

For factory (manufacturing) overhead costs, the variances are calculated for the factory as a whole, or separately for each producing cost centre and service cost centre. For effective control separate variances for each cost centre will normally be essential. The overhead variance is the difference between standard overhead specified and the actual overhead incurred Overhead variance is analysed into variable overhead variance (VOW) and Fixed Overhead Variance (FOV). The latter is further analysed as under.



The calculation of variable overhead variance is fairly simple as the standard is specified as a rate per unit. By multiplying the actual units produced by the standard rate specified, the standard variable overhead is obtained. This is composed with the actual variable overhead incurred to work out the variable overhead variance.

Fixed Overhead Variance:

This variance depends upon two factors, viz, the Fixed Expenses Incurred, and the volume of production obtained. The volume of production in turn dependent upon the capacity at which factory works, the number of days it works and the efficiency at which it works. Hence in the analysis of fixed overhead variance; the variances due to each one of these factors is separately looked out. The calculation of variances for fixed overheads is the most complicated of all variances.

As fixed overheads are to be charged to production on some indirect basis, the most important problem regarding fixed overhead variances is that of recovery which depends on two fundamental factors-output during a period or output expressed in standard hours and fixed overhead expenses during the corresponding period.

By way of interplay of these factors, the standard fixed overhead cost per unit is arrived at by dividing the budgeted fixed overheads during the same period. In the context of fixed overhead cost per unit the standard cost of production is evaluated so that a deviation of this figure from the budgeted would indicate an over-recovery or under-recovery in respect of fixed overheads. Deviations might be the result of either a change in overhead costs or a change in output. Therefore, fixed overheads, variances may be calculated either on the basis of units of output or on the basis of standard hours. Though both the methods are used to emphasis usually on the fixed method.

VARIANCES BASED ON OUTPUT

(a) Fixed Overhead Cost Variance:

This is the difference between the standard cost and actual cost relating to fixed overheads. The variance may be expressed as

$$SC - AC$$

Standard cost would be obtained by multiplying actual output by standard rate and the actual cost is obtained by dividing the budgeted output into budgeted fixed overheads.

(b) Fixed Overhead Expenditure Variance (FOEV):

It is the difference between budgeted expenditure and actual expenditure incurred. This variance is a measure of efficiency in spending. The variance may be expressed as:

(BC) Budgeted Cost – Actual Cost (AC)

Reasons for F.O.E.V.:

This variance reveals the efficiency in spending. If the actual overhead exceeds budgeted cost, there has been excess spending in respect of one or a number of elements which go to make up total overhead cost. The cause of the variance can be obtained by comparing the actual and budgeted for each overhead account, eg. Rent A/c, Lighting A/c, Insurance A/c etc. The actual expenditure on Analysis shows more or less than the budgeted. Understanding may not always be a mark of efficiency and should be investigated with the same thoroughness as used for finalising over-spending.

(c) Fixed Overhead Volume Variance (FOVV):

It denotes the variance in overhead recovery due to the budgeted quantity of products being greater or less than actual production. This variance may be sub-divided into efficiency variance and capacity variance. Sometimes calendar variance, which arises due to the actual number of days worked being different from those budgeted, is also studied as part of volume variance. The formula is

Standard Rate (Actual Quantity – Budgeted Quantity)

or SR (Aq-Bq.)

Reasons for FOVV:

An analysis of the reasons for the overhead volume variance involves consideration of the factors which bring about either idle time or overtime of plant and facilities. In the case of idle time, the principal causes may be as follows.

(a) Failure of the sales department to obtain sufficient orders.

(b) Machine breakdown

(c) Defective materials

(d) Labour Trouble

(e) Power failures

(f) Ineffective planning leading to faulty instructions.

(g) Non-delivery of essential materials.

(b) Fixed Overhead Efficiency Variance (FOEV) :

This variance attempts to measure efficiency for the output actually achieved. It is closely related to labour efficiency variance because, if the workers have been efficient the production will be above standard with the result that overheads would be over recovered. The variance signifies the deviation of standard quantity from the actual quantity produced. Its formula is

Standard Rate (Actual quantity – Standard quantity)

or SR (AQ-SQ).

Reasons for FOEV :

There is great similarity between FOEV and LEV. Both arise from consideration of the actual and standard times for a given out put. Not unnaturally, therefore, the reasons for the labour efficiency variance already outlined in the earlier lesson, may be the same for the fixed overhead Efficiency Variance.

(e) Fixed Overhead Capacity Variance (FOCP)

This variance reveals the over utilisation or under utilisation of plant and equipment. The formula for calculating the variance is

Standard Rate (Standard Quantity – Budgeted Quantity)

or SR (SQ-BQ).

(f) Fixed Overhead Calendar Variance (FOCL)

This portion of the overhead variance is the difference between number of working days in the budget period and the number of working days in the period to which the budget is applied. If for example, a businessman has an accounting period, which covers 22 working days (4weeks x 5 days) and the budgeted or standard costs are fixed with this number of days in mind then obviously, if this number of days in mind then obviously, if this number is not worked out or is exceeded the overhead charges will not be as expected. This may be result of unexpected public holidays being declared so that the work in the unit is stopped. It is also sub-variance of FOW. The formula is:

Standard Rate (Revised Budgeted Quantity – Budgeted Quantity)

Definitions of Terms Used:

1. Budgeted Cost: It is obtained by multiplying the quantity budgeted for production with the standard cost per unit.

2. Standard Output: This connotes the quantity which should have been produced during the actual hours worked.

3. Actual Output: It is the actual quantity produced in the number of hours actually worked.

4. Actual Cost: it denotes actual quantity produced at actual cost per unit.

5. Standard Cost: The term refers to actual quantity produced at standard cost per unit.

6. Budgeted Output: This represent quantity to be produced during the budget period with the number of hours budgeted to be worked.

Interdependence of Variance:

All variances are, in some way, related and interdependent. Though they are classified under different heads as explained in the preceding pages they are not to be treated separately and in isolation.

A labour rate variance, may be favourable and Its "twin" the labour efficiency variance, unfavourable. The reason may be simply that a lower grade of labour has been employed, at a rate below standard and because of their lack of skill, the employees have failed to achieve standard efficiency. A similar occurrence could take place for material; and cheaper grade may be used with a reduction in efficiency usage.

The employment of lower-paid workers may affect the material and overhead variances. Lack of skill may lead to a high rate of spoilage and therefore an unfavourable material usage variance. Both the Fixed overhead Efficiency Variance and the Fixed Overhead Variance could be brought about by the same substandard workmanship. Longer hours taken, for a specified output would result in an unfavourable efficiency variance. These are but a few of the many instances which Illustrate the interdependence of variances. If the variance analyst is to guide management along the paths to greatest efficiency, he should be well aware of this close relationship.

Example:

From the following data calculate the different fixed overhead variances.

Budgeted Fixed overheads for January is	50,000
Budgeted output for january	25,000 units
Standard time for 1 unit	5 hours
Actual hours worked	1,27,500
Actual fixed overheads for the month	Rs. 55,000
Units produced during the month	26,000 units

Solution :

(a) Fixed Overhead Cost Variance = SC − AC

$$\text{Standard cost per unit} = \frac{\text{Budgeted Fixed Overhead}}{\text{Budgeted output}}$$

$$= \frac{50,000}{25,000} \text{ Rs. } 2$$

Standard Cost of output = Actual Output x Standard cost per unit = 26,000 x 2 = Rs.52,000

FOCV = 52,000 − 55,000 = Rs.3,000 (A)

(b) Fixed Overhead Expenditure variance

= Budgeted Cost – Actual Cost
= Rs.50,000 – Rs.55,000
= Rs. 5,000 (A)

(c) Fixed Overhead Volume Variance

= SR (AQ – BQ)
= Rs.2 (26,000 – 25,000)
= Rs.4,000 (F)

(d) Fixed Overhead Efficiency Variance

= SR (AQ-SQ)
= 2 (26,000 – 25,500)
= 2 x 500 = Rs.1,000 (F)

Standard Quantity = Actual Hours x Standard Quantity per hour = 1,27,500 x = 2550 Units.

(e) Fixed Overhead Capacity Variance

= SR (SQ – BQ)
= 2 (25,500 – 25,000)
= 2 x 500 = Rs.1,000 (F)

Proof:

FOCV = FOEXV + FOEV + FOCAV

3,000 (A) = 5,000(A) + 1,000 (F) + 1,000 (F).

Fixed Overhead Variances Based on Standard Hours:

Fixed overhead variances can also be calculated on the basis of standard hours. Though the method differs, the ultimate results concur with the earlier method based on standard output.

1. FOCV : This is the difference between standard cost and actual cost. Standard cost is the product of standard hours and standard rate per hour while standard hours are the quotient obtained by dividing standard output per hour into actual units produced; standard rate is the result of budgeted overheads being divided by budgeted hours. The formula is SC – AC.

2. FOEXV: This is the amount by which actual cost deviates from budgeted cost and is therefore, expressed as: BC – AC.

3. FOVV: It is obtained by multiplying standard rate per hour with the difference between standard hours and budgeted hours. The algebraic expression for variance is

Standard Rate (Standard Hours – Budgeted Hours)
or SR (SH – BG)

4. FOEV: It is the product of the difference between standard hours and actual hours as well as the standard rate, the formula is

Standard Rate (Standard Hours – Actual hours)
or SR (SH – AH)

5. FOCAV: This is the difference between actual hours and budgeted hours expressed in standard rate. The formula is

Standard Rate (Actual Hours – Budgeted Hours)
SR (AH – BH)

Example:

Using the data given in the previous example calculate different fixed overhead variances based on standard hours.

Solution:

(a) FOCV: = SC – AC

= Rs.52,000 – 55,000 = Rs.3,000(A)

Budgeted Hours = Budgeted Output x standard for one unit of output

= 25,000 units x 5 hours per unit

= 1,25,000 hours

Standard Rate: = Budgeted overhead / Budgeted Hours.

$= \frac{50,000}{1,25,000} =$ 40 ps. per hour

Standard hours: = Actual Output x Std. Production per hour.

= 26,000 x 5 – 1,30,000 hours.

Standard Cost: = Standard Hours = Standard rate per hour

= 1,30,000 x 40 ps. = Rs.52,000

(b) FOEXV: = BC – AC

= 60,000 – 55,000 = Rs.5,000(A)

(c) FOEV: = SR (SH – BH)

= 40 Ps.(1,30,000 – 1,25,000

= 40 Ps. x 5,000 = Rs.2,000 (F)

(d) FOEV : = SR (SH – AH)

= 40ps. (1,30,000 – 1,27,500 hours)

= 40 ps. x 2,500 hours. = Rs. 1,000 (F)

(e) FOCAV: = SR (AH – BH)

= 40 ps. (1,27,500 – 1,25,000)

= 40 ps. x 2,500 = Rs. 1,000 (F)

Proof:

FOCV = FOEXV + FOEV + FOCAV

Rs. 3,000 (A) = Rs.5,000 (A) + Rs.1,000 (F) + Rs.1,000 (F)

Example:

From the following data of a manufacturing company compute the various fixed overhead variances.

Actual number of days worked	22	
Actual man worked during month	8,600	
Units produced	850	
Actual fixed overhead incurred		Rs. 3,600

The following additional information is obtained:

Budgeted number of working days	20
Budgeted man hours per month	8,000
Standard man hours per unit produced	10
Standard fixed overhead rate per man hour	Re.0.50

Solution:

$$\text{Standard Cost per unit} = \frac{\text{Budgeted Fixed Overhead}}{\text{Budgeted output}}$$

$$\text{Budgeted output} = \frac{\text{Budgeted Hours}}{\text{Standard Hours per unit}}$$

$$= \frac{8000}{10} = 800 \text{ units}$$

Budgeted Fixed Overhead = Budgeted hours × Std. rate per hour

= 8,000 × 0.50 = Rs. 4,000

SC per unit = $\frac{\text{Rs. } 4{,}000}{800 \text{ units}}$ = Rs 5 per unit

Standard Cost of output = Actual x Std. cost unit

= 850 x Rs. 5 = Rs. 4,250

RBQ = $\frac{\text{Budgeted output}}{\text{Budgeted days}}$ × Actual days

= $\frac{8{,}600 \times 22}{20}$ = 880

Standard Quantity = AH x SQ per Hour

= 8,600 × $\frac{1}{10}$ = 860

(a) FOCV = SC - AC
= Rs. 4,250 - 3,600 = Rs. 650 (F)

(b) FOEXV = BC - AC
= Rs.4,000 - 3,6000 = Rs.400 (F)

(c) FOVV = SR (AQ - BQ)
= Rs. 5 (850 - 800 units)
= Rs.5 x 50 units = Rs.250 (F)

(d) FOVE = SR (AQ - SQ)
= Rs.5 (850 - 860)
= 5 x 10 = Rs.50 (A)

(e) FOCAV = SR (SQ - RBQ)
= Rs.5 (860 - 880) = Rs.5 x 20 - Rs.100 (A)

(f) FOCALV = SR (RBQ - BQ)
= 5 (880 - 800) = 400 (F)

Proof:

FOCV = FOEXV + FOVV

650 (F) = 400 (F) + 250 (F)

Sales Variance :

Earlier we have seen the cost variances. Cost variances affect profit but to get a complete picture for the overall change in profit, one must analyse sales as profit is also affected by change in sales price, sales quantity or sales mix or change in standard allowances, rebates etc. There are two different methods of computing and presenting sales variances.

(i) Based on Sales Margin:

This shows the effect of a change in sales price or quantity on sales margin i.e., profit. Management is normally interested in knowing the sales margin variances.

(ii) Based on Sales Value:

This method shows the change in sales value due to quantity or price. This is the earliest method and is not-generally used, because it does not clearly show the effect of change in sales on profits, on which the management is primarily interested.

Whatever method is followed, the cost units sold must be taken at standard cost so that actual cost does not enter at all into sales variance analysis. This is so because sales function is relatively independent of production from the point of view of control and hence it is logical to exclude all production cost variances while calculating sales variances. When costs have been standardised. the next problem is analysis of sales variances, and in doing so we must start from standard costs of products and not from actual costs.

SALES MARGIN METHOD

The important variances are:

(a) Total Sales Margin Variance:

It is the difference between the standard margin appropriate to the quantity of sales budgeted for a period and the margin between the standard cost and the actual selling price of the sales affected. It is thus the difference between total actual margin and total budgeted margin. The total actual margin is obtained by deducting standard cost of actual sales from actual sales.

Total Sales Margin Variance = Total Actual Sales Margin – Total Budgeted Sales Margin

= (Actual Sales – Standard cost of sales) – Budgeted Sales Margin.

(b) Sales Margin Price Variance:

It is the difference between the actual profit and standard profit Symbolically the variance is

= Actual Sales Margin = (Actual Units x Std. Margin)

= Actual Sales Margin = Standard Sales Margin.

(c) Sales Margin Volume Variance:

This is the amount by which standard profit differs from budgeted profit. It is that portion of the total margin variance which is due to the difference between budgeted and actual quantities of each product, both valuing at standard margin. This is thus an ordinary usage variance and is obtained as follows:

VOLUME VARIANCE = (Actual units sold x Std. margin per unit of actual mix) - Budgeted units x standard margin unit of std. mix.

This variance is sometimes called Quantity Variance or total quantity variance. But in order to analyse this variance further into quantity and mix variances, the name "volume variance" has been used to show the total effect of change due to quantity or mixture.

(d) Sales Margin Mixture Variance :

It is that portion of the sales margin volume variance which is due to the change in actual mix from the budgeted sales mix, based on actual quantities. It may be mentioned that is mix variance can be obtained by quantity technique or value technique. Under quantity technique, change in margin is related to quantity in actual mix where as under value technique, change in margin is related to sales value or respective products. Only quantity technique is used in this variance.

MIXTURE VARIANCE = Total actual Quantity sold (Standard margin per unit of standard mix Standard margin per unit of actual mix)

(e) Sales Margin Quantity Variance :

This variance is that portion of the sales margin volume variance which is due to the difference between the budgeted margin and the actual quantity at standard margin per unit of standard margin per unit of standard mix.

(f) Sales Margin due to Sales Allowance:

It is that portion of total sales margin which is due to the difference between budgeted rebates, discounts etc., on the sales effected and the actual rebates, discounts etc., allowed on those sales where different classes of customers and individual buyers are allowed various rates of quantity and trade discounts, standard margin is obtained after deducting the standard allowances from cross margin on sales.

Example :

From the following data, compute the sales margin variances.

Budgeted Sales;

A 600 units at Rs. 10 (Std. cost Rs.6)

B 400 units at Rs.5 (Std. cost Rs.3)

Actual Sales :

A 1000 units at Rs. 9 each

B 500 units at Rs.4 each.

Solution :

STANDARD MARGIN ON STANDARD MIX

	Units	Rate	Rs.
A	600	4	2,400
B	400	2	800
	1,000		3,200
Standard Margin per unit Standard Mix $\frac{3,200}{1,000}$ = Rs. 3.20			

STANDARD MARGIN ON ACTUAL

	Units	Rate	Rs.
A	1000 Units	4	4,000
B	500	2	1,000
	1,500		5,000
Standard Margin per unit Standard Mix $\frac{5,000}{1,500}$ = Rs. $3\frac{1}{3}$			

ACTUAL MARGIN ON ACTUAL MIX.

	Units	Rate	Rs.
A	1000	4	3,000
B	500	2	500
	1,500		3,500

(a) Sales Margin price Variance

= Actual Sales Margin - (Actual Units sold x Std. Margin)
= 3,500 - (1,500 x 3)
= 3,500 - 5,000 = Rs. 1,500 (A)

(b) Sales Margin Mixture variance

= Actual units sold (Standard margin per unit of Standard mix - Standard margin per unit of actual mix)

$$= 1{,}500\ (3\frac{1}{3} - 3\frac{1}{3}) = \text{Rs. } 200\ (F)$$

(c) Sales margin Quantity Variance

= Actual Quantity x Standard margin per unit on Standard (mix) - Standard Quantity x Std. margin on Standard (Mix).

= (1,500 x3.20) - (1,000 x 3.20)

= 4,800 - 3,200 = 1,600 (F)

(d) Standard Margin Volume Variance

= (Actual Quantity x Std. margin per unit on actual mix) (Budgeted Quantity x Std. Margin per unit)

$$= (15{,}000 \times 3\frac{1}{2}) - (1{,}000 \times 3.20)$$

= 5,000 − 3,200 = Rs. 1,800 (F)

(e) Sales Margin Variance

= Actual Margin on Sales - Std. Margin on budgeted sales

= (3,500 - 3,200) = Rs. 300 (F)

Proof :

Sales Margin Variance = Sales Margin Price Variance + Sales Volume Variance

300 (F) = 1,500 (A) + 1,800 (F)

Sales Value Method:

Sales Value Variances may be calculated by substituting "Sales value and selling price" in lieu of "Sales margin". This method is still found in practice because the variance under this method enable the manager to see the effects of the various factor on his overall sales value figures. The formula for different sales value variances are :

(a) Total Sales Value Variance = Actual Sales Value Budgeted Sales Value.

(b) Sales Volume Variance = (Actual Quantity x Std. Price of actual mix) - Budgeted Sales Value.

(c) Sales Mix Variance = Actual Quantity (Std. Price of actual mix-Std. price of std. mix)

(d) Sales Price Variance = Actual Sales at Std. Price (Actual Quantity x Std. Price of actual mix)

(e) Sales Quantity Variance = (Actual Quantity x Std. Price of std. mix) - Budgeted Sales.

Example :

From the following data calculate sales value variances.

Budgeted Sales	A 300	units at Rs. 10	Rs. 3,000
	B 200	units at Rs. 5	Rs. 1,000
	750		5,500

Solution:

Standard Price per unit of Std. Mix. $= \frac{4,000}{500} =$ Rs. 8

Std. Price per unit of actual mix. $= \frac{(500 \times 10) + (250 \times 5)}{750}$

$= \frac{5,000 + 1,250}{750}$

$= \frac{6,250}{750} =$ Rs. $8\frac{1}{3}$

(a) Sales price Variance = 5,500 - 6,250 = Rs. 750 (A)

(b) Sales Mix Variance = (750x8) - (750x) = 250 (F)

(c) Sales Quantity Variance = (750x8) - 4,000 = 6,000
4,000 = 2,000(F)

(d) Sales Volume Variance = $(750 \times 8\frac{1}{3}) - 4,000$
= 6,250 - 4,000 - 2,250 (F)

(e) Sales Volume Variance = 4,000 - 5,500 = 1,500 (F)

Proof :

Sales Value Variance = Sales Price Variance + Sales Volume Variance

[1,500(F) = 750(A) + 2,250(F)]

3,000 / Rs.673 / 92 – 1 – 21

Profit of Loss Variance:

profit (or loss) Variance is the difference between the budgeted Profit (or loss) and the actual profit (or loss). This variance comprises the sum of variances appropriate to standard cost of sales, the sales margin variances and variances on any charges which have not been included in standard cost of production.

Management is primarily interested in profit (or loss) variances and as such a suitable statement must be prepared and presented to management. From this management will get clear and complete picture of the company's results for each month, quarter or year.

Presentation of Standard Cost Statement:

The statement that is ultimately prepared in a standard costing system is the statement of réconciliation between the standard profit and the actual profit. This is merely consolidated summary of all the variances discussed so far. A specimen form of such a statement is given below:

STANDARD COST STATEMENT FOR JANUARY 1981

Item	Standard	Variance F	Variance A	Actual
Standard Sales Value				
Salves Variances				
Price				
Quantity				
Mix				
Actual Sales Value				
Less: Std. Cost of Sales				
Direct Material				
Mix				
Actual Sales Value				
Less: Std. Cost of Sales				
Direct Material				
Direct Labour				
Variable O.H.				
Fixed O.H.				
Standard Net Profit				
Cost Variances				
Material :				
Price				
Mix				
Yield				

Labour :

Rate

Efficiency

Mix

Variable O.H. :

Expenditure

Efficiency

Calendar

Capacity

Total of Cost Variance

Actual Net Profit

Accounting Entries:

The principle of accounting in standard costs is to

[i] Debit work-in-progress A/c at Standard costs

[ii] Credit all expense control A/c at actual costs and

[iii] Keep all variances under separate A/cs and close them to costing P & L A/c.

For example to record Material Cost Variances, the entry required is:

1.	Work in Progress A/c	Dr.	340
	Material Mix Variance A/c (A)	Dr.	30
	Material yield Variance A/c (A)	Dr.	20
	To Material Price Variance (F)	Dr.	20

To Stores Ledger Control A/c		370
Material Price Variance	Dr.	18
P & L. A/c	Dr.	32
To Material Mix Variance		30
To Material Yield Variance		20

(Variance A/cs are closed by transfer to P &L A/c). Similarly for other cost variances entries can be recorded likewise.

Test Yourself:

1. What is a Standard Costing? How does it differ from budgetary control? In what way standard costing is superior to historical costing?

2. What are the different types of standards? Explain how the standards are to be set for materials and labour?

3. What is an attainable standing? How does it differ from ideal standard? Explain the method of setting overhead standards.

4. What is a Variance? What are the different material variances? Explain the reasons for such variances?

5. What is variances Analysis? How do you account for the different labour variances in a manufacturing concern.

6. What are the advantages of having a system of Standard Costs? Have they any significance in effective Budgetary Control?

7. What do you understand by:

(a) Usage Variance

(b) Mix Variance

(c) Yield Variance

Give examples of each of these variances. As the management accountant of a manufacturing concern, what steps would you advice to remedy adverse variances of this kind?

8. In a manufacturing process, the following standards apply:

Standard Price:	Raw Material x Rs.1 per kg. Raw Material Y Rs.5 per kg.
Standard Mix:	75% x 25% Y, (by weight)

Standard Yield (weight of product as percentage of weight of Raw material) 90%

In a batch 500 kg of chemical 'B' was produced from:

x 140 kg at Rs. 588
y 220 Kg at Rs. 1056
z 440 Kg at Rs. 2860

Explain the variances for 100 kg of production.

12. Standard for 100 kg. of output is:

Material A	45	Kg. at Rs.2 per Kg.
Material B	40	Kg. at Rs.4 per Kg.
Material C	25	Kg. at Rs. 6 per Kg.
	110	
Less: Std. Loss	10	
Production	100 Kg.	

Actual Production 2000 kg. usuage (actual)		
Material	Kg.	Rate Rs.
A	1000	1.9
B	850	4.2
C	450	6.5
Prepare a State of Material Variance		

13. A company is manufacturing a special type of tile 12" x 8" x 1/2". The standard mix of a compound which is to produce 12,000 sft. tiles of 1/2" thickness is:

1200 Kg of Material 'A' at 30 paise
500 Kg of Material 'B' at 60 paise
800 Kg of Material 'C' at 70 paise

During one period 1,00,000 tiles were produced. Actual data

A 7,000 Kg at 32 paise
B 3,000 Kg at 65 paise
C 5,000 Kg at 75 paise

15,000

Prepare a statement of material variances.

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LESSON : 14

BUDGETARY CONTROL

Introduction :

Planning is an important function of management. It is an all- pervasive function in the sense that every aspect of a business calls for thinking in advance anticipate problems and be prepared to meet them. Often those organisations which plan their activities prove to be more efficient than those without plans. The final shape of a plan is the budget. Budget as a tool of control and per formance evaluation of the various divisions of an enterprise is now being universally recognised.

Evolution of Budgetary Control:

Budgeting, contrary to the common belief, seems to be an old practice. Even in the days of the phrase forecasting and acting on the forecasts seem to have been practiced. Joseph, in Egypt, made a budget of corn supplies and planned Pharaoh's investment and consumption policy in the light of it. In England Budgetary Control seems to have been practiced for more than 200 years in Govt. Departments, where it is necessary to equate revenue (income) with expenditure. On a smaller scale all of us are familiar with some sort of budgeting on an informal basis whenever we have to balance our income with our expenditure. In the USA the practice dates back to 1921.

Even in the humbler sphere of business budgeting is of respectable antiquity. As early as 1825, a French author referred to the use of accounting date for future use, in this book on agricultural accounting. He insisted upon the need to establish monthly budgets and annual budgets. In the past 70 years this vital concept has been tried in businesses for cost control and it has reached the state of an art. It is also recognised as an imperative step for efficient use of resources. It puts pla..ning where it belongs-in the forefront of the manager's mind.

Definition:

A budget is defined as an account of probable future income and expenditure by the Longman Dictionary of Business English. This is too narrow a definition. In management it is defined as an account of the probable cost of carrying out a plan of programme for a special purpose. They are simply our best estimates of the effect of our plans in monetary terms. They provide a standard against which our actual performance is measured. Charles T. Horn-gren defines a budget as "a formal quantitative expression of management plan.

A Dictionary for Accountants (Eric L.Kyhler) defines a budget as "Any financial plan serving as an estimate of and a control over future operations. Hence, any estimate of future costs. It is any systematic plan for the utilisation of manpower, material or other resources.

Webster's Dictionary gives the meaning of the term budget as a "statement of the financial position of a sovereign body for a definite period of time based on the estimates of expenditures during the period and proposals for financing them. It is a plan for the coordination of resources and expenditure.

The Terminology of Cost Accounting of the Institute of Cost and Management Accountants of England defines a budget as " a financial and/or quantitative statement prepared prior to a defined period of time of the policy to be pursued during that period for the purpose of attaining a given objective.

In the words of Shilling Law a budget is a predetermined detailed plan of action developed and distributed as a guide to current operations and as a partial basis for the subsequent evaluation of performance. They are the expression, largely in financial terms of management's plans for operating and financing the enterprise during specific period of time. They are instruments of management control in three ways; i) as the means of organising and directing a large segment of the management planning process; ii) as continuing reminders of current plans and programmes for the day-to-day management; iii) as the benchmarks against which to measure actual performance.

Budgetary Control is defined by the ICMA of England as "the establishment of budgets relating the responsibilities of executives to the requirements of a policy and the continuous comparison of actual with budgeted results. either to secure by individual action the objectives of that policy or to provide a firm basis for its revision".

Budgetary control is planned to assist management in the allocation of responsibilities and authority, to aid in making estimates and plans for the future, to assist in the analysis of the variations between estimated and actual results and to develop bases of measurement standard with which to evaluate the efficient of operations.

Because of its projections regarding future profits of an enterprise a budget is often called a profit plan.

Budget Period :

Depending upon the period for which the plan is being drawn a budget may be for a long term or a short term. However a budget for a short period pertaining to the near future is likely to be more realistic because the projections will be less uncertain. The father you try to look into the future, more and more factors will be uncertain. Therefore your projections also will be less accurate. the period for which the budget is prepared is called the budget period. The year upto which the budget is drawn up is called the budget horizon. Even if a long-term budget is prepared it is necessary to break it down into budgets for shorter periods. For example if the budget covers the five years, it is usual to break it into five annual budgets. Even an annual budget is divided into 12 monthly or 52 weekly budgets. Where market conditions are changing, forecasts cannot be accurate. In such circumstances the budget period should be short. As all businesses have seasonal variations in sales and production the budget should cover the normal business period.

Many concerns are now using two periods for budgetary control, a long period usually from one to three years and a short period ranging from one to three months. The long term budget consists of general estimates including seasonal variations which help in the preparation of general operating plans and policies. The monthly or quarterly budgets contain detailed estimates of operating costs and income. This helps the comparison of actual with budgeted figures and the analysis of the deviations. Short budget periods facilitate adjustments in the original budget figures to suit the changing conditions.

The adoption of the 12 month calendar as the budget period has been found to have some disadvantages. The length of the month may be 28,29, 30 or 31 days. With the exception of February each month has four weeks of 7 days plus two or three days. Further the date of a given month fálls on a different day of the week from the preceding year and this complicates the preparation of payroll and other records. The comparison of accounting data and performance must be between identical periods of time to be of the greatest value. What now happens is a thirty-day-month budget is compared with à 28.29 or 31 day month performance involving expensive and time consuming reconciliations of monthly statements and budgets before successful comparisons are possible. To avoid these complications many American and Canadian companies have adopted a 13 month calendar for all internal record but use the regular 12 month calendar for external purposes. This helps in valuable comparisons to be made because of the standardisation and equalisation of the budget periods periodic statistical and accounting reports can be more easily prepared and analysed because the periods are equal length and the closing period end with the same day of the week. The payroll periods of equal length and a the disadvantages of spilt payrolls are avoided. Further the most complicated calendar variances; need not be calculated.

Many companies in the USA are adopting variations of the 13 month budget period. Where the year happens to be a leap year usually the last month is added with the extra day.

Elements of Budgetary control:

The method of operation of a system of budgetary control can best be understood by comparing it with the routine of a captain of an aircraft on his fights each day. Before commencing his flight he collects information about his destination, the route he has to take, the weather forecast for the entire route, the working of each part of the aircraft, the fuel position, the staff assigned to him, the number of passengers he has to carry, their destinations, security check on them, their luggage, any special condition or factor, the time of departure the expected time of touchdown at the destination, etc. During the flight he will be in continuous communication with the groundstaff and will be monitoring his position and check whether everything is functioning properly. He will record any unusual condition and if necessary he may even have to deviate from his usual route.

From the above it can be seen that budgetary control involves the following steps;

1) Establish a plan or a target for each of the divisions in the organisation, and coordinate them to the plant for the entire organisation.

2) Record the actual performance through regular feedback reports.

3) Compare the actuals with the targets and measure the differences.

4) Analyse the differences, establish the causes for them. These causes could be either controllable or non-controllable.

5) Act either to remedy the cause or to alter the original plan.

Objectives of Budgetary control:

The main objectives of budgetary control are:

1) To involve all levels of executives in the framing of the plan so that there can be an assembly of various ideas.

2) To coordinate the activities or the various departments and direct them towards the achievement of the enterprise objectives.

3) To provide for centralised control.

4) To decentralise responsibility to each managerial level. Hence budgetary control is sometimes called Responsibility Accounting.

5) To provide a reference point and a guideline for managerial action especially when unforeseen factors affect the working of the business.

6) To plan the entire income and expenditure so that maximum profits can be reaped.

7) To direct the capital expenditure in the most profitable manner.

8) Anticipate the shortage of working capital funds and provide the required funds by identifying the sources of such funds in advance.

9) To provide yardsticks to measure the actual performance.

10) To show the top management the areas where their attention is needed urgently. In this the principle of exception is followed. Only the deviations are informed and the regular routine performance are not reported.

Organisational set up for budgetary control:

The success of budgetary control depends very much on the existence of an efficient organisation. As already mentioned, it is not a mere accounting exercise but an excellent tool for coordination at all levels. There must be whole hearted cooperation from everyone concerned without which it will turnout to be a futile exercise.

The following preliminaries have to be satisfied before the introduction of budgetary control.

1) A number of BUDGET CENTRES have to be created. Usually each department will be a budget centre. The departmental heads will be given the responsibility to prepare the budget for their own department. Each functional department will prepare a departmental budget, e.g., purchases production, sales etc.

2) There must be accounting records to an adequate extent. It is necessary to keep a chart of accounts in keeping with the budget centres.

3) Everyone must be told what is expected of him. Communication must be absolute and total. Each are must know what a budget is and what it is supposed to achieve for the organisation and what his role is. He must be convinced that he will be able to carry out the task assigned to him.

4) An organisation chart is to be prepared showing the responsibilities of various individuals in the budgetary control setup. An organisation chart showing the persons involved together with their respective roles is to be drawn up. The usual practice is to set up a budget committee consisting of all the

departmental heads. It will be headed by the chief executive. A Budget Officer will be named who will usually be the Cost Accountant or the Management Accountant or the Accounts Officer, as the case may be. His job will be to regularly sit with the other departmental heads and get the budgets from the various departments. He has to coordinate the various budgets and consolidate all of them into the Master Budget in the form of the projected profit and Loss Account and the Balance Sheet for the budget period. The Budget Committee will meet every now and then to discuss the rough budget figures supplied from the various departments. If the committee feels that particular department's budget has to be revised, the concerned department head will take the budget back to his department for the purpose. The procedure will go on till all the figures in all the budgets are approved by the Budget Committee. Thereafter the Budget Officer will prepare the Master Budget and place it before the Committee for approval. Once this is done the budget becomes a commitment of everyone to achieve the target set for his department..

The main functions of the Budget Committee may be summed up as follows:

i) To supply post data for forecasting.

ii) To issue instructions regarding budget requirements, last date for the receipt of the budgets form the departments, to decide the date for the next meeting.

iii) To decide the general policies of the management regarding the budget and to suggest changes where necessary.

iv) To periodically review, amend and approve the budgets.

v) To ensure that the budgets are submitted in time ....

vi) To compare the budgeted figures with the actuals from time to time and to analyse the variances.

vii) Generally to coordinate the entire budget programme.

viii) To prepare a BUDGET MANUAL; This is publication to facilitate the understanding by everyone, of the implications of the entire budget programme. It is defined by the ICMA as "a document which sets out the responsibilities of the persons engaged in the routine of, and the forms and records required for budgetary control". It is usually kept in loose-leaf form so that additions and deletions can be easily made whenever required. Loose leaf method also facilitates the issue of the appropriate actions to the executives. For easy reference an index is also provided. The usual information included in a budget manual are;

1) The general objectives of the system.

2) The description of tl.e system.

3) The statement of the general policies, and the procedures to be followed.

4) A clear demarcation of the duties and responsibilities of the various persons.

5) The reports and the statements to be prepared and the persons who should give and the pers ns who should receive them.

6) A description of the code to be adopted for the various accounts and the statements.

7) The time schedule for the submission of various statements. It must be remembered that the budget manual once prepared need not be considered to be unalterable. In fact it should be altered as and when needed. If must be continuously updated by constantly reviewing the working of the system prescribed.

8) Before the budget is drawn up it is necessary to identify the factors which are likely to prove to be constraints on the freedom in fixing the targets to be included in the budget. These factors are called the 'key factors' or limiting factors' or 'principal budget factors'. The extent of their influence must be clearly assessed so that the entire organisation is keyed up to meet the crisis when it comes. As has been already pointed out these 'governing factors influence the entire budgeting programme and help the management to recognise the limitations of the organisation. The usual 'governing factors influence the entire budgeting programme and help the management to recognise the limitations of the organisation. The usual 'governing factors' are sales, production, capacity, finance etc.

9) The budget committee under the chairmanship of the chief executive should meet as often as necessary for continuous dialogue between those who are preparing the budget so that any problem faced by them can be sorted out in time without waiting for the finalisation of the budget itself. The top management must be give its wholehearted cooperation to the programme of budgetary control without which it can never take off, let alone, succeed.

10) The budget officer or coordinator as the case may be is usually given the job of collecting the necessary figures from the various departments and coordinating them into the master budget. This is possible only if the fullest cooperation is extended to the programme by everyone involved in it. For this the chief executive must sufficiently motivate the various departmental heads

11) The procedures and the routines must be clearly established everyone must be made to understand them clearly. Communication must be unambiguous and total. The process of communication should incidentally help in wiping out mutual suspicious and result in better understanding among the executives of the various departments. In a way this is an indirect benefit of the budgeting process itself.

In short, unless the organisation is totally prepared to understand the implications of the budgetary control programme, the programme would have been hastily introduced and can never achieve its objectives.

Forecasts and Budgets:

Sometimes these two terms are misunderstood to be synonymous. The following are the differences between the two;

i) Forecasts are mere statements of probable events. They merely project into the future on the basis of th past and the possible conditions in the future. Budgets, on the other hand, are planned events and they are a programme of action to be actively pursued by the organisation.

ii) Forecasts may not have anything to do with the actual situation obtained with the organisation. They may be made by people outside the enterprise and as such may not be realistic for the enterprise. Budgets are conditioned by the local conditions and therefore prove more useful. Thus budgets become useful devices of control.

iii) Budgets are commitments made by the departmental heads as they are consulted before the targets for their departments are finalised: Hence they commit themselves to achieving the goals they have set for themselves. Forecasts, on the other hand do not involve any such commitment.

iv) Forecasts may be made for purposes other than budgeting. For, example, forecasts can be made about the consequences of following alternative courses of action. However, budgets will have only one course of action. Further economic forecasts of a general nature may not have anything to do with budgeting.

v) Forecasting usually precedes budgeting. It is on the basis of the forecasts the targets and plans for the next budget period are drawn up. Forecasting has become a highly sophisticated technique and makes use of mathematical and statistical tools. The results of using such technique to make forecasts which are closer to reality.

Differences between budgets and standards :
(Budgetary Control and Standard Costing)

Both are tools in the hands of a cost accountant to help the management in its control function. Controlling as a managerial function consists in fixing a goal, comparing the performance against it, analysing the variances and taking corrective measures after establishing the causes for the deviation if any. As control tools they have common features in their approach. But the following are the differences.

i) They differ in their basis purpose. Budgets are anticipated or expected costs used for forecasting requirements for a budget period. Standard costs relate to the various functional costs estimated on the basis of actual observations under given conditions of capacity, efficiency, duration, etc.

ii) Budgetary Control can be operated even without standard costing. However, wherever standard costing is used, it is usually combined with budgetary control.

iii) Budgets are complete in the sense that they cover all the activities of the firm such as sales, purchase, production, distribution, research and developments, etc. Standard costing is confined to production and related manufacturing costs only.

iv) While budgets include both income and expenditure standard costing includes only expenses.

v) Budgets fix the ceilings above which the expenses should not normally rise while the standards are the minimum targets that are to be attained at specified levels of efficiency.

vi) Variances from standards have to be analysed with greater care than those from the budgets. While variances from budget call for explanation from the particular department variances from standards have to be probed deeply to find further savings in the costs.

vii) Budgets are Indices pointing out the path of keeping out of difficulties, while standards are pointers to further efficiency.

Advantages to be derived from Budgetary Control:

1) The process of preparing a budget forces everyone, in the firm to think about the future. This makes everyone to anticipate the problems and prepare themselves to be ready to face the situation with possible solution, if any. Thus the firefighting type of management is replaced by a fully prepared management. Therefore there is no surprise element in a crisis ~~and all~~ keep cool to find a solution.

2) As every body has to take stock of the existing position budget preparation can be a time to examine each item carefully to see whether it can be given up or modified for greater economy.

3) Budget can be used as a motivator, if it is prepared with the consent of every one. It can be an excellent example of 'Management by exception, and 'Management by consent' rather than 'Management by imposition'. As all departments give their own. targets it becomes what may be called 'Participative pative management'. Planning at the grass root level is a good description of the system.

4) Budgets are a means of delegating responsibility from top management to lower levels. Thus the actions of the lower level executives can be easily controlled. Because of this budgetary control is called 'responsibility accounting; as was stated.

5) It spreads cost-consciousness all-round. The analysis of variances and identifying their causes will result in all the

department people becoming aware of the costs of their decisions and actions.

6) As a general administration policy guide, budgetary control helps attaining cooperation and coordination of different departments. The interdepartmental rivalry is eliminated and team spirit is encouraged.

7) It acts as a guide to each of the functional departments and efficiency is achieved because of the self-search and also because of the analysis of variances. Everybody is watchful because the reports generated by the system focus on the actions of individuals in the departments.

Limitations of Budgetary Control:

1. As has been pointed out, the success of the budgetary control system depends on the attitude of top management to its implementation half hearted implementation will not yield the expected results. This of course is not the defect of the system itself.

2. The process of fixing the responsibility is difficult in practice. Even reasonable accuracy is difficult to obtain, especially in fit industries. Under such circumstances budgets prove to be mere estimates.

3. There is a tendency to treat the annual budgets, once fixed as sacrosanct and many managements are an willing to alter it. But budgets to be reflective of the actual business situations must be altered to suit the needs of the changing situations. Without this, they become static and will be unrelated to reality and people will lose faith in its efficacy.

4. The team spirit, which it is supposed to create, often remains unachieved. In spite of the best efforts, many departmental heads may not be willing to co-operate and may take steps to disrupt the budget programme to discredit others. In all organisations informal groupism exists and this will affect the budgetary control set up.

5. Budgets become an annual ritual in many enterprises and budgets are drawn up rather mechanically. The coordinator is asked to supply the past period budget and the actuals for each department and on that basis certain percentage are added to the figures or certain items are kept static. Finally an additional 'margin' is added for 'contingencies' and this becomes the budget. Even at the 'Budget Committee 'level a certain percentage may be added for contingencies to the totals and thus 'padding' becomes universal. At the end of the budget period the actual expenses may be smaller than the budget and the departmental heads expect to be congratulated for spending less than their 'budgeted' expenses. An efficient top manager will immediately identify those departments which have been achieving less expenditure than the budget and will investigate the causes for such consistent 'savings' year after year. This becomes a 'budget game'.

6. In many organisations, another type of game is played in the implementation of budgetary control' system. Many departmental heads are aware of that if they spend less than their 'budget' the allocation for the next period will be drastically cut. Therefore they manage to spend their budgeted funds by scrambling for last minute expenses on various items which may not actually be necessary for them. On the basis of this they may demand a higher allocation in the next period.

'Never underspend your budget' is a well known slogan.

7. Similarlly while giving the physical targets the usual practice is to give a lower target and show a better achievement at the end. 'Never over state your performance' seems to be the other slogan for many other executives. Because, even if they are praised for this better performance their targets for the next period will be higher than the previous period. Therefore, it is the individuals who fix the target and decide the performance to be repeated. Thus the budgetary reports may not reflect the actuals and the entire data base on which the budgets are drawn up and operated may become unrelated to the actuals. Soon the budgetary control programme may loose its respectability

8. External factors cannot be budgeted with certainty and some emergency in the external environment may totally upset the budgets of the business. Under such circumstances, the budgets may become totally useless.

9. Some executives may discredit the budgetary control system by pointing out its deficiencies and may argue for standard costing as a superior device for control. They may argue for comparison of part of actuals with present actuals without an elaborate system of cross linked budgets. However a good budget is a better yard stick of performance than the past actuals or even the actuals of other forms.

10. Budgets being usually drawn up by the 'Accounts' people lead often to a clash between the finance people and the other departments. How ever, these clashes can be averted by the top management acting in time.

Types of Budgets:

Budgets can be classified on the basis of the different requirements.

1. Responsibility budgets-in this system the emphasis is on the control aspect through fixing the responsibilities on particular individuals.

2. Programme Budget-For each programme the management draws up a separate budget. This may be a short term or a long term budget depending upon the duration of the programme.

3. Operating Budget-This is the most popular annual budget broken into sub-period budgets fixing targets for different function over the entire period.

4. Functional Budget: These are the budgets for the sales production, purchase, labour, research and development, capital expenditure function of an enterprise.

5. Financial Budget: This consolidates the results obtained in the operating budget and the functional budgets from the point of view of cash flows profit and loss account and balance sheet.

6. Capital Budget: It assesses the capital needs and evaluates the capital expenditure projects on the basis of expected cash flows from the projects.

7. Performance Budgets: For individual sectors or areas budgets are prepared.

8. Flexible Budgets: These are the dynamic budgets which recognize the need for budgeting for various levels capacity levels which unknown factors may upset the actual capacity worked. More of it in the next chapter.

9. Zero Based Budget (ZBB) : They are of recent origin. They do not have the past budgets or actuals as the starting point for drawing up the new budgets. The data have to be collected afresh. This helps in avoiding the 'bias' towards the past figures. Everyone has to examine critically the activity, the expenditure etc., and come up with the budget. It is refreshing exercise in that nobody is tied down by the historical figures. The figures burden of past mistakes over estimation etc., will not be carried forward into the subsequent periods.

Recording and Reporting the Results:

The essence of the budgetary control programme is the reporting of the achievements. These reports provide a feed back on how the budgets are implemented. On the basis of these performances the cost controller will draw up variance reports, pointing out the deviations from the budgets and the factors which have caused the deviations. These deviations should be further analysed for establishing the causes and this will help in identifying whether they are controllable or non-controllable. Controllable causes show the need for corrective or remedial action. If the causes are non-controllable adaptive actions are taken to revise the targets. These actions are the key to the success of the whole budgetary control programme.

Here is an example of such a report which is usually known as a Departmental Operating Statement'.

Apart from the general form of the Departmental Operating Report giving the figures in absolute terms certain ratios are used for measuring deviations from the budget and these are incorporated in the budget reports. The commonly used ratios are;

1. **Standard capacity usage Ratio**

The ratio compares the maximum possible working hours in the budget period with the budged number of working hours in the period. It is usually expressed in percentage and indicates the use of the actual production facilities during the period. The formula for the ratio is.

Budgeted working hours Max. hours in the budgeted period of budgeted capacity

$$\frac{\text{Budgeted working hours}}{\text{Max hours in the budgeted period of budgeted capacity}} \times 100$$

2. **Actual usage of budgeted capacity ratio:**

This measures the relationship between the actual working hours and the budgeted working hours in the budgeted period. It indicates the extent of use of the production facilities actually made available during the budget period. The formula for this ratio is:

$$\frac{\text{Actual working hours}}{\text{Budgeted working hours}} \times 100$$

3. **Actual Capacity usage ratio:**

It compares the actual hours worked with the maximum possible hours in the budget period. The formula is

$$\frac{\text{Actual hours worked}}{\text{Max. Possible hours}} \times 100$$

4. Activity Ratio:

It compares the number of standard hours equivalent of actual production with the budgeted standard hours. It measures the actual level of activity at which the organisation is operating. The formula is :

$$\frac{\text{Actual production in std. hours}}{\text{Budgeted production in std. hours}} \times 10$$

5. Efficiency Ratio:

This compares the standard hours equivalent of actual production with the actual hours spent. It measures the efficiency level at which the firm has operated. The formula is

$$\frac{\text{Actual production in std. hours}}{\text{Actual hours worked}} \times 100$$

6. Calender Ratio :

It compares the actual number of working days in a period and the number of working days in the relative budget period. It indicates whether the maximum number of days in budget period have been actually used. The formula is :

$$\frac{\text{Actual working days in the period}}{\text{Working days in the budget period}} \times 100$$

SALES BUDGET

This budget shows the planned or targeted sales for the budget period both in quantitative and revenue terms relating to each product, area- wise, with all the important details in each area. The data for the preparation of the sales budget are usually available from the past records, but fresh information regarding the possibilities in the market is gathered through sales forecasts. It is important to be aware of the impact of the various market forces at work, on the company's sales programme.

Sales forecasts indicate the quantity of anticipated sales level at various price level for the different products. There are many sophisticated sales forecasting techniques. As no method gives totally reliable information, usually all the methods are adopted. The information available within the company viz., the past records of the company are modified in the light of fresh information gathered through: i) reports by salesmen, ii) market research, iii) analysis of part trends in sales, iv) a study of the general economic, industrial and trade conditions both internationally and nationally and interpreting then to understand their implications for the firm's business and v) other factors; such as a change in the company's sales promotion policy and techniques, the entry of competitors, product profitability, production capacity, government policy and controls, etc.

In the light of these factors and also the information regarding any limiting factors, the optimum sales mix is calculated and the budget committee should regard this as the budget, to give maximum profits in the budget period. Any other figure for the sales budget will result only in a lower profit.

In this process, the sales-staff play a vital role. They will be the persons who are intimately connected with this activity and the information gathered and supplied through the area salesmen, the divisional and zonal sales managers and the judgment of each one of them regarding the predictions for the budget period will all go to concretise the sales budget figures.

Illustration 1:

MULTI PRODUCTS LTD.; markets two products X and Y in two areas Madras and Madurai. The following information is available from its records Budget for the current period:

Product	Madras	Madurai
X	1,000 at Rs. 6/-	3,000 at Rs. 6/-
Y	700 at Rs. 20/-	1,000 at Rs. 20/-
Actual Sale		
X	3,000 at Rs. 6/-	4,000 at Rs. 6/-
Y	1,000 at Rs. 20/-	800 at Rs. 20/-

A study of the markets reveals that X is popular with the people but there is a feeling that it is underpriced. If the price is raised by Rs.2/- per unit of X, still the market will remain unaffected for it. On the other hand it is understood that the main reason for the poor sales of Y in both areas is its price. The sales high people have informed that a reduction of Rs.1/- per unit of Y will bring about higher sales for it. The company has decided to adopt these price changes. As a result the following projections have been made: percentage increase over current actuals.

Product	Madras	Madurai
X	Plus 20%	Plus 10%
Y	Plus 25%	Plus 15%

Show the sales budget for the next period.

Solution:

MULTI PRODUCTS LTD.

SALES BUDGET

For the period ....

Prepared by

Checked by..................

AREA		Budget for future period			Budget for current period			Actuals for current period		
		Units	Price Rs.	Value Rs.	Units	Price Rs.	Value Rs.	Units	Price Rs.	Value Rs.
Madras	X	3,600	8	28,800	1,000	6	6,000	3,000	6	18,000
	Y	1,250	19	23,750	700	20	14,000	1,000	20	20,000
Total		4,840		52,550	1,700		20,000	4,000		38,000
Madurai	X	4,400	8	35,200	3,000	6	18,000	4,000	6	24,000
	Y	920	19	17,480	1,000	20	20,000	800	20	16,000
Total		5,320		52,680	4,000		38,000	4,800		40,000
SUMMARY	X	8,000	8	64,000	4,000	6	24,000	7,000	6	42,000
	Y	2,170	19	41.230	1,700	20	34,000	1,800	20	36,000
Total		10,170		1,05,230	5,700		58,000	8,800		78,000

SELLING AND DISTRIBUTION COST BUDGET

This shows the cost of selling and distribution of the produced quantity shown in the sales budget. It is usually prepared as an adjunct of the sales budget after consulting the sales manager, distribution manager, advertising manager and the accounts officer.

Illustration : 2

MULTI NATIONALS LTD., decide to prepare a Selling and Distribution Costs Budget for 1984 on the basis of the Sales Budget (as in illustration1) The following information is available. Selling and Distribution Cost Budget for 1983.

Costs	Madras	Madurai	Total
Direct Selling Costs:	Rs.	Rs.	Rs.
Salesmen's Salaries	4,000	7,000	11,000
Salesmen's Commission	5,000	8,000	13,000
Conveyance	3,000	3,000	6,000
Sales Office Expenses:			
Staff Salaries	4,000	5,000	9,000
Rent, Rates, etc.	4,000	500	900
Miscellaneous Expenses	1,000	1,000	2,000
Advertising Costs:			
Press, Radio & TV	5,000	6,000	11,000
Others	2,000	2,000	4,000
Distribution Costs:			
Warehouse Wages	3,000	4,000	7,000
Drivers Wages	2,000	3,000	5,000
Cleaners Wages	1,500	2,000	3,500
Van repairs and Maintenance	1,000	1,500	2,500
Miscellaneous Expenses	500	1,000	1,500
Total	36,000	44,000	80,000

During the next period:

1) Salesmen's salaries are to go up by 20% and one more salesman is to be employed at Rs.350-p.m. in Madurai area.

2) Salesmen's commission is to be 10% more than the current period.

3) Conveyance expenses will be increased by 20% owing to increased transport cost.

4) Staff salaries to go up by 20%. One more clerk at Rs.300/- p.m. will be employed in each area.

5) Rent, rates, etc., are expected to be increased by 5%

6) Advertising through press, radio and TV will be increased by 20% and other advertising costs by 5%

7) Where house wages, drives wages and cleaners, wages will go up by 10%

8) Van repair and maintenance costs are likely to go up by 20%

9) Miscellaneous expenses in general will be increased by 10%

Prepare the Selling and Distribution Costs for 1984.

MULTI-PRODUCTS LIMITED
SELLING AND DISTRIBUTIONS COSTS BUDGET FOR 1984

COSTS	BUDGET - 1983 Rs.			BUDGET - 1984 Rs.		
	Madras	Madurai	Total	Madras	Madurai	Total
Direct Selling Costs						
Salesmen's Salaries	4,000	7,000	11,000	5,000	12,000	17,000
Salesmen's Commission	5,000	8,000	13,000	5,500	8,800	14,300
Conveyance	3,000	3,000	6,000	3,300	3,300	6,600
Sales Officer Expenses						
Staff Salaries	4,000	5,000	9,000	8,400	9,600	18,000
Rent, Rates, etc.	400	500	500	420	525	945
Miscellanous Expenses	1,000	1,000	2,000	1,100	1,100	2,200
Advertising Costs						
Press, Radio and TV	5,000	6,000	11,000	5,500	6,600	12,100
Others	2,000	2,000	4,000	2,100	2,100	4,200
Distribution Costs						
Warehouse Wages	3,000	4,000	7,000	3,300	4,400	7,700
Drivers' Wages	2,000	3,000	5,000	2,200	3,300	5,500
Van Repairs & Maintenances	1,500	2,000	3,500	1,650	2,200	3,850
Miscellaneous Expenses	1,000	1,500	2,500	1,200	1,800	3,000
	500	1,000	1,500	550	1,100	1,650
Total	32,400	44,000	76,400	40,200	56,825	97,045

Production Budget:

This budget shows the forecast of production in units for the budget period. It is followed by the various cost budgets for materials, labour, over heads, etc. The production Budget must indicate the targets of production for each product, each department for each month. A carefully prepared production BUDGET will lead to better inventory control, improved production schedules, better co-ordination between production and sales departments and reduced costs leading to increased profits. The following factors are to be kept in mind while preparing the production budget.

(a) Production capacity in the various departments may be measured in terms of units or available hours.

(b) The practical operating capacity may be considerably lower than the licensed, maximum and installed capacities. This may be due to shutdowns and breakdowns on account of shortage of power, machinery breakdown, labour turnover, absenteeism, idle time owing to various causes etc.

(c) The related tools must be readily available close at hand.

(d) The time lag between input and output. This is important in scheduling the material purchase.

(e) Bottleneck departments and limiting factors must be identified and the flow of production must be kept even.

(f) Seasonality of the market for the product must be kept in mind. production will be increased just before the busy season and will be slackened just before the slack season.

The production Budget should be linked with the Sales Budget and the policy of the management regarding the quantities to be held in stock will influence the production targets. Production targets are calculated by taking into account the opening inventory of finished stocks, the sales budget for the period and the closing inventory to be carried over to the next period.

Illustration : 3

From the following prepare a production Budget for 1984 of the Multiproducts Ltd.

Products	Estimated	Stock	Sales as per Sales Budget (units) (see Ill.1)
	1.1.1984	31.12.1984	
X	1,000	1,250	8,000
Y	300	500	2,170

Solution :

MULTI NATIONALS LTD.

PRODUCTION BUDGET for 1984

Items	X Units	Y Units	Remarks
Sales Budgeted	8,000	2,170	
Required Stock on 31.12.1984	1,250	500	
Total	9,250	2,670	
LESS: Opening Stock	1,000	300	
PRODUCTION ESTIMATED	8,250	2,370	

Production Cost Budget:

In this budget the physical units of production given in the production Budget are broken down into the elements of cost viz. materials cost, labour cost and overheads and each element is budgeted. Thus the production cost Budget is usually divided into sub- budgets and then a summary is prepared to show the overall budgeted cost of production. The production Cost Budget is usually prepared in the following form.

MULTI PRODUCTS LTD.

PRODUCTION COST BUDGET for 1984

BUDGET UNITS	X	Y	Total
	Rs.	Rs.	Rs.
Elements of Cost :			
Materials cost (from materials cost budget)			
Labour cost (from labour cost budget)			
Factory overheads (from factory overheads budget)			
Total			

Material Purchase Budget

This budget shows the total quantity of various materials to be used in a budget period for each product along with the total costs for each. It helps the purchase Department in scheduling the purchases so that they are available whenever needed. This budget precedes the purchases Budget. Also this budget helps the Stores Department in fixing the maximum and minimum levels for various materials and components. The Materials Budget is prepared in three stages. In the first stage the quantity of material required for each product/cost centre etc. is determined. This is decided after allowing for all normal losses. This quantity is multiplied by the rate to give the cost of materials for each product or cost centre. Then the cost of individual materials are added up to give the total cost of materials for the budget period.

This is usually prepared in the following form.

MATERIAL PURCHASE BUDGET

	Material X	Material Y
Estimated Consumption	XXX	XXX
ADD: Estimated Closing Stock	XXX	XXX
	XXX	XXX
LESS: Estimated Opening Stock	XXX	XXX
Estimated Quantity to be Purchased	XXX	XXX

If the purchase prices of Raw Materials are given we can also calculate the purchase Cost.

Illustration :4

Prepare a purchase Budget for 1985 of Multiproducts Ltd based on the following informations.

	A Kg.	B Kg.	C Kg.
Estimated Consumption during 1985	25,980	29,490	43,620
Estimated Stcok on 1.1.1985	5,000	6,000	8,000
Estimated Stock on 31.12.1985	6,000	6,500	9,000
Price Per Kg.	Rs.5/-	Rs.3/-	Rs.2/-

Solution :

MULTI PRODUCTS LIMITED

Purchases Budget for 1984

Items	Material A (Kg.)	Material B (Kg.)	Material C (Kg.)
Estimated consumption during 1984	25,980	29,490	43,620
ADD: Closing Stock to be carried	6,000	6,500	9,000
	31,980	35,990	52,620
LESS: Opening Stock	5,000	6,000	8,000
Estimated quantity to be purchased	26,980	29,990	44,620
Prices per kg.	Rs.5/-	Rs.3/-	Rs.2/-
Cost of Purchase Rs.	1,34,940	89,970	89,240

Direct Labour Budget:

Similar to the Materials Budget, this budget shows the different grades of labour required and the number of labourers required plus the total wages expenses to be incurred during the period. The standard time for each product along with the standard rates of wages have to be fixed before a correct estimate of the wages to be paid can be made. After this the total wages payable for each category of labour for each product are totalled to give the total wages to be paid during the budget period. This will also help in fixing the labour cost per unit during the budget period. The labour cost budget is easier to prepare than the materials budget as there is no question of opening or closing stocks.

Personnel Budget:

This is different from Direct labour Budget in that, it includes both direct and indirect labour costs to be incurred during the budget period. The details shown will include total manhours required for each category of labour in each department and also total labour costs for the departments product wise and also the overall labour cost. The costs will include estimated overtime wages to be paid training costs, costs of amenities, etc. The preparation of the personnel Budget helps management to test the efficiency of the personnel department to focus attention on the manpower costs and see whether economies can be made in this sphere. It can also provide an opportunity of evolving suitable yardsticks to measure the manpower requirements of the various departments. This search may throw up incidentally the cause of high labour turn over, if any. The proportion between direct labour can be calculated. As wages necessiate holding of cash to be disbursed, it helps in preparation of the Cash Budget.

Production Overheads Budget:

Before this budget can be prepared the overheads should be classified into fixed, variable and semi-variable categories.

This is a laborious and tedious task as many overheads are difficult to classify. However by studying the behaviour of these in the past, it is possible to classify them though not with great accuracy. Usually standing order numbers are given to identify the expenses. The second requisite is to departmentalise the cost to fix the responsibility and for control purposes. Each department may be asked to prepare a budget of its own and then all these can be consolidated into one budget. In this process the Service Department Overheads have to be redistributed to the production Departments on a predetermined basis to give the final estimated manufacturing overheads for each department productwise, during the budget period.

Administration Costs Budget:

This budget indicates the total estimated expenditure on top management general office, accounts, legal finance etc. departments during the budget period. Budgetary figures can be decided on the basis of past records. As most of these fall in the category of fixed costs, they can be easily calculated providing for anticipated changes.

Other functional budgets:

In a large organisation, there may be many more functional budget depending on the various departments with different functions. Some of the more important budget are briefly described here:

1) **Maintenance Cost Budget:**

This is prepared where the maintenance cost sale usually substantial. Routine maintenance costs may be predicted with some certainty. However sudden breakdowns calling for emergency maintenance costs to be incurred cannot be estimated with accuracy. Each department may be required to submit its own anticipated maintenance expenditures which may then be consolidated into the maintenance expenditures Cost Budget. The Budget may be conveniently be divided into three parts; the costs of preventive maintenance, the cost of emergency repairs and the costs of the Maintenances Department.

ii) Plant Utilisation Budget:

This budget indicates the total machine capacity required in each department to meet the production targets given in the production budget. It will point out whether there is enough capacity in each department and also whether there is overloading or underloading in any department. Further, it will also point out whether fresh equipments are to be ordered, etc., which is important to prepared the Capital Expenditure Budget. The capacity and under-capacity indicated in this budget may be useful to the management in planning the future capacities in the various departments.

iii) Research and Development Cost Budget :

This is defined as the planned outlay on Development and Research in so far as this work is not regarded as directly incidental to the existing manufacturing and trading operations of the business. The budget defines in terms on money the permissible limits within which activities are to be pursued and the directions which they are to take (ICMA -LONDON). The budget shows the estimated total expenditure on fixed assets during the budget period. As the amount to be laid out are large. the decisions involved in this budget are to be made by top management only. The budget is preceded by several calculations and evaluation of the alternatives available.

Annual Capital Expenditure Budgets are usually compiled from the long term forecasts. The Budget coordinator first receives the requests from the various departments. Then he consolidates them after approval from the authorised person. Capital Expenditure Budgets may be prepared under the following convenient heads:

i) Projects already in progress requiring fresh expenditure.

ii) Projects which have been approved for the budget period.

iii) Projects which been approved earlier but not yet started.

Priorities are decided on the basis of pre-determined criteria. The objects of the Capital Expenditure Budget are:

i) To correct capacity imbalances-According to the ICMA publication on Budgetary Control and Standard Costing.

"The plant utilisation Budget may indicate that in certain places the present capacity is fully loaded, whereas in other places there are considerable idle facilities, where this is so, the capacity of the factory as a whole may be capable of much expansion with relatively little capital expenditure".

ii) To decide the financial allocation.

iii) To finalize the depreciation figures and maintenance costs to be included in the Maintenance Cost Budget.

The influence of the capital Expenditure Budget will be felt in all the other budgets as the production Budget cannot be finalised without the capacity considerations, manufacturing overheads should include power, depreciation and maintenance costs, equipment or capital expenditure for the office or the selling and distribution function will include the annual estimated expenses on capital expenditure in connection with these respective functions. The cash budget will include the scheduled payments for the capital expenditure decisions,

Cash Budget:

This is also called Financial Budget' or Ways and Means Budget. The purpose of this budget is to portray the estimated cash receipts the cash payments and the cash balances, usually monthly, over the entire budget period It shows the period during which payments may be more than the receipts and hence the periods of cash shortage. The management may decide in advance the methods of raising the cash inflow through various methods including borrowings and reduction sales to ensure enough various methods including borrowings and reduction sales to ensure enough cash inflow to meet the overflow. Similarly surplus cash periods known in advance will help in making arrangements for profitable alternative uses for the cash.

A cash budget is usually preceded by a cash forecast. A cash forecast is an estimate of available cash in a future period. The sources of cash inflow will be usually, cash sales, debt collections, borrowings, sale of assets. interest and dividends on investment etc. Payments will usually include recurring payments to creditors wages salaries payments for fixed assets payment of loans interest and dividend payments, payments for operating expenses taxation payments etc.

There are three methods of preparing the cash forecast:

1. Receipts and Payments Method:

This is usually adopted to prepare on short term forecast based on the figures in the functional budgets including the Capital Expenditure Budget. Sales budget shows the monthly sales. These will have to be split into cash sales and credit sales. Credit sales figures cannot be incorporated into the cash budget straightway. The amounts collected from the debtors depends on the debt collection period allowed by the firm and on this basis only cash from accounts receivables can be included in the cash budget. The cash forecast starts with the opening balance of cash on hand and ends with the closing balance at the end of the period. It is usual to show the monthly total receipts and payments and also the surplus of deficiency at the end of each month.

Illustration : 5

ABC Co. wished to arrange overdraft facilities with its Bankers during the period April to June 1985 when it will be manufacturing mostly for stock. prepare a cash budget for the above period from the following data indicating the extent of the bank facilities the company will require at the end of the each month.

(a)

	Sales Rs.	Purchases Rs.	Wages Rs.
February	1,80,000	1,24,800	12,000
March	1,92,000	1,44,000	14,000
April	1,08,000	2,43,000	11,000
May	1,74,000	2,46,000	10,000
June	1,26,000	2,68,000	15,000

(b) 50% of the Credit Sales are realised in the month following the sales and the remaining 50% in the second month following creditors are paid in the month following the month of purchase.

(c) Cash at Bank on 1-4-1985 (Estimated) Rs.25,000.

Solution :

Cash Budget for 3 months (April to June 1985)

Items	April Rs.	May Rs.	June Rs.
Opening Bank Balance	25,000	53,000	(-)51,000
ADD: Receipts :			
1. Collection from Debtors	1,86,000	1,50,000	1,41,000
	2,11,000	2,03,000	90,000
LESS: Payments :			
1. Payment to Creditors	1,44,000	2,43,000	2,46,000
2. Payment of Wage	14,000	11,000	10,000
	1,58,000	2,54,000	2,56,000
Closing Bank Balance	53,000	--	--
Bank Overdraft	--	(-)51,000	(1)1,66,000

Workings :

COLLECTIONS FROM DEBTORS

APRIL	50% of Feb.	90,000	MAY	50% of Mar.	96,000
	50% of Mar.	96,000		50% of Apr.	54,000
		1,86,000			1,50,000
JUNE	50% of April	54,000			
	50% of May	87,000			
		1,41,000			

Note : It is assumed that wages are paid on monthly basis the first of very next month.

Illustration : 8

From the following figures extracted from the functional budgets of EMPTY LTD Prepare a monthly cash budget for the 6 months ending September 1983 In the months .of deficient an overdraft may be arranged with the bankers for the required amount.

Items	Jan. Rs.	Feb. Rs.	Mar. Rs.	Apr. Rs.	May Rs.	June Rs.	July Rs.	Aug. Rs.	Sep. Rs.
Sales	6,000	7,000	6,000	5,000	4,500	6,500	7,000	7,500	8,000
Materials	2,500	3,000	3,000	2,500	1,800	3,000	3,000	4,000	3,000
Wages	1,000	1,100	1,100	1,000	900	900	1,000	1,000	1,000
Overheads									
Manufacturing	800	900	900	800	700	700	800	800	900
Admin.	300	400	400	300	300	300	400	400	400
Selling	400	400	400	500	400	300	300	300	400
Distribution	300	400	400	400	300	200	200	300	400

Notes:

a) A plant is to be purchased for Rs.6,000/- to be paid in 6 equal instalments starting from June.

b) The purchase instalments at Rs.500/- p.m. on equipment purchased last year are to be provided.

c) 10% commission is to be paid on sales following the month of sales.

d) Cash sales of Rs.400/- p m. is included in the above figures (on which no commission is payable).

e) Interest on 6% Debentures (Rs.10,000) is due for payment on 1st July for the half year ending 30th June.

f) Interest on investments Rs.800/- is to be received in August.

g) Advance tax instalment of Rs.800/- is payable in August.

h) Rs.4,000/- share cash amount will be received in April.

i) Debtors are allowed 3 months to pay.

j) Creditors are to be paid 2 months after purchases.

k) Wages and overheads are to be paid one month in arrears. The estimated cash balance on hand on 1st April was Rs.10,000.

Solution : **CASH FORECAST FOR 6 MONTHS ENDING SEPTEMBER**

Items	April Rs.	May Rs.	June Rs.	July Rs.	August Rs.	September Rs.
RECEIPTS :						
Balance	10,000	12,700	12,700	11,650	7,300	3,900
Debtors	5,600	6,600	5,600	4,600	4,100	6,100
Cash Sales	400	400	400	400	400	400
Interest Receivable	--	--	--	--	800	--
Share Call	4,000	--	--	--	--	--
Total	20,000	19,700	18,700	16,650	12,600	10,400
PAYMENTS :						
Materials	3,000	3,000	2,500	1,800	3,000	3,000
Wages	1,100	1,000	900	400	1,000	1,000
Admn. Overhead	400	300	300	300	400	400
Mfg. Overhead	900	800	700	700	800	800
Selling Overhead	400	500	400	300	300	300
Distribution Overhead	400	400	300	200	200	300
Income Tax	--	--	--	--	800	--
Interest on debentures	--	--	--	3,000	--	--
Hire purchase instalment	500	500	500	500	500	500
Commission on Sales	600	500	450	650	700	750
Plant Account	--	--	1,000	1,000	1,000	1,000
Total	7,300	7,000	7,050	8,350	8,700	8,050
Cash on Hand	12,700	12,700	11,650	7,300	3,900	2,350

ii) Balance Sheet Change or Balance Sheet Forecast method:

This is useful for long term cash forecast. The anticipated changes if. the assets and liabilities are either added or deducted. Reduction in the asset items is interpreted as cash inflow and increase is treated as an outflow. Similarly increase in the liability items is treated as cash inflow and any reduction is treated as a cash payment. To the figure so arrived, net profit before depreciation, without deduction of non-cash charges like provision for bad debts provision for taxes losses on sale of asset, deferred revenue expenses written off, capital losses written off etc. are added. The method does not show the actual expenses incurred and also it shows only the cash at end. It does not show the changes in cash from month to month which is more important for management decisions.

iii) Profit Forecast Method:

This is also suitable for long term cash forecast. It is based on the view that it is the profits which provide the cash. The net profit before depreciation and other non-cash charges is the starting point for each period. To this is added capital receipts decrease in assets etc. period wise. From this is deducted the expenses prepayments, capital expenses, increase in assets etc. The final adjusted figure combined with the cash balance at the beginning for each period will show the cash balance at end.

Flexible Budget :

ICMA -LONDON Terminology defines a flexible budget as 'a budget which by recognising the difference between fixed semi fixed and variable cost is designed to charge in relation to the level of activity attained.

"A fixed Budget is a budget which is designed to remain unchanged irrespective of the level of activity actually attained.

Sometimes the term "Multiple Budget" is used to describe one which is prepared for several different volumes of production and sales.

Flexible budgets are also called variable' or 'sliding scale budgets.

A fixed budget is not adjusted to the changed business condition. Therefore it should be only for a short term to be realistic and relevant. Fixed budgets are useful for fixed costs which do not change in the short term nor over if particular level of activity. However, a particular level of activity is assumed to prevail throughout the budget period.

As this is a static budget. it has only limited use as control device. The variances between the budget and the actuals cannot be properly explained. If the variance is too large it must be shown whether it was due to change in the level of activity alone or other causes, this involves a tedious and laborious analysis of the variances. Fixed budgets will be useful only where stable conditions can be assumed to exist throughout. However as uncertainties regarding inputs into production are increasing every day, fixed budgets have lost their importance as tools of control.

Flexible budgets on the otherhand may be prepared in advance for varying levels of capacity or may be adjusted to current conditions arising out of seasonality of the business. It is more elastic and is definitely more practical and useful. It is dynamic and rejects the actual conditions.

The most important aspect of preparing a flexible budget is the identification of the costs as fixed, semi-fixed and variable ones. A budget allowance the amount of overhead cost which a budget centre is expected to incur during the given period in relation to the level of activity attained is set in advance for each item of expenditure. Fixed expenses do not change upto a certain level of activity once changed they remain constant upto another level. They may change in a stepped fashion.

Variable costs are predetermined at a fixed rate per unit and the budget allowances can be easily calculated for all levels of activity. The semi-variable overheads are carefully analysed item by item to decide the fixed components of the costs. After this is done a statement of the budget allowances so determined for various assumed levels of activity is prepared to show the profits under different capacities. Thus it becomes a services of fixed budgets for successive capacity variations. If the actual capacity achieved falls between two levels of capacity budgeted for, the various figures can be suitably adjusted for actual capacity attained. For those which do not correspond to the capacity attained the budget allowances can be calculated by interpolation. Small differences in capacity may be ignored in such cases. For example, if the budget gives the figures for say 90% and 100% capacities it is possible to interpolate for 95% capacity achieved. But fixed costs need not be calculated for say 91% capacity achieved. The figures given for 90% capacity may be adopted for 91% or 92% even Flexible budgets may be prepared in the form of a graph with different lines for different expenses.

Advantages of Flexible Budgets:

i) It is possible to ascertain costs with greater accuracy. Over/under recovery of overheads can be minimised.

ii) As the various costs are determined and anticipated at different capacity levels it is easier to control them.

iii) Management by exception can be followed more easily at each capacity level actually achieved.

iv) By experimenting with the various capacity levels the company can determine the optimum level of activity at which the profits will be maximum.

Illustration: 7

With the following data at 60% activity prepare a budget at 80% and 100% activity.

Production at 60% Capacity 600 Units

Materials	Rs. 100 per Unit
Labour	Rs. 40 per Unit
Expenses	Rs. 10 per Unit

Factory Expenses Rs.40,000 (40% Fixed)

Administrative Expenses Rs.30,000 (60 % Fixed)

Solution :

FLEXIBLE BUDGET

Particulars	Capacity Levels		
	60% (600 units) Rs.	80% (800 units) Rs.	100% (1000 units) Rs.
Variable Expenses :			
Material	60,000	80,000	1,00,000
Labour	24,000	32,000	40,000
Expenses	6,000	8,000	10,000
Factory Expenses	24,000	32,000	40,000
Administrative Expenses	12,000	16,000	20,000
Total (a)	1,26,000	1,68,000	2,10,000
Fixed Expenses :			
Factory Expenses	16,000	16,000	16,000
Administrative Expenses	18,000	18,000	18,000
Total (b)	34,000	34,000	34,000
Total Cost (a) + (b)	1,60,000	2,02,000	2,44,000

Illustration: 8

Following are the data available form the records of a manufacturing company. Prepare a flexible budget for the year showing the profit under 60% 75% 90% and 10% capacities.

Items	(Per Annum) Rs. (in lakhs)
Fixed Expenses :	
Wages and Salaries	19
Rent, Rates and Taxes	13
Depreciation	14
Misc. Admn. Expenses	12

Semi-Variable Expenses (at 50%) capacity	(per annum) Rs. (in lakhs)
Maintenance & repairs	7
Indirect Labour	16
Sales Department Salaries	8
Mic. Admm. Expenses	6
Variable Expenses (at 50% capacity)	
Materials	44
Labour	40
Other Expenses	16

Assume the fixed expenses to remain unchanged for all levels, semivariable remain unchanged between 45% and 75% capacity increasing by 10% between 65% and 80% increasing by 20% between 80% and 100% capacity. (from the 50% capacity levels.)

Sales at 100% capacity are Rs. 500 lakhs

Solution :

Flexible Budget

Items	(Rs. in lakhs) Period :............ Capacity				
	50%	60%	75%	90%	100%
Sales (A)	200	240	300	360	400
Variable Expenses	Rs.	Rs.	Rs.	Rs.	Rs.
Materials	44	52.8	66	79.2	98
Labour	40	48	60	72	80
Other Expenses	16	19.2	24	28.8	32
Semi-Variable	Rs.	Rs.	Rs.	Rs.	Rs.
Maintenance & Repairs	7	7	7.7	8.4	8.4
Indirect Labour	16	16	17.6	19.2	19.2
Sales department salaries	8	8	8.8	9.6	9.6
Misc. Admn. Expenses	6	6	6.6	7.2	7.2
Fixed Expenses	Rs.	Rs.	Rs.	Rs.	Rs.
Wages and Salaries	19	19	19	19	19
Rent, Rates & Taxes	13	13	13	13	13
Depreciation	14	14	14	14	14
Misc. Admn. Expenses	12	12	12	12	12
Total (B)	195	215	248.7	282.4	302.4
Profit (A - B)	5	25	51.3	77.6	97.6

Master Budget:

the master budget is the summary budget which is prepared when the functional budgets have been finalised and approved by the budget committee. This budget will integrate all the subsidiary operating budgets in their summarised form. It will incorporate the sales, production cost, operating expenses and the resulting profit for the budget period. The master budget as prepared and approved by the budget committee is sent to the Board of Directors for approval. If the Board finds the profits as projected by the profit and loss account to be unsatisfactory it may suggest changes and refer the budget back to the budget committee.

Illustration: 9

A Glass manufacturing Company requires you to calculate and present the budget for the next year from the following information:

Sales :	
Toughened Glass	Rs.3,00,000
Bent Toughened Glass	Rs.5,00,000
Direct Materials Cost	60% of Sales
Direct Wages	20 workers a Rs.150 per month
Factory Overhead	
Indirect Labour	
Works Manager	Rs.500 per month
Foreman	Rs.400 per month
Stores and Spares	2-1/2% on Sales
Depreciation on Machinery	Rs.12600
Light and Power	Rs.5000
Repairs and Maintenance	Rs.8000
Other Sundries	10% on direct wages
Administtration Selling & Distribution Expenses	Rs.14,000 per year

Solution :

Master Budget for 19...

	Rs.	Rs.	Rs.
Sales Budget :			
Budgeted Sales :			
Toughened Glass		3,00,000	
Bent Toughened Glass		5,00,000	8,00,000
Production Cost Budget :			
Direct Material 60% of Sale		4,80,000	
Direct Wages (20 x 150 x 12)		36,000	
Prime Cost		5,16,000	
Factory Overhead :			
Variable:			
Stores	20,000		
Light & Power	5,000		
Repairs & Maintenance	8,000	33,000	
Fixed :			
Works Manager Salary	6,000		
Foreman	4,800		
Depreciation	12,600		
Sundries	3,600	27,000	
Works Cost		5,76,000	
Admin. Selling & Distributive Expenses		14,000	5,90,000
Expected Profit			2,10,000

Illustration: 10

From the following information relating to 1984 and conditions expected to prevail in 1985 prepare a budget for 1935.

State the assumptions you have made.

1984 Actuals :	**Rs.**
Sales	1,00,000 (40,000 Units)
Wages	53,000
Variable overheads	16,000
Fixed overheads	10,000
1985 Prospects :	
Sales	150,000 (60,000 Units)
Raw Materials	5% price increase
Wages	{10% increase in wage rates 5% increase in productivity}
Additional Plant	{One Lathe Rs.25,000 One Drill Rs. 12,000}

Solution :

Master Budget for 1985

Particulars	Actual for 1984 Rs.	Budget for 1985 Rs.
Sales (a)	1,00,000	1,50,000
Variable Costs :		
Raw Material	53,000	83,475
Wages	11,000	17,286
Variable Overheads	16,000	24,000
	80,000	1,24,761
Fixed Overheads	10,000	13,700
Total Cost (b)	90,000	1,38,461
Profit (a) - (b)	10,000	11,539

Notes:

(i) Raw material Cost $\frac{53,000 \times 105}{100} \times \frac{150}{100} \times$ Rs. 83,475

(ii) Wages $\frac{11,000}{40,000} \times 60,000 \times \frac{110}{100} \times \frac{100}{105} = 17,286$

(iii) Depreciation on additional plant is provided 10% per year.

(iv) Depreciation on additional plant is provided at 10% per year.

(v) Fixed overheads 10 000 +3 700 = 13.7000.

TEST YOURSELF

1. Explain the term 'Budgetary Control', make out a case for the introduction of budgetary control in business.

2. Distinguish between budget and forecast.

3. A Company expects to have Rs.25,000 in bank on 1st may, 1974 and requires you to prepare an estimate of cash position during the three months may, June and July, 1974.

The following information is supplied to you.

Month	Sales	Purchases	Wages	Factory Expen.	Office Expen.	Selling Expen.
	Rs.	Rs.	Rs.	Rs.	Rs.	Rs.
March	50.000	30.000	6.000	5.000	4.000	3.000
April	56.000	32.000	6.500	5.500	4.000	3.000
May	60.000	35.000	7.000	6.000	4.000	3.500
June	80.000	40.000	9.000	7.500	4.000	4.500
July	90.000	40.000	9.500	8.000	4.000	4.500

Other information :

a) 20% of sales are in cash, remaining amount is collected in the month following that of Sales.

b) Suppliers supply goods at two months credit.

c) Wages and all other expenses are paid in the month following the one in which they are incurred.

d) The Company pays Dividends to Shareholders and Bounds to workers of Rs.10,000 and Rs.15,000 respectively in the month of may.

e) Plant has been ordered and is expected to be received in June, if will cost Rs.80,000 to be paid in June.

f) Income-tax Rs.25,000 is payable in July.

4. Draw up a flexible budget for overhead expenses on the basis of the following data and determine the overhead rate at 70% 80% and 90% plant capacity.

	70% Rs.	80% Rs.	90% Rs.
Variable overheads			
Indirect labour		12,000	
Stores including spares		4,000	
Semi-variable overheads :			
Power (30% fixed 70% variable)		20,000	
Repairs and maintenance (60% fixed 405 variable)		2,000	
Fixed overheads			
Depreciation		11,000	
Insurance		3,000	
Salaries		10,000	
Total overheads Rs.		62,000	

Estimated direct labour hours 1,24,000 hours.

5. The following data relates to the working of a factory at Wardha for the year 1976:

	Capacity Worked 50%	Rs.
Fixed Costs:		
Salaries	84,000	
Rent and rates	56,000	
Depreciation	70,000	
Other Administrative Expenses	80,000	2,90,000

Variable Costs :

Materials	2,40,000	
Labour	2,56,000	
Other Expenses	38,000	5,34,000

Possible sales at various levels of working are:

Capacity	Sales in Rs.
60%	9,50,000
75%	11,50,000
90%	13,75,000
100%	15,25,000

Prepare a flexible budget and show the forecast of profit at 60,60,90.

....

LESSON : 15

CAPITAL BUDGETING

Every business concern has to face very often the problem of capital expenditure decisions. Hence the forecasting and budgeting of capital expenditure is a vital part of policy-making, management and budgetary control. Capital expenditure is one which is intended to benefit future periods and normally includes investment in fixed assets and other development projects. It is essentially a long-term function, for a decision to buy land, buildings or plant and machinery would influence the activity of the business for a considerable period of time. Hence a keen watchfulness and positive awareness of capital expenditure needs is essential at all times. Further, the advent of mechanisation and automation has resulted in management being confronted with ever more frequent and difficult problems. Despite the fact that various techniques have been developed to help management to make its decision more effectively, the final choice still remains for want of relevant information which only a well-established capital budgeting system could generate. In addition, a basic understanding of the technique would enable the management for its efficient application. The present lesson and the one that follows would make clear the relevant aspects of capital budgeting.

CONCEPT OF CAPITAL BUDGETING

Capital budgeting normally refers to the long-term planning for proposed capital outlays and their financing. It is the decision- making process by which firms evaluate the acquisition of major fixed assets whose benefits would be spread over several time periods. Succinctly, it involves a current investment in which the benefits are expected to be received beyond one year in future. The use of one year as a line of demarcation is, however, somewhat arbitrary. The main exercise in capital budgeting is to judge whether or not an investment proposal provides a reasonable return to investors which would be consistent with the investment objective of the business. hence, capital budgeting involves the generation of investment proposals; the estimates of costs and benefits (cash flows) for the investment proposals; the evaluation of net benefits and the selection of projects based upon an acceptance criterion.

IMPORTANCE OF CAPITAL BUDGETING

The capital investment involved is usually very large It will have several far-reaching implications on the activities of business and may even affect seriously the very financial stability or flexibility of the business. It is these implications which capital budgeting go so much importance.

It shows the possibility of expanding the production facilities to cover additional sales shown in the sales forecast. In fact, the economic life of the asset acquired represents an indirect sales forecast for the duration of the economic life. Any error in this regard may result in over-investment or under-investment in fixed assets, i.e. excess production capacity or inadequate capacity. It also enables the cash forecast to be completed.

Capital budgeting may allow alternative forms of assets to be considered as replacements for assets which are wearing out or are in danger of becoming obsolete. In other words, it would lead to better timing of asset purchases and improvement in quality of assets purchased. It helps to match efficiently the need for capital goods with their availability. It also assists in formulating a sound depreciation and asset replacement policy.

Capital expenditure decisions involve substantial funds which may not be immediately and automatically available. A well-established capital budget would enable to decide in advance the sources of finance and ensure their availability at the right time.

OBJECTIVES OF CAPITAL BUDGETING

It may be said that the overall objective of capital budgeting is to allocate the available investible funds among the competing capital projects in order to maximise the total profitability. This is made possible by employing the various evaluation techniques for the selection of investment projects which contribute the maximum towards the overall investment objective. In the case of public enterprises, capital budgeting may also assure other objectives such as promotion of employment, development of backward region etc.

Control of capital expenditure is the next important objective of capital budgeting. This is achieved by forecasting long before the financial requirements and thereby enabling to plan in advance to raise funds at the right time. The objective of preparing capital budget is to plan and then compare the actual capital expenditure with the budgeted figure for controlling costs.

The next important objective of capital budgeting is to determine the funds required for long-term projects and to see that such estimates fall in line with the company's financial policies. It also aims to compromise between the availability of funds and needs of the capital projects.

TYPES OF INVESTMENT PROJECTS

Investment projects may be classified in a number of ways. The following kinds of investment projects are commonly used by both private and public sector business units in their capital expenditure forecasts:

(a) Expansion of existing product lines or markets:

(b) Expansion into new product lines or markets:

(c) Replacement and modernisation schemes:

(d) Projects for the utilisation of scraps and utilisation of surplus installed capacity:

(e) Cost reduction projects.

The projects stated above are generally profit-oriented and therefore they may be evaluated on their costs and benefits. But there are investments which are under taken by all business units and on which it would be difficult to measure a return, such as:

(1) Safety precautions: Provision of safety devices and equipment may be demanded by various legal requirements.

(2) Welfare Projects: Provision of sports facilities for employees may boost employee morale which cannot so easily be evaluated financially.

(3) Service Department Projects: The provision of buildings or equipment for non-manufacturing departments may be essential but the return can not be evaluated.

(4) Research and Development: this may be initiated to improve company methods or products. It would be very difficult to measure the return for a considerable time.

(5) Education Projects: Provision of a company training course may be instrumental in improving the efficiency of employees but here again results would be difficult to evaluate.

RELEVANT COSTS FOR CAPITAL BUDGETING

Generally costs and benefits in the form of cash flows are more relevant for capital budgeting than the conventional accounting of costs and benefits which normally encounter a number of measurement problems owing to factors such as method of depreciation, valuation of inventories, write off, etc. Different types of investment decisions call for different kinds of costs. Not all costs which are used in conventional accounting are relevant for investment decision- making. A few items of relevant costs are as follows:

Future Costs: Future costs are the projected or estimated costs. They are relevant for all types of investment decisions. Past costs, though not relevant for decision-making, are useful to the extent that they furnish a starting point for future cost projections. While calculating these costs, factors such as market conditions, economic conditions, political situations, general trend in the price levels, probabilities relating to the future productions and sales, economic life of the project, etc., are to be taken into account.

Opportunity Costs: In simple terms, opportunity cost refers to the benefits of the best alternative foregone. As the investment in a project involves commitment of firm's investible funds, it becomes relevant to consider the opportunity of getting some benefits employing the resources on some other alternative. For example, in an expansion scheme, the economic value of the space required rather than its book value is relevant; in a replacement decision, the realisable value rather than the book value may be relevant as a reduction of the cost of replacement. This type of cost is relevant for all types of investment decisions. Imputed cost is a division of opportunity cost; it is a cost which is not actually incurred but would be incurred in the absence of self-owned factors, e.g. cost of retained earnings, rent on company-owned facilities, etc.

Incremental or Differential Cost: It is the additional cost due to a change in the volume of business or nature of business activity. Hence, it is useful for such types of decisions as adding a new machinery, new product, changing a distribution channel, etc. Sometimes this cost is considered synonymous to marginal cost. But marginal cost has much limited meaning referred to the cost of an added unit of output.

Interest Cost : Accounting reports normally ignore the imported interest on capital which is relevant for decision-making purposes. Interest cost constitutes the minimum acceptance criterion for capital investment projects undertaken for profit. A firm must at least recover its money cost before it can realise a profit on its own investment.

Depreciation and Income Taxes : Depreciation is normally excluded while calculating cash flows for the investment appraisal and evaluation. But it is included for calculating accounting rate of the project. Payment of taxes results in cash flows and therefore, is an important element in capital investment decisions, Income taxes have a number of effects on capital investment decisions. Hence, the complexities of tax laws and applicable legal decisions emphasise the need for special skill in this area.

Secondary cost and Benefits: These costs and benefits are particularly relevant for the capital budgeting decisions in public enterprises. They are external to project implementing body and therefore are called external costs and benefits which are imposed on other sectors, government, society or the economy as a whole during the construction and operation of the project and for which nothing is paid or received. There are two types of externalities. viz. technological and pecuniary. The smoke and dust or pollution and noise created etc. are examples for technological externalities. Pecuniary externalities are such as increasing factor rates or hire, cutting down prices of substitute products, etc. Secondary benefits are the increase in profits that can be attributed to the increased activity of processors, merchants and others who handle the project's output and input. The major problems associated with these costs and benefits are their identification and measurement. However, for easy identification, they should be related to the socio-economic objectives assigned to the project. To measure these costs and benefits, shadow prices or imputed prices should be used.

CAPITAL EXPENDITURE CONTROL

The control of capital expenditure is growing in importance as mechanisation and automation are introduced and extended. However, formal capital budgeting is still undeveloped as it is of comparatively recent origin. Any system of capital expenditure control shall have the following features:

Planned Development : Capital expenditure should be carefully planned to include developments in each site or department to ensure that each unit in the group or company is developing in step with the overall plan.

Preparation of capital budget will be essential even when companies do not operate a complete system of budgeting control. Capital appropriations and payments must be planned well in advance.

Control of progress : A progress record is necessary to show the progress of each capital project. The budget and actual expenditure will be compared for the analysis and control. These reports are also useful to observe that overall programme remains within the limits set by the policy of the company.

Post-Completion Audits: This is an important step of capital expenditure control. Post-completion audits of projects determine whether actual value is in accordance with the one determined at the time of authorisation. This review can be very important because it may reveal inefficiency in the system, and it would provide experience which would enable repetition of mistakes to be avoided.

Forms and Procedures : There should be a routine for controlling capital expenditure. A procedure should be adopted for the various stages of requesting, authorisation, progress and audit. Forms should be available which can be completed and presented to management whenever necessary.

LIMITATIONS OF CAPITAL BUDGETING

The different methods of project appraisal need a continuous attempt at systematic analysis of available alternative proposals as well as improvement of procedure of evaluation. However, sophistication and refinement of procedure cannot ensure the right choice if the data are wrong. Most of the decisions are taken under conditions of uncertainty and risk and therefore, reliable data, skillful management and judgement are imperative. For example, an investment may have a direct or indirect effect on employee morale which is very difficult to quantify. In the case of public sector investment appraisal, it is more difficult to identify and measure secondary costs and benefits which are essential.

METHODS OF RANKING INVESTMENT PROPOSALS

The final step in the capital budgeting system involves evaluating the profitability of the alternative projects and selecting the best one. A business firm may face situation where more investment proposals may be available than the investible funds. Some proposals may be good, others different and yet others poor. Hence, a ranking procedure is to be evolved so that the available funds could be allocated among different proposals in a profitable manner. Essentially, the ranking procedure envisages relating of stream of future benefits to the cost of investments. Among the various methods, the following are commonly used by many business concerns.

TRADITIONAL OR NON-TIME VALUE TECHNIQUES

(a) Payback period

(b) Average rate of return

(c) Discounted cash flow method.

MODERN OR TIME VALUE TECHNIQUES

(1) Net present value

(2) Benefit/cost ratio

(3) Internal rate of return.

PAYBACK PERIOD

Business units, while selecting investment project, would consider the recovery of cost as the first and foremost concern, even though earning maximum profits is their ultimate goal. Payback period normally refers to the length of time required for recouping the initial investment in full with the help of the stream of annual cash flows generated by the project. It is also called as 'payout' or 'payoff' or 'recoupment' period, symbolically expressed :

$$\text{Payback period (PB)} = \frac{I}{C}$$

Where C = original cost or original investment and I = annual cash inflows. In the case of uneven cash inflows, it may be expressed as below :

$$PB\ (I) = \sum_{n=0}^{I} X_n = 0$$

Where X represents cash flows during periods 0,1,2,......,n. The cash flows for the purpose of PB calculations would be savings of earnings after payment of taxes but before depreciation. To illustrate, if a cash outlay of Rs.30,000 is expected to yield a constant net cash flow (cash earnings − cash expenses) of Rs. 12,000 p.a. for a period of 5 years, PB is years (Rs. 30,000 ÷ Rs.12,000).

Selection Criteria : Among the mutually exclusive or alternative projects whose PBs are lower than the cut off period, the project with the shorter PB would be selected. In case there are budget constraints, the procedure would be to rank the projects in the ascending order of PBs and select the first 'X' number of projects which the budget provision permits.

However, with a view to make the selection process more realistic, a cut off period or a minimum payback ratio could be set up and all investment proposals for which the PB is greater than this cut off period be returned. Payback ratio is the inverse of the payback period. For a payback period of 4 years, the payback ration is. The larger the payback ratio, the better is the project.

Illustration 1:

	Project A Rs.	Project B Rs.
Cash outlay	15,000	15,000
Cash Flows (savings before depreciation but after taxes) :		
I Year	5,000	4,000
II Year	5,000	4,000
III Year	5,000	4,000
IV Year	2,000	3,000
V Year	2,000	7,000
VI Year	2,000	9,000
	21,000	31,000
Payback Period	3 Years	4 Years

Standard payback or cut off period is 5 years.

The PBs of A and B are 3 years and 4 years respectively and thus project A is adjudged superior to project B in terms of PB criterion since it is also shorterthan the cut off period.

Illustration 2:

Empees Ltd. is considering the purchase of a new machine. The two alternative models under consideration are 'Laxmi' and 'HMT' From the following formation, prepare a profitability statement for submission to the Board of Directors:

	Machine 'Laxmi' Rs.	Machine 'HMT' Rs.
Cost of Machine	90,000	1,50,000
Estimated Life (in years)	10	15
Estimated Savings in scrap p.a.	600	9,000
Additional cost of supervision p.a.	7,200	9,600
Additional cost of maintenance p.a.	9,200	6,600
Cost of indirect material p.a.	3,600	4,800
Estimated Savings in Wages:		
Wages per worker p.a.	360	360
Workers not required	150	200

Assume the taxation rate to be 50 per cent of profits Suggest which model can be purchased. giving reasons for your answer.

Solution:

PROFITABILITY STATEMENT

Estimated Savings p.a. :

	Machine 'Laxmi' Rs.	Rs.	Machine 'HMT' Rs.	Rs.
Scrap	6,000		9,000	
Wages	54,000	60,000	72,000	81,000
Less: Estimated Additional				
Cost p.a. :				
Cost of Supervision	7,200		9,600	
Cost of Maintenance	4,200		6,600	
Indirect Materials	3,600	15,000	1,800	21,000
Savings before Depreciation		45,000		60,000
Less: Depreciation		9,000		10,000
Savings after Depreciation		36,000		50,000
Less: Taxes at 50%		18,000		25,000
Net Savings p.a.		18,000		25,000
Annual Cash Flow:				
Net Savings	18,000		25,000	
Depreciation	9,000	27,000	10,000	35,000

$$\text{Payback period} = \frac{\text{Cost of machine}}{\text{Savings p.a.}}$$

Machine 'Laxmi'	Machine 'HMT'
Before Tax Rs.90,000 ÷ 2 years = Rs.45,000	Rs.1,50,000 ÷ 2-1/2 years = Rs.60,000
Before Tax Rs.90,000 ÷ 3-1/2 years = Rs.27,000	Rs.1,50,000 ÷ 4 years = Rs.35,000

Thus, machine 'Laxmi' clearly recommends itself for purchase. However, the information provided and conclusion desired may be supplemented with some additional calculations as regards profitability beyond payback period.

	Machine 'Laxmi'	Machine 'HMT'
Profitability beyond payback period :		
Before Tax :		
Life beyond payback period	8 Years	12-1/2 Years
Savings p.a.	Rs. 45,000	Rs. 60,000
Total Savings	Rs.3,60,000	Rs.7,50,000
After Tax :		
Life beyond payback period	6-2/3 Years	10-5/7 Years
Savings p.a.	Rs. 27,000	Rs. 35,000
Total Savings	Rs.1,80,000	Rs.3,75,000

MERITS OF PAYBACK PERIOD

(1) It is easy to operate and simple to understand.

(2) This method is preferred on the ground that return beyond three or four years are so uncertain as to disregard them altogether in a planning decision.

(3) It is appropriate for industries with a high rate of technological obsolescence in which the receipts beyond PB are regarded as totally uncertain

(4) This method is also useful to a concern which is short of cash and is eager to get back its cash invested in capital expenditure project.

(5) As the method considers the cash flows during the payback period of the project, the estimates would be reliable and the results may be comparatively more accurate.

DRAWABACKS OF PAYBACK PERIOD

Despite its simplicity and easy operation, this method suffers from several drawbacks:

(1) The PB is more a liquidity concept than a profitability one, for it by laying accent only on the recovery of cash outlay, over- stresses the importance of liquidity; its connotation confined to recovery at the cost of profitability.

(2) It does not consider the earnings beyond the payback period. This may lead to wrong selection of investments. Profitable projects with long gestation periods or projects which would generate high returns only after a certain period of time may be rejected under this method.

(3) The most serious limitation of this method is that it ignores the time value of money. In other words, it

provides only for the recovery of the original investment and not for the cost of funds committed to such investments. But in practice the investible funds of a business is not free of cost.

AVERAGE RATE OF RETURN METHOD (ARR)

ARR is considered to be an improvement over the PB, for it considers the earnings of a project during its entire economic life. It is also known as 'Return on Investment Method'. Return on investment is the ratio of earnings after depreciation to original cost or investment. The formula for ARR Which may be used for calculating profitability on new investment is as follows:

$$ARR = \frac{\text{Average earnings or return}}{\text{Average Investment}} \times 100$$

For replacement and expansion schemes, the formula is :

$$ARR = \frac{\text{Average additional profits p.a.}}{\text{Average amount invested}} \times 100$$

The average return is computed by adding all the earnings after depreciation and dividing them by the project's economic life. Average investment is the simple average of the values of asset which in most cases would be zero. Though sometimes initial investment is used. average investment is more logical.

Selection Criteria : The decision rule is that a project with highest rate of return on investment is selected on condition that such rate is above the standard rate set or the cut off rate. In the case of replacement or expansion, the return on investment on additional investment should be above the cut off rate. The company's cost of capital or any other desired rate may be set as the cut off rate. Risk and inflation could be provided for by adding certain percentages to the desired rate of return. In case of alternative projects, one with the highest ARR is to be selected.

Illustration 3:

Calculate the average rate if return for project 'A' and 'B' from the following information:

	Project A	Project B
Investment (Rs)	25,000	37,500
Expected Life (in Years)	4	5
Net Earnings (after Depreciation and Taxes) :		
Years	Rs.	Rs.
1	2,500	3,750
2	1,875	3,750
3	1,875	2,500
4	1,250	1,250
5	--	1,250
	7,500	12,500

If the desired rate of return is 12%, which project should be selected?

Solution :

		Project A	Project B
Average Return	=	$\frac{\text{Rs. } 7{,}500}{4}$	$\frac{\text{Rs. } 12{,}500}{5}$
	=	Rs. 1,875	Rs.2,500
Average Investment	=	$\frac{\text{Rs. } 25\,000 + 0}{2}$	$\frac{\text{Rs. } 37{,}500 + 0}{2}$
	=	Rs.12,500	Rs.18,750
Average Rate of Return	=	$\frac{\text{Rs. } 1{,}875 \times 100}{\text{Rs. } 12{,}500}$	$\frac{\text{Rs. } 2{,}500 \times 100}{\text{Rs. } 18{,}750}$
	=	15%	13.33%

Both the projects satisfy the minimum required of return. However, if the projects are mutually exclusive of alternative, project A will be selected as its ARR is higher than Project B. If they are not mutually exclusive and there are no budget constraints, both the projects will be selected.

MERITS

(1) This method is also easy to understand and simple to operate.

(2) ARR method takes into account earnings over the entire economic life of the project even though estimates of distant future may be subject to wide margin of errors.

(3) This is really a profitability concept since it considers not earnings after depreciation, i.c. excess of earnings over original cost of investment.

(4) Another strong point of this method is that projects which widely in character could be compared under this system. For example, an investment in new products could be compared with one that results in cost reduction or a current investment proposal could be compared with one that will be ready only by next year.

DEMERITS :

(1) The most severe criticism of this method is that it ignores time value of money.

(2) Normally a host of variants are to be resolved relating to its components, viz. earnings and investment cost. For example, it may be the gross, net or average investment which is to be considered for computation. This may produce different rates of return for any one proposal.

(3) Another problem in connection with the method is regarding reasonable rate of return on investments. Some stipulate a minimum rate so that, if projects do not satisfy this rate they are summarily excluded from consideration.

DISCOUNTED CASH FLOW TECHNIQUES (DCF)

As investments are essentially current capital outlays incurred in anticipation of future returns, the timing of expected future flows is extremely important in the investment decision. It any economy capital or funds invested has value and therefore the time value of money is an important concept. For example, investors place higher value on an investment project that promises returns over the next four or five years than the one that could generate similar returns only after some years onwards. It is felt that DCF techniques provide a more objective basis for evaluating and selecting investment projects. Also they enable one to isolate differences in the timing of cash flows to their present values. Because these techniques take into account both the magnitude and the timing of expected cash in each period of a project life, they are considered more refined as well as realistic. Generally, these techniques consider the net cash flow as representing the recovery of original investment plus a return on capital invested. There are mainly two types of DCF techniques, viz. Net present value (NPV) and Internal Rate of Return (IRR).

DISCOUNTING

Discounting is the basic operation of any DCF method. The basic problem is that a rupee today is worth more than the same rupee in a year's time regardless of inflation because the rupee now can be invested to grow to a large sum in the future. This is achieved by discounting. Discounting involves reducing the value of future returns to make directly comparable to present sums. Suppose that we could invest any money today in the capital market for a return of 10 per cent. Then rupee one today would be worth rs. 1.10 in one year's time. Conversely, we can say that Rs. 1.10 in a year's time is worth re.1 now at a rate of interest of 10 per cen*. Therefore, Re.1 in a years time is worth 91 paise (1/1.1) now, since 91 paise could be invested to get Re.1 in a year's time. Consider Rs.1.21 in two years' time. To find its present value, if we can invest at 10 per cent, we divide it by $(1+0.1)$ twice, i.e present value of Rs.1.21 $=1.21(1+0.1)^2$ $(1 + 0.1)^2$ $=$Re.1. This is the logic of discounting and can be applied in all cases to make inter-temporal comparison of money sums. All we do it to multiply the future sum by a discount factor which depends on the time period involved and the rate at which sums can be invested. In the case of Rs.1.21 in two years; time as money could be invested at 10 per cent, the discount factor was 1 (1.1) =0.826. However to avoid the tedium a of calculating a new discount factor for every situation, a table is discount factors is produced as appendix to this lesson, showing the relevant multiples for common time periods and rates of discount.

DISCOUNT RATE

The rate at which the future cash flows are reduced to their present value, is termed as discount rate. Discount rate otherwise called as time value of money, is some interest rate which expresses the time preference for a particular future cash flow. Realistic capital investment appraisal depends on two factors, viz. discounting period and a suitable discount rate.

THE DISCOUNTING PERIOD

Normally the economic life of project is used as the discounting period. However, the length of discounting periods depends on such factors as the life of equipment with the largest life span, technological change, availability of raw material, market stability, etc.

CHOICE OF DISCOUNT RATE

The choice of an appropriate discount rate which represents the time preference for money flows, is essential as well as critical task for both discounting methods. It is needed to find present value and as a yardstick against the internal rate of return can be compared. Normally the market rate of interest is considered for discounting. However, the actual discount rate depends on the circumstances in which the firm is. If the firm must borrow to finance investments, the borrowing rate will be the appropriate discount rate.

Alternatively, if surplus funds are used, the appropriate rate would be the rate at which the firm could lend the funds. If a project is to be financed partly from surplus funds and partly by borrowing, the discount rate should reflect the weighted average cost of both the components. As there are multiple market rates in the capital markets and they are influenced by a number of factors such as risk involved. amount and guarantee offered, etc., there is difficulty in adapting these rates for discount purposes. Hence, the cost of capital computed based on the overall capital structure of the company or on the basis of financial pattern would be an appropriate discount rate. Theism also be comparable with the firm's overall investment objective.

NET PRESENT VALUE TECHNIQUE

NPV may be define as the excess of present value of project cash inflows (stream of benefits) over that of outflows (cash outlays). The rate of discount employed for obtaining the present value of cash flows is some desired rate of return which is mostly equivalent to the cost of capital. Symbolically, NPV may be represented as follows : For a conventional investment where all cash outflows take place in the base year.

$$NPV = \frac{B_0}{(1+r)} + \frac{B_1}{(1+r)} + \frac{B_2}{(1+r)^2} + \ldots + \frac{P_n}{(1+r)^n} \; C_0$$

$$NPN = \sum_{t=1}^{n} \frac{B_t}{(1+r)_t} \; C_0$$

For a non-conventional investment where the cash outflows take place more than one year.

$$NPV = \left[\frac{B_0}{(1+r)} + \frac{B_1}{(1+r)} + \frac{B_2}{(1+r)^2} + \ldots + \frac{B_n}{(1+r)^n} \right]$$

$$\left[\frac{C_0}{(1+r)} + \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_n}{(1+r)^n} \right]$$

$$\text{NPN} = \sum_{t=0}^{n} \frac{B_t}{(1+r)_t} \quad \sum_{t=0}^{n} \frac{C}{(1+r)^t}$$

Where 'B$_t$' represents cash inflows in periods 0, 1, 2, 3, ... t;

'C$_t$' represents the cash outlays in periods 0. 1, 2, 3, ..., t;

'r' represents the desired discount rate; where 'r' is equal to firm's cost of capital., it may be replaced by 'K'.

DECISION RULES

The decision rules under NPV method are as follows: In the case of mutually exclusive or alternative project, accept a project that has the highest NPV. In the case of independent investments, accept a project if its NPV is greater than or equal to zero; reject, if its NPV is negative.

In case there are budget constraints, for instance capital rationing, the project selection could be effected by ranking projects having positive NPV in the descending of their NPVs and then choosing the first 'X' number of projects that budget provision permits.

Illustration 4:

The Bata Co.Ltd. is considering the purchase of a new machine. Two alternative models (X and Y) are available, each costing Rs. 80,000. Earnings after taxation are expected to be as follows:

Year	Cash Flows Model X	Model Y
	Rs.	Rs.
1	8,000	24,000
2	24,000	32,000
3	32,000	40,000
4	48,000	24,000
5	32,000	16,000

The company has a target of return on capital of 10 per cent and on this basis, you are required to compare the profitability of the models and state which alternative you consider financially preferable.

Solution :

PROFITABILITY STATEMENT
(at 10 per cent discount factor)

Year	PV of Re.1 at 10% Rs.	X		Y	
		Cash Flows Rs.	PV Rs.	Cash Flows Rs.	PV Rs.
1	0.909	8,000	7,272	24,000	21,816
2	0.826	24,000	19,824	32,000	26,432
3	0.751	32,000	24,032	40,000	30,040
4	0.638	48,000	32,784	24,000	16,392
5	0.621	32,000	19,872	16,000	9.936

Total	1,03,784	1,04,616
Cost	80,000	80,000
NPV	23,784	34,616

From the above statement it is evident that Model Y is a more profitable investment. Though Model X would earn Rs.8,000 more than model Model Y, cash inflow is small to begin with and, even though it rises steeply until the last year of its working life, the present value of cash flow is less that of Model Y.

This salient feature of the NPV technique is that it indicates the value added to the total assets of the firm by undertaking the proposed investment at a particular rate of discount. NPV criterion automatically allows for the recovery of the initial investment and interest cost of investment. Suppose a machine costing Rs. 60,000 generates a total discounted proceeds of Rs.68,175 at 10 per cent discount rate. It indicates that the NPV of Rs. 8,715 (Rs.68,175 -Rs.60,000) would be the value added to the total assets of the firm in present value terms. This is after providing for the initial investment of Rs.60,000 and for interest cost at 10 per cent p.a.

MAJOR WEAKNESS OF NPV TECHNIQUE

Under the NPV criterion, the profitability is referred to the capital required to produce it. The project may have a high NPV but can still be unattractive, for it requires a very high capital outlay. For example, two projects A and B costing Rs.60,000 and Rs. 80,000 respectively generate total discounted proceeds - Project A rs. 61,500 and project B Rs.92,500. Of these two, the project B would be recommended by NPV criterion since its NPV is higher by Rs.1,000. However, the profitability is not attractive compared to the additional investment required for undertaking project B. Another weakness, though it is not unique to the NPV method alone, is the selection of the discount rate.

With a view to establish the profitability of projects relating to investment required, NPV criterion could be supplemented with an additional tool called 'Benefit/Cost Ratio' or 'Profitability Index'.

BENEFIT/COST RATIO (BCR)

BCR is the ratio of gross discounted benefits to gross discounted costs. This may be used as an extension of NPV and expressed in coefficient or in percentage. Symbolically this could be expressed as follows.

BCR = B/C or (B/C) X100

or

Profitability Index (PI) =

(B/C)/C or (B- C)/CX100

where B = gross discounted benefits, and
C = gross discounted cost.

From the calculations of the previous illustration, the BCR could be found out as follows:

	Model X Rs.	Model Y Rs.
Gross discounted value of benefits	1,03,784	1,04,615
Gross discounted cost	80,000	80,000

$$BCR = \frac{Rs.\ 1,03,784}{Rs.\ 80,000} = \underset{(app.)}{1.297} \qquad \frac{Rs.\ 1,04,616}{Rs.\ 80,000} = \underset{(app.)}{1.308}$$

$$PI = \frac{Rs.\ 23,784}{Rs.\ 80,000} = \underset{(app.)}{0.297} \qquad \frac{Rs.\ 24,616}{Rs.\ 80,000} = \underset{(app.)}{0.308}$$

The salient feature of BCR is that it would also explain NPV position of an investment, that is, if BCR is equal to one, then NPV will be equal to zero; if it is greater than one, then NPV will be positive; and if it is lesser than one, then NPV will be negative.

DECISION RULES

The decision rules under BCR are as follows:

In the case of independent investments, accept a project if the BCR is greater than one and reject other wise.

In the case of mutually exclusive or alternative investments, accept that project with the largest BCR provided it is greater than one; reject others. Under conditions of capital rationing or budget constraints, rank the projects in the descending order of their BCRs and choose the first 'X' number of projects that the budget provision permits.

BCR is particularly useful for ranking the projects on the basis of their profitability. However, it also suffers mostly from the same weaknesses of NPV technique.

Illustration 5.

Calculate the NPVs and BCRs of Machine 'A' and Machine 'B' from the following information:

		Machines A Rs.	Machines B Rs.
Initial investment		40,000	60,000
Expected Life		5 Years	5 Years
Salvage Value		2,000	2,000
Cash Inflows :	Year	Rs.	Rs.
	1	10,000	40,000
	2	20,000	20,000
	3	20,000	10,000
	4	5,000	6,000
	5	5,000	4,000
		60,000	80,000

The management determines 10% as the desired rate of return for the proposed investment project. Discount factors at this rate are given below :

Year	1	2	3	4	5
PV	0.909	0.826	0.751	0.683	0.621

Solution :

PROFITABILITY STATEMENT

Cash Outlays		Discount Factor at 10%	Machines A Cash Flow Rs.	A PV of Cash Flow Rs.	Machines B Cash Flow Rs.	B PV of Cash Flow Rs.
Initial Investment at year 0		1	40,000	40,000	60,000	60,000
Cash Inflows :						
	Year					
	1	0.909	10,000	9,090	40,000	36,360
	2	0.826	20,000	16,520	20,000	16,520
	3	0.751	20,000	15,020	10,000	7,510
	4	0.683	5,000	3,415	6,000	4,098
	5	0.621	2,000	3,105	4,000	2,484
Salvage Value	5	0.621	5,000	1,242	3,000	1,163
Total Present Value				48,392		68,835
NPV				8,392		8,835

$$BCR = \frac{(\text{Gross discounted cash inflows})}{(\text{Gross discounted cash outflows})}$$

$$= \frac{\text{Rs. } 48{,}392}{\text{Rs. } 40{,}000} = \underset{(\text{app.})}{1.21} \qquad \frac{\text{Rs. } 8{,}893}{\text{Rs. } 60{,}000} = \underset{(\text{app.})}{1.15}$$

The NPV of Machine B is greater than that of Machine A. Hence Machine B is preferable as per NPV criterion. When the two machines are mutually exclusive so that only one machine can be undertaken, the NPV fails to establish which is a better project, since NPV is expressed in absolute rather than in relative terms. Hence, for making a better comparison between the two projects, BCR could be considered. BCR of Machine A is greater than that of Machine B and therefrom Machine A is profitable, Irrespective of the fact the NPV of Machine B is greater than that of Machine A. Machine A should be selected since it has a higher profitability index.

INTERNAL RATE OF RETURN (IRR)

Some of the other names of this method are discounted cash flow method, yield method or time adjusted rate of return method. This method is used when the cost of investment and the annual cash inflows are known while the unknown rate of return is to be calculated. The IRR is defined as that rate of interest when used to discount the cash flows of an investment, reduces its NPV to zero. Or, it is the rate of discount which equates the aggregate discounted benefits with the aggregate discounted costs. Symbolically, the IRR may be For a conventional investment.

For a conventional investment :

	(Aggregate discounted Cash Inflows)	Minus	(Aggregate initial Investment)
NPV	$= \sum_{t=1}^{n} \frac{B_t}{(1+1)} - C_0 = 0$		
OR	$= \sum_{t=1}^{n} \frac{B_t}{(1+1)} = C_0$		

For a non-conventional investment :

	(Aggregate discounted Cash Inflows)	Minus	(Aggregate initial Investment)
NPV	$= \sum_{t=1}^{n} \frac{B_t}{(1+1)} = \sum_{t=0}^{n} \frac{C_t}{(1+1)}$		

where B_t represents cash inflows during the years 1,2,3,.........t,

C_t represents cash outflows during the years 1, 2, 3, ...t,

and i represented the IRR.

IRR indicates the maximum rate of return that a project could contribute, mainly based on the internal cash inflows generated by it. In other words, its most staightforward interpretation is as 'break- even' rate of interest, showing the highest rate of interest with the project not making a loss.

However, in order to judge the profitability of the project, IRR should be compared with a cut off rate of return which might be the company's cost of capital.

DECISION RULES

In the case of independent investment, accept the project if its IRR is greater than cut off rate; if lower, reject.

In the case of mutually exclusive projects, accept the project with the largest IRTR provided it is greater than the cut off rate; reject others.

In case there are budget constraints, rank the projects in the descending order of their IRRs and choose the first 'X' number of projects which the budget provision permits.

METHOD OF CALCULATING IRR

Trial and Error Yield method: As the name suggests, the object is to find the expected yield from the investment. The procedure involved may be summarised as follows:

(a) List the annual sales and cost, other than depreciation and, deducting the latter from the former, obtain the net cash flow.

(b) Obtain the capital cost. This will already be at present value because the amount is spent at the base year.

(c) Calculate the present value of the net cash flow by using an appropriate rate of interest. This rate is found by trial and error from present value tables. The object is to make the cash flow equal to the capital cost.

(d) Carry out this procedure for each project being considered and then rank the projects in order of preference.

Illustration 6:

Govindan Ltd. is planning to increase its present capacity, and is considering the purchase of a new machine. Machines X and Y are available at a price of Rs.40,000 and Rs.45,000 respectively. The machines are mutually exclusive projects. Earnings before taxation have been estimated as follows:

Year	Cash Flow	
	Machine X Rs.	Machine Y Rs.
1	11,000	8,000
2	15,000	12,000
3	20,000	18,000
4	16,000	24,000
5	8,000	15,000
	70,000	77,000

Compute IRR of each machine and ascertain the most profitable project.

Solution:

Year	Cash Flows	Trial No.1		Trial No.2		Trial No.3	
		Disct. Factor (20%)	PV	Disct. Factor (24%)	PV	Disct. Factor (22%)	PV
	Rs.	Re.	Rs.	Re.	Rs.	Re.	Rs.
0	(40,000)	1.000	(40,000)	1.000	(40,000)	1.000	(40,000)
1	11,000	0.833	9,163	0.800	8,866	0.820	9,020

2	15,000	0.694	10,410	0.650	9,750	0.672	10,080
3	20,000	0.579	11,580	0.524	10,480	0.551	11,020
4	16,000	0.482	7,712	0.423	6,768	0.451	7,216
5	8,000	0.402	3,216	0.231	272	0.370	2,960
	30,000		2,081		(1,408)		296

From the above statement it is ascertained that in the first trial at 20% the present value of cash inflows exceeds the present value of cash outflows by Rs.2,801. In the second trial at 24% the present value of cash inflows is than the cash outflows. In the third trial at 22%, cash inflows are marginally greater than cash outflows. Thus approximately 22% is the rate of return which equates the prevent value of cash inflows with cash outflows. If greater accuracy is required, interpretation may be used to find the required discount rate rather than the calculation of unlimited trials. To apply interpretation techniques, a simple method is as follows (based on the above illustration).

22 % + (298/1704) 2 + = 22.34 %

PROFITABILITY STATEMENT - MACHINERY

Year	Cash Flows Rs.	Trial No.1 Disct. Factor (16%) Re.	Trial No.1 PV Rs.	Trial No.2 Disct. Factor (20%) Re.	Trial No.2 PV Rs.	Trial No.3 Disct. Factor (18%) Re.	Trial No.3 PV Rs.
0	(45,000)	1.000	(45,000)	1.000	(45,000)	1.000	(45,000)
1	8,000	0.862	6,896	0.833	6,664	0.847	6,776

2	12,000	0.743	8,916	0.694	6,664	0.718	8,619
3	10,000	0.941	11,538	0.579	6,328	0.609	10,962
4	24,000	0.552	13,248	0.482	10,422	0.516	12,384
5	15,000	0.476	7,140	0.402	11,658	0.437	6,555
	32,000		2,738		(1,988)		293

At 18%, the present value of cash inflows are marginally greater than cash outflows. Approximately 18% is the rate of return which equate the present values of cash inflows with cash outflows. By interpolation,

$$IRR = 18\% + (293/2281)\ 2\% = 18.26$$

These profitability statements show that using the IRR method, Machine X yields a rate of 22%, while Machine Y yields approximately 18%. On this basis, machine X would be recommended.

FORMULA FOR INTERPOLATION

$$A + \left[\frac{C}{C - D}\right] + (B - A)$$

where A = Discount Factor of the low trial

B = Discount factor of the high trial

C = Present value of cash inflow of the low trial

D = Present value of cash inflow of the high trial

ESTABLISHING THE TRIAL RATE

In the trial and error method, difficulty is often experienced in the selection of first trial rate. Once the first trial rate has been established, the second rate is determined by considering whether the cash inflow is greater or lesser than the cash outflow. If the net flow is negative, the second trial rate will be lower than the first rate; if it is positive, the second trial rate will be higher than the first rate. However, the problem is to select the first trial rate. As an aid to this selection, a rough guide is as follows:

$$\frac{\text{Average earnings or savings p.a}}{\text{Investment}} \times 100$$

ADVANTAGES OF DCF TECHNIQUES

(1) DCF techniques are superior to other methods since they consider earnings of a project over its entire economic life and also time value of money flows.

(2) They are more objective, for their conclusions are not directly influenced by decisions regarding depreciation methods, capitalisation versus expense decisions and considerations.

(3) DCF method automatically gives more weight to units of money which are nearer than those which are distant. But other methods treat distant units of money unrealistically with the same weight as present units.

(4) DCF method enables a ready comparison to be made between projects having different lives and different timings of cash flows by facilitating comparison to be made at the same point of time.

(5) By comparing the rates of return of projects with cost of capital ratios decisions can be taken quickly and safely.

GENERAL CRITICISMS AGAINST DCF TECHNIQUES

(1) DCF methods are considered difficult to understand and operate. Comparing with the complexities associated with some modern statistical and mathematical tools which have gained acceptance in recent times. DCF techniques are simple to understand. Further with the development of mechanised accounting and computer technology, it has become very easy to operate.

(2) The second criticism is aginst the assumption of fixed reinvestment rate throughout the life of the project. This assumption results mainly because of the use of compound interest as the basis of these methods. However, in practice, the proceeds could be reinvested at some varying rates instead of a fixed rate.

(3) The third-criticism is that they fall to take account of risk and uncertainly. However, recent developments in risk analysis which could be incorporated into DCF methods have made this criticism almost as irrelevant.

(4) It is complained that IRR and NPV methods give rise to conflicting issues while taking decisions. Recent theoretical refinements have mitigated most of the weaknesses of these methods.

NPV vs. IRR

NPV and IRR are the species of the same genus, ie., DCF techniques are based on the concept that money has a time value. However, there are certain fundamental differences between them. Firstly, the present value treats the discount rate as a known factor while the internal rate of return treats it as an unknown factor. In most cases, the business's known cost of capital is used for discounting under NPV method.

Secondly, the IRR method seeks to find the maximum rate of interest at which funds invested in any given project could be repaid with earnings generated by that project. But the present value method arrives at the amount to be invested in a given project, so that its anticipated earnings could recoup the amount with interest at the market rate.

Thirdly, at a particular discount rate, there can be only one NPV and not multiple net present values of a project. But in the case of unconventional investment where the capital outlays are incurred in different time periods, a project may have more than one IRR which may lead to baffling contradictions.

Fourthly, the present value method explicitly recognises availability of a market for funds and assumes that business will use the market rationally to enhance their earnings that the IRR method does not recognise the existence of such a capital market.

ANALYSIS OF UNCERTAINTY AND RISK

The decision situations are generally classified into three types, viz. certainty, risk and uncertainty. Certainty situation is one where the future occurrence of a particular outcome such as future cash flow or discount rate could be expected with certitude, But in practice all investment decisions are undertaken under conditions of risk and uncertainty. Risk refers to a situation where the probability distribution of a particular outcome could be objectively known in advance, while uncertainty refers to a situation where such probability distribution could not be objectively known but only guessed.

However, in the case of investment decisions such a theoretical distinction is highly hypothetical and may not serve much useful purpose in practice. Even the best estimates of financial manager regarding the probability of the expected cash flows materialising and their magnitude are only subjective guesses.

Some of the factors which add to the degree if uncertainty of an investment are the possibilities of (a) the process or the product, becoming obsolete, (b) declining demand for the product, (c) change in government policy on business, (d) price fluctuations, (e) foreign exchange restrictions, and (f) inflationary tendencies, etc. Uncertainty could be managed by two ways: first, by employing some modern quantitative techniques such as Systems Analysis and Operations Research, Marketing Research, Net Worth Analysis, etc., which could make the estimates for the appraisal more realistic, and second, by applying some techniques for handling risk at capital investment appraisal state. A discussion on the various quantitative techniques is beyond the scope of this lesson. However, a few risk handling techniques would be briefly discussed below.

TECHNIQUES OF RISK ANALYSIS

The techniques used to handle risk may be classified into two groups as follows:

a) Conservative methods such as shorter payback period, risk-adjusted discount rate, and conservative forecasts or certainty equivalents etc., and

(b) Modern methods such as sensitivity analysis, probability analysis, decision tree analysis, etc.

SHORTER PAYBACK PERIOD

According to this method, projects with shorter pay back period are normally preferred to those with longer payback period. It would be more effective when it is combined with a"cut off period". Cut off period denotes the risk tolerance level of the firm. For example, a firm has three projects. A, B and C for consideration with different economic lives say 15,16 and 8 years respectively and with payback periods of, say 6,7 and 5 years. Of these three, project C will be preferred, for its payback period is the shortest. Suppose the cut off period is 4 years. then all the three projects will be rejected.

RISK ADJUSTED DISCOUNT RATE

Under this method, the cut off rate or minimum required rate of return (mostly this is firm's cost of capital) is raised by adding what is called risk premium to it. When the risk is greater, the premium to be added would be greater. For example, if the risk free discount rate (say, cost of capital) is 10 per cent and the project under consideration is riskier one, then the premium of say 5 per cent is added to the above risk free rate. The risk adjusted discount rate would be 15 per cent which may be used either for discounting purposes under NPV or as a cut off rate under IRR.

CONSERVATIVE FORECASTS

Under this method, the estimated risk from cash flows are reduced by employing initiative correction factor or certainty equivalent coefficient, which is calculated by the decision-marker subjectively or objectively. Normally this coefficient reflects the decision- marker's confidence in obtaining a particular cash flow in a particular period. For example, the decision-maker estimates a net cash flow of Rs.10.000 next year, but if he feels (subjectively) that only

Rs.6,000 of such flow is a definite sum, then the said coefficient would be 0.6. This may also be determined (objectively) by relating the desirable cash flows with estimated cash flows as follows:

$$\text{Certainty Equivalent Coefficient} = \frac{\text{Desirable Cash Flows}}{\text{Estimated Cash Flows}}$$

For example, if the estimated cash flow is Rs.80,000 in period 't' and an equally desirable cash flow for the same period is Rs.60,000 then the certainty equivalent is 0.75 (60,000/80,000). Besides the computation of certain cash flows, in order to provide for more risk, the economic life over which cash flows are estimated, may be reduced simultaneously.

SENSITIVITY ANALYSIS

Sensitivity analysis consists of examining the magnitude of change in the rate of return (or the NPV) for the project, for a small change in each of its components which are uncertain. The best practical way to do this is to select these variables whose estimated value may contain significant errors or element of uncertainty and then to calculate the effect of errors of different sizes on the present value of the project. Some of the key variables are cost price, project's life, market share, etc. The common operating mechanism would be to vary each strategic variable by certain fixed percentages in both positive as well as negative direction in turn, say =10% or =5% etc., and study the effect of change on the rate of return (or on NPV).

The advantages of a sensitivity analysis are obvious. The main limitations are: a) Unless the combined effect of changes in a set of inter-related variables is examine, single variable sensitivity testing could be worse than useless-it might lead to wrong conclusions. Hence, it is a highly difficult task (b) The second lies in the fact that the values of the variables (components) are generally changed in an arbitrary manner (5% or 10%) to examine the effect on the returns. Unless it is done in a meaningful manner, it might mislead the investor. (c) The third is that it ignores the changes associated with the different possible values of the components.

PROBABILITY ANALYSIS

Probability may be described as a measure of some one's opinion about the likelihood that an event (cash flow) will occur. The likelihood of occurrence normally ranges from 1 to 0 i.e. 100 per cent certainty to 100 per cent uncertainty. In this analysis, in the place of one single estimate and its associated probability are calculated. a probability distribution in its simplest from could be with a few estimates such as 'optimistic', 'pessimistic' and 'most likely. The real problem, however, is how to obtain this probability distribution. Two types of probabilities -objective and subjective - are normally used for decision-making under uncertainty. Objective probability is the probability estimate which is based on a very large number of observations, say, for example, on the objective evidence of 100 trials conducted repeatedly under independent identical situations.

TEST YOURSELF

1. Explain the meaning and objectives of capital budgeting.

2. Explain the importance and features of capital expenditure control.

3. Discuss the salient features of (a) payback method and (b) return on investment method.

4. Exe Ltd. is considering the purchase of a new machine which would carry out some operation at present performed by manual labour. Two models are available.

The following estimates are made by expects:

	Machine 'Kirloskar'	Machine 'Batliboi'
Estimated Working Life (years)	5	5
Cost of Machine	1,00,000	1,50,000
Saving in Scrap p.a	5,000	6,000
Indirect Materials p.a	3,000	3,500
Savings in Labour p.a.	50,000	60,000
Cost of Maintenance p.a.	11,500	7,500

The taxation rate may be regarded as 40% of profits. Which model can be recommended for purchase? Give reasons for your answer.

5. The following two projects A and B requires investment of Rs2,00,000 each. The income returns after taxes for these projects are as follows:

Year	Project A	Project B
1	80,000	20,000
2	80,000	40,000
3	40,000	40,000
4	20,000	40,000
5	--	60,000
6	--	60,000

Using the following criteria determine which of the projects is Beferahle;

(i) Pay each period

ii) Average Rate of Return

iii) Present value approach if the Company's cost of Capital is

(a) 10% and (b) 6%

LESSON : 16

COST OF CAPITAL

The objectives of this iesson are to enable you to understand and meaning of cost of capital, its importance in financial management, the different forms of costs of capital and the classification of costs based on specific sources of capital. Through this lesson you will also be able to calculate the costs of debt and Preference Share capitals of different business units.

Meaning :

The Cost of capital is an important concept in formulating a firm's capital structure. It is widely used as the very basis of capital budgeting decisions and evaluating the alternative sources of finance. The cost of capital is usually referred to as the 'cut off' rate. It is a yardstick to assess the acceptability or otherwise of particular capital projects. It is the minimum required rate of earnings for capital expenditure.

It is the minimum rate which should be earned by a firm investing its current funds efficiently in long-term activities. In simple language the cost of capital denotes the cost that should be incurred by a firm when it decides to employ various sources of finance such as.

(1) Equity Share Capital,

2) Preference Share Capital,

3) Debt Capital, and

4) Retained Earnings.

In the light of the wealth maximisation objective, the cost of capital can be termed as the rate of return the funds used should produce to justify their use within the firm.

Importance of Cost of Capital:

The development of the concept is recent phenomenon. The concept is very important and relevant in the field of managerial decision as this dynamic concept is affected by a Company's capital structure, its financing plans for the future and any changes in its rate of earnings.

The main purpose for measuring the cost of capital lines in its importance as a tool of decision criterion in capital budgeting decisions. In case of the net present value method, a project will be accepted if it has a positive net present value when the cash flows are discounted at the cost of capital. In this context the cost of capital is the discount rate. In case of internal rate of return method, the project will be accepted if its rate of return is greater than capital. In this situation the cost of capital is treated as the minimum rate of return expected or required on investment projects. The main force behind the investment projects which are being accepted by the firm is the maximisation of the market value per share. The cost of capital indirectly helps the firm to maintain the market value per share at its current level. If the firm to maintain the market value per share at its current level. If the firm earns more than the cost of capital, the market value per share is expected to increase. In this sense, the cost of capital is a target for allocating the firm's finds in profitable manner.

Cost of capital is also significant in designing the capital structure of a firm. The term capital structure if frequently used to indicate the long-term sources of funds employed in a business enterprise. The capital structure influences the cost of capital and any change in the former results in change in the latter and vice versa. Though an optimal capital structure, a firm would be able to minimise the cost of capital and maximise the market value of the firm.

Further, the cost of capital is helpful in evaluation of expansion projects, evaluation of the financial performance of the top management through the comparison of the projected overall cost of capital and the actual cost incurred in raising the required funds.

In addition the cost of capital is a vital factor in management decision about the method of financing at a given time. Costs of various sources of capital at a given time influence the management's decision in favour of a certain capital. Moreover the measure of capital is also important in many areas of decision making such as working capital policy, dividend policy, financial plants for the future etc.

Forms of Costs of Capital:

Earlier we have stated that the cost of capital is the minimum rate of return required on new capital projects in order to keep the per share market value unchanged. However, there are various, concepts of cost of capital which are used in different contexts. Those are given below:

1. **Marginal Cost of Capital:**

   It is the average cost of new funds raised by the firm.

2. **Specific Cost of Capital:**

   It is the cost associated with particular component of capital structure.

3. Average Cost of Capital:

It is the average of various costs or the weighed average of the costs of each component of funds employed by the concern, the weights being the proportion of each component in the capital structure.

4. Combined Costs of Capital:

It includes the costs of capital from all sources: debt, equity, preference capitals and retained earnings. It is also called the weighted cost of capital.

5. Spot Cost of Capital:

It is the cost prevailing in the market at a certain time.

6. Future Cost of Capital:

It is the Projected cost of capital used in designing the capital structure to minimise the future cost of capital and to control it.

7. Opportunity Cost of Capital:

It is the rate of return on other investments available to the firm in addition to the one currently being used.

Classification of Costs Based on Specific Sources of Capital:

Firms used funds from different sources over time. A firm cannot continually finance with debt without having adequate equity basic. Similarly it cannot use solely common stock or retained earning for investment projects. A mix of these sources of capital are used by the firm in order to minimise the Cost of

Capital and maximise the wealth of the firm. There are four basic sources of long-term funds for the business firm. Those are shown on the left hand side of the balance sheet. The basic sources of long-term debt, preferred shares, equity shares and retained earnings. Although all firms will not necessarily use each of these methods of financing, each firm is expected to have funds from some of these sources in its financial structure. The costs that are incurred for obtaining these sources are called with specific source name viz., cost of Debt, Cost of Equity Share Capital, Cost of Preference Share Capital and Cost of Retained earnings. The techniques for measuring the cost of debt and preference share capital and the computation of costs of equity and retained earnings and weighed average cost of capital are given in the following pages. The measurement of cost of capital is most complex and controversial problem in finance area. There are widespread differences as to how these costs should be measured. We examine some of the techniques that are widely used for measuring specific costs of these sources of financing which are presented in the following pages.

Measurement of Cost of Debt:

Long-term Debt debentures and long-term loans. Debentures are debt instruments carrying promises to pay interest and repay principal on maturity. The cost of debt is the rate of return expected by the lenders. Debentures may be issued at par or at premium at discount. Debentures are considered to have been issued at par if the sale proceeds are equal to the face value of Debentures. Debentures may be sold at a premium (more than the face value) in order to align the actual interest yield with the yields prevailing in the market. Debentures sold for less than their face value, or at a discount, have stated interests rates (coupon rates), below the prevailing rates for similar-risks debt instruments. Also the under writing costs and brokerage costs at the time of issue of these debentures reduce the net proceeds from the sale of a debentures at a premium, at a discount or at a par face) value. Only the net proceeds from the sale of debentures are to be considered for calculating the cost of long- term debt.

Example:-

The Xerox company is interested in selling Rs.10 lakhs worth of 20-year 7 per cent debentures, each with a face value of Rs.100. Since similar-risk debt instruments are yielding more than 6 per cent. The firm must sell the debentures for Rs.93 in order to compensate for the low coupon rate. The underwriting firm guaranteeing the issue of debentures receive a fee of 2 per cent of the face value of the debenture which is Rs.2 per debenture bond. The net proceeds to the firm from sale of each debenture are,therefore, Rs. 96 (Rs.98-Rs.2)

Debt Issue at Par:

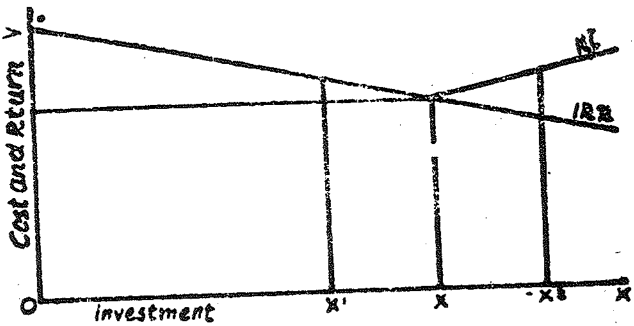
The market yield on debt is considered as the cost of debt. It is nothing but the rate of interest on debt before tax. For example, if the above concern sells a new issue of 6 per cent 20-year debentures to raise Rs.10 lakhs and realises the full face value Rs.100, the before tax cost would be 6 per cent.

$$\text{Before tax cost of Debt Capital} \quad K_1 = \frac{\text{Interest}}{\text{Principal}}$$

$$= \frac{60,000}{10,00,000} \times 100 = 6\,\%$$

Before tax cost of debt, i.e., this example, is equal to the rate of interest or the return earned by the lenders. The main point to be noted while employing debt capital, is that the common shareholders' earnings are not diluted. To avoid this the earnings of the firm should be equal to or greater than the interest rate.

From the figure it is clear that the firm will be able to maximise its share value upto the point X where the rate to return intersects the cost of debt line, K. To some extent the cost of debt will be uniform. But after a certain level it increases



whereas the rate of return decreases with the successive investment projects. If the firm employs X amount of funds, then cost tend to increase as the rate of interest is more than the return from investment. Therefore, it results in dilution of the shareholders, earnings. Thus the market value per share will be affected adversely. This is demonstrated below in the example.—

Example:

The Xerox company has sales of Rs. 10 lakhs, operating cost of Rs. 9 lakhs and no debt. The Company's marginal income tax rate is 55 per cent. Its income statement is shown in the BEFORE column below. Then it borrow Rs. 1 lakh at 6 per cent and invest the funds in assets whose use causes sales to rise by Rs. 70,000 and operating expenses by Rs. 10,000. Hence profits interest rises by Rs. 60,000. The new situation is shown in the AFTER column. Earnings after taxes are unchanged as the investment just earns its cost of debt capital.

INCOME STATEMENTS

	Before Rs.	After Rs.
Sales	10,00,000	10,70,000
Operating Costs	9,00,000	9,10,000
Earnings before interest and taxes (EBIT)	1,00,000	1,60,000
Interest	--	60,000
Earnings before Tax (BBT)	1,00,000	1,00,000
Taxes @ 55%	55,000	55,000
Earnings after Tax (EAT)	45,000	45,000

If the above firm earns less than Rs.1.60,000 after the new investment, the earnings to shareholders will be reduced and as a result per share earnings decline.

The Cost of debt which we have seen earlier is before tax-cost. But interest is allowed as a deductible expenditure for the purpose of calculating income tax on firm's net earnings. Therefore, the cost of debt should be calculated after-tax only. Higher the interest charges, lower will be the amount of tax payable by the firm. As a result the after-tax cost of debt would be lower than the before-tax cost of debt. Thus, in the case of profitable concerns, the cost of debt is the contractual interest rate adjusted further for tax liability of the firm, in the case of unprofitable concerns, the true cost of debt is the before tax cost. The after tax cost of debt $= (K\,I\ -\ t)$

where is the tax rate.

Suppose the Xerox Company issues 6% debentures at face value Rs.100 and its marginal income tax rate is 55 per cent the effective cost of debentures will be:

$$K_i = 6.00\ (1 - 0.55) = 6 \times 0.45 = 2.7\,\%$$

In the calculation of average cost of capital, it is the after-tax cost of debt which should be used and not the before-tax cost. We note that the 2.7 per cent after tax cost in our example represents the marginal or incremental cost of additional debt. It does not represent the cost of debt already employed. In view of tax deductibility, the explicit cost of debt is considerably cheaper than the cost of another source of financing having the same K but where the financial charges are not deductible for tax purposes.

DEBT ISSUED AT A PREMIUM OR DISCOUNT

Debenture bonds may be issued to the public at a price which differs from the face value. The sale value may be more or less than the face value. In both the cases, the yield on debt will not be equal to the coupon rate of interest. If the debentures are issued at premium or discount, the formula for calculating its cost is as follows:

$$\text{After-tax cost of debt} = \frac{(1 - t)\left[R + \frac{1}{n}(F - P)\right]}{\frac{1}{2}(F + P)}$$

Where F = Face value

P = Price at which the debenture is sold,

N = Number of years to maturity,

R = Fixed interest charges,

$\frac{1}{n}(F + P)$ = Amortisation of premium or discount

Example :

Suppose the Xerox company sells its 20-year debentures at Rs. 90 each, whose face value is Rs. 100. The coupon rate of interest is 6% and the marginal tax rate 0.55%. The after-tax cost of debt is:

$$= \frac{(1 - 0.55)\left[6 + \frac{1}{20}\ (100 - 90)\right]}{\frac{1}{2}(90 + 100)}$$

$$= \frac{0.45\left[6 + \frac{1}{20}\ (10)\right]}{\frac{1}{2}(190)}$$

$$= \frac{0.45 \times 6.5}{95} = 3.08\%$$

Example :

Suppose the Xerox Company sells its 20-year debentures of Rs. 100 at Rs. 110 each (i.e. at 10% premium). If the coupon rate of interest is 6% and the marginal tax rate is 0.55%. The after-tax cost of debt is

$$= \frac{(1 - 0.55)\left[6 + \frac{1}{20}(100 - 110)\right]}{\frac{1}{2}(110 + 100)}$$

$$= \frac{0.45\left[6 + \frac{1}{20}(10)\right]}{\frac{1}{2}(210)}$$

$$= \frac{0.45 \times 6.5}{105} = 2.79\%$$

PERPETUAL DEBT

Sometimes firms may issue debentures which are irredeemable. In such a case the debenture capital forms as a permanent source of firm's capital. The cost of debt in such circumstances may be calculated by dividing the price at which the bond is sold into fixed interest charges, adjusted for tax effect.

$$\text{After-tax cost of debt} = \frac{R}{P}(1 - t)$$

where R = Coupon rate of interest
t = Tax Rate
P = Price at which the bond or debenture is sold.

Example:

The Xerox Company issues 6% irredeemable debentures of Rs.90 each and the tax rate is 55%. The cost of debt is:

$$= \frac{6}{90}(1 - 0.55)$$

$$\text{After-tax cost of debt} = \frac{6}{90} \times 0.45 = 3\%$$

COST OF PREFERENCE CAPITAL

Preference share capital represents a special type of ownership interest in the firm. Preferred shareholders must receive their stated dividends prior to the distribution of any earnings to equity shareholders. Since preference share is a form of ownership, and a business firm is viewed as a 'going concern', the proceeds from the sale of preference shares are expected to be held for an indefinite period of time. Though it is not compulsory for the firm to pay dividend on preference shares except for cumulative preference shares, just to avoid complications, it is generally paid. Otherwise, the firm's credit standing may be damaged. Therefore, the dividend on preference shares is paid regularly except under tight cash position and when the firm earns losses. Thus the dividend on preference shares is a fixed cost to the firm like interest on debt. Their rate of dividend per share is fixed well in advance at the time of the issue. So the cost of preference share capital, Kp, is found by dividing the annual preference share dividend, dp, by the net proceeds from the sale of preference shares, Np,. The net proceeds represents the amount of money to be received net of any underwriting or sales expenses to market the stock. For example, if a preference share is sold at Rs. 100 per share on which Rs.3 is paid as underwriting commission, the net proceeds from sale are Rs.97. The formula for calculating the cost of preference share capital is:

$$K_p = \frac{d_p}{N_p}$$

Since the preference share dividends are paid out of the firm's after-tax cash flows, a tax adjustment is not required. Therefore, the cost of preference share capital is greater than the cost of debt.

Example:

The Xerox Company is contemplating an issue of 8 per cent preference shares of Rs. 90 at Rs.85 each. The cost of issuing and marketing the shares is Rs.3 per share. The cost of preference share capital is:

$$K_p = \frac{7.20}{82} = 8.78\,\%$$

Here the annual dividend, dp is Rs.7.20 (i.e 8% of rs. 90) and the net proceeds Rs.82 is obtained after deducting Rs.3 from Rs.85. If the dividend on the preference shares of Xerox Company had been stated in rupees, the calculations would have been greatly simplified since the rupee dividend could have been substituted into the above formula.

Example :

The Xerox Company issues Rs.8 preference share capital which is irredeemable. The face value of share is Rs. 100 but they are issued at Rs. 105. The cost of issuing and marketing the share is Rs.3 per share. Calculate the cost of preference share capital.

Formula for finding cot of preference share capital is :

$$K_p = \frac{d_p}{N_p}$$

In this problem, d_p = Rs. 8

N_p = Rs. 105 – 3 = Rs. 102

Therefore, $K_p = \frac{8}{102} = 7.84\,\%$

COST OF EQUITY SHARE CAPITAL

The cost of equity share capital is not nearly easy to calculate as either the cost of debt or the cost of preference share capital. The rate of dividend on equity share is not fixed like the interest rate on debt or dividend on preference shares Shares. The income from equity investments, i.e. dividends, depends on the level of profits and the dividend influences the market value of equity shares. The cost of equity capital is the minimum rate of return that a company must earn on the equity financed portion of its investments in order to maintain the market price of the equity share at the current level. In other words, the required rate of return which equates the present value of the expected dividends with the market value of share is the cost of equity capital.

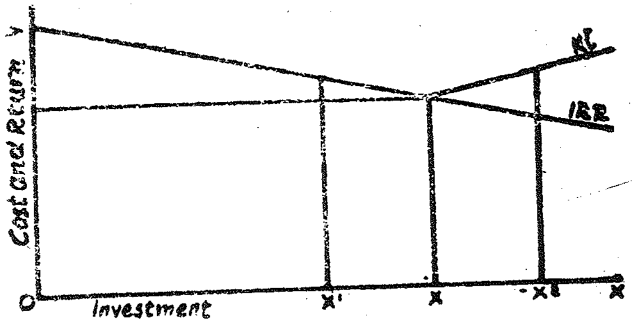
The cost of equity capital includes the cost of external equity or the new issue of ordinary shares and the retained earnings. There are two important factors to be considered while calculating the cost of equity capital. They are (1) Unpredictable nature of future dividends and (2) The nature of earnings and dividends, unlike the interest on debt or the dividend on preference share. are expected to grow.

Further, with the maximisation of shareholders' welfare as the main objective, the management should ensure that the existing shareholders are well off in respect of the expected return on their investment and the market value of share, as they were before. While raising funds through the issue of equity shares for any new investment project, the expected return from the investment should be computed, as the insufficiency of the return to meet the cost of capital results in dilution of equity shareholder's share. The company should not issue the new equity shares, if the new investment does not generate sufficient earnings.

There are several approaches for computing the cost of equity share. The are:

(i) D/P Ratio Approach,

(ii) . E/P + Growth Approachment.



Therefore, it results in dilution of the share-holders, earnings. Thus the market value per share will be affected adversely. This is demonstrated below in the example.

Example:

The Xerox Company has sales of Rs.10 lakhs, operating cost of Rs. 9 lakhs and no debt. The Company's marginal income tax rate is 55 per cent. Its income statement is shown in the BEFORE column below. Then it borrow Rs. 1 lakh at 6 per cent and invest the funds in assets whose use causes sales to rise by Rs. 70,000 and operating expenses by Rs. 10,000. Hence profits interest rises by Rs.60,000. The new situation is shown in the AFTER column. Earnings after taxes are unchanged as the investment just earns its cost of debt capital.

INCOME STATEMENTS

	Before Rs.	After Rs.
Sales	10,00,000	10,70,000
Operating Costs	9,00,000	9,10,000
Earnings before interest and taxes (EBIT)	1,00,000	1,60,000
Interest	--	60,000
Earnings Before Tax (EBT)	1,00,000	1,00,000
Taxes @ 55%	55,000	55,000
Earnings After Tax (EAT)	45,000	45,000

If the above firm earns less than Rs.1,60,000 after the new investment, the earnings to shareholders will be reduced and as a result per share earnings decline.

The Cost of debt which we have seen earlier is before tax-cost. But interest is allowed as a deductible expenditure for the purpose of calculating income tax on firm's net earnings. Therefore, the cost of debt should be calculated after-tax only. Higher the interest charges, lower will be the amount of tax payable by the firm. As a result the after-tax cost of debt would be lower than the before-tax cost of debt. Thus, in the case of profitable concerns, the cost of debut is the contractual interest rate adjusted further for tax liability of the firm, in the case of unprofitable concerns, the true cost of debt is the before tax cost. the after tax cost of debt $= (K\ 1 - t)$

where is the tax rate.

Suppose the Xerox Company issues 6% debentures at face value Rs. 100 and its marginal income tax rate is 55 per cent the effective cost of debentures will be:

$$K_i = 6.00\ (1 - 0.55) = 6 \times 0.45 = 2.7\%$$

In the calculation of average cost of capital, it is the after-tax cost of debt which should be used and not the before-tax cost. We note that the 2.7 per cent after tax cost in our example represents the marginal or incremental cost of additional debt. It does not represent the cost of debt already employed. In view of tax deductibility, the explicit cost of debt is considerably cheaper than the cost of another source of financing having the same K but where the financial charges are not deductible for tax purposes.

DEBT ISSUED AT A PREMIUM OR DISCOUNT

Debenture bonds may be issued to the public at a price which differs from the face value. the sale value may be more or less than the face value. In both the cases, the yield on debt will not be equal to the coupon rate of interest. if the debentures are issued at premium or discount, the formula for calculating its cost is as follows:

$$\text{After-tax cost of debt} = \frac{(1-t)\left[R + \frac{1}{n}(F-P)\right]}{\frac{1}{2}(F+P)}$$

Where F = Face value

P = Price at which the debenture is sold,

N = Number of years to maturity,

R = Fixed interest charges,

$\frac{1}{n}$ (F + P) = Amortisation of premium or discount

Example:

Suppose the Xerox Company sells its 20-year debentures at Rs.90 each, whose face value is Rs.100. The coupon rate of interest is 6% and the marginal tax rate 0.55%. The after-tax cost of debt is:

$$= \frac{(1 - 0.55)\left[6 + \frac{1}{20}(100 - 90)\right]}{\frac{1}{2}(90 + 100)}$$

$$= \frac{0.45\left[6 + \frac{1}{20}(10)\right]}{\frac{1}{2}(190)}$$

$$= \frac{0.45 \times 6.5}{95} = 3.08\,\%$$

Example:

Suppose the Xerox Company sells its 20-year debentures of Rs.100 at Rs. 110 each (i.e. at 10% premium). If the coupon rate of interest is 6% and the marginal tax rate is 0.55%. the after-tax cost of debt is:

$$= \frac{(1 - 0.55)\left[6 + \frac{1}{20}(100 - 110)\right]}{\frac{1}{2}(110 + 100)}$$

$$= \frac{0.45 \left[6 + \frac{1}{20} (10) \right]}{\frac{1}{2} (210)}$$

$$= \frac{0.45 \times 6.5}{105} = 2.79 \%$$

PERPETUAL DEBT

Sometimes firms may issue debentures which are irredeemable. In such a case the debenture capital forms as a permanent source of firm's capital. The cost of debt in such circumstances may be calculated by dividing the price at which the bond is sold into fixed interest charges, adjusted for tax effect.

$$\text{After-tax cost of debt} = \frac{R}{P} (1 - t)$$

where R = Coupon rate of interest

t = Tax Rate

P = Price at which the bond or debenture is sold.

Example :

The Xerox Company issues 6% irredeemable debentures of Rs.90 each and the tax rate is 55% the cost of debt is:

$$= \frac{6}{90} (1 - 0.55)$$

$$\text{After-tax cost of debt} = \frac{6}{90} \times 0.45 = 3 \%$$

COST OF PREFERENCE CAPITAL

Preference share capital represents a special type of ownership interest in the firm. Preferred shareholders must receive their stated dividends prior to the distribution of any earnings to equity shareholders. Since preference share is a form of ownership, and a business firm is viewed as a 'going concern', the proceeds from the sale of preference shares are expected to be held for an indefinite period of time. Though it is not compulsory for the firm to pay dividend on preference shares except for cumulative preference shares, just to avoid complications, it is generally paid. Otherwise, the firm's credit standing may be damaged. therefore, the dividend on preference shares is paid regularly except under tight cash position and when the firm earns losses. Thus the dividend on preference shares is a fixed cost to the firm like interest on debt. their rate of dividend per share is fixed well in advance at the time of the issue. So the cost of preference share capital, Kp is found by dividing the annual preference share dividend, dp, by the net proceeds from the sale of preference shares, Np. The net proceeds represents the amount of money to be received net of any underwriting or sales expenses to market the stock. For example, if a preference share is sold at Rs. 100 per share on which Rs.3 is paid as underwriting commission, the net proceeds from sale are Rs.97. The formula for calculating the cost of preference share capital is:

$$K_p = \frac{d_p}{N_p}$$

Since the preference share dividends are paid out of the firm's after-tax cash flows, a tax adjustment is not required. therefore, the cost of preference share capital is greater than he cost of debt.

Example :

The Xerox Company is contemplating an issue of 8 per cent preference shares of Rs.90 at Rs.85 each. The cost of issuing and marketing the shares is Rs. 3 pr share. The cost of preference share capital is:

$$K_p = \frac{7.20}{82} = 8.78\%$$

Here the annual dividend, dp is Rs.7.20 (i.e. 8% of Rs.90) and the net proceeds Rs.82 is obtained after deducting Rs.3 from Rs.85. if the dividend on the preference shares of Xerox Company had been stated in rupees, the calculations would have been greatly simplified since the rupee dividend could have been substituted into the above formula.

Example ;

The Xerox Company issues Rs.8 preference share capital which is irredeemable. The face value of share is Rs.100 but they are issued at Rs.105. The cost of issuing and marketing the share is Rs.3 per share. Calculate the cost of preference share capital.

Formula for finding cost of preference share capital is :

$$K_p = \frac{d_p}{N_p}$$

In this problem, $d_p = \text{Rs. } 8$

$$N_p = \text{Rs. } 105 - 3 = \text{Rs. } 102$$

Therefore, $K_p = \frac{8}{102} = 7.84\%$

COST OF EQUITY SHARE CAPITAL

The cost of equity share capital is not nearly as easy to calculate as either the cost of debt or the cost of preference share capital. The rate of dividend on equity share is not fixed like the interest rate on debt or dividend on preference shares. The income from equity investments, i.e. dividends, depends on the level of profits and the dividend influences the market value of equity shares. The cost of equity capital is the minimum rate of return that a company must earn on the equity financed portion of its investments in order to maintain the market price of the equity share at the current level. In other words, the required rate of return which equates the present value or the expected dividends with the market value of share is the cost of equity capital.

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Further, with the maximisation of shareholders welfare as the main objective, the managment should ensure that the existing shareholders are well off in respect of the expected return on their investment and the market value of share, as they were before. While raising funds through the issue of equity shares for any new investment project, the expected return from the investment should be computed, as the insufficiency of the return to meet the cost of capital results in dilution of equity shareholder's share the company should not issue the new equity shares, if the new investment does not generate sufficient earnings.

There are several approaches for computing the cost of equity share. they are:

(i) D/P Ratio Approach,

(ii) E/P + growth Approach,

(iii) E/P Ratio Approach,

(iv) Realised yield Approach.

(i) Dividend-Price Ratio (D/P) Approach:

Under this approach, the return represents what investors expect when they invest savings in a company.

$$K_e = \frac{D}{P}$$

where, K_e = Cost of equity shares,
D = Current dividend rate,
P = Current market price of equity share.

Example:

The Xerox Company has issued 50,000 equity shares of Rs.100 each, the current market price being Rs.90 and the dividend rate Rs.4.50. Calculate the cost of equity capital.

Applying the formula $K_e = \frac{D}{P}$

$$= \frac{4.5}{90} = 5\ \%$$

is the cost

From the above example, it is clear that the cost of capital is an important factor in determining the market price of the equity capital.

Since $K_e = \frac{D}{P}$

$$P = \frac{D}{K_e}$$

Though it is compulsory on the part of the management to declare dividend on equity shares to maintain the market price of the equity shares at the current level, the declaration becomes indispensable.

This method has got three limitations. they are:

(1) It ignores the earnings on retained earnings,

(2) It ignores the fact that market price changes may be due to retained earnings also and not on account of high dividend, and

(3) The growth of the firm.

(ii) D/P + Growth Approach:

Under this approach, the expected rate of return has to be higher than the present earnings per share relative to present market value per share as the earnings and dividends on ordinary share capital are generally expected to grow.

Therefore, $K_e = \frac{D}{P + G}$

where K_e = Cost of equity share,

D = Current dividend rate,

P = Current market price of share,

G = Expected growth rate of returns.

Example :

The Xerox Company has issued 5,000 equity shares of Rs. 100 each. Its current market price is Rs. 95 and the current dividend rate is Rs. 4.50 per share, the dividend are expected to grow at a rate of 8%. Compute the cost of equity capital.

$$K_e = \frac{4.5}{95} + 8\%$$

$$= 4.74 + 8\% = 12.74\%$$

The cost of equity is thus the current yield plus the growth rate. This approach is based on certain assumptions. they are:

(1) The market value of the share is a function of the expected dividends.

(2) The initial dividend is greater than zero.

(3) The growth rate of dividend is constant for ever.

(4) The dividend-pay out ratio is constant.

The main limitation of this approach is the difficulty in determining or predicting the expected rate of return by a shareholder.

The above formula is to be altered whenever there are cost of issue or flotation costs. Such costs include the brokerage fees, commission for underwriting of shares, etc. The net proceeds expected to be realised by the company would be the market price less the flotation costs. This requires an adjustment in the calculation of cost of equity. If the company pays 'f' fraction of the share price as flotation cost, the cost of new issue of ordinary shares will be:

$$K_e = \frac{D}{P(1-f)} + G$$

It is quite important that the cost of ordinary share be adjusted for any under pricing or underwriting costs in determining the cost of new issues will be greater than the cost of existing issues as long as 'f' in the above equation is greater than zero.

Example:

The Xerox Company has issued 5,000 equity shares of Rs. 100 each. The market price of the share is Rs. 110. The current dividend rate is Rs. 5 per share and they are expected to grow at the rate of 4 percent. The cost of flotation for the issue is Rs. 2.50 per share. Compute the cost of new equity share capital.

$$K_e = \frac{D}{P(1-f)} + G$$

where

D = Rs.5

P = Rs.110

f = Rs.2.50 per share or 2.77%

G = 4%

$$K_e = \frac{\text{Rs. } 5.00}{\text{Rs. } 110\ (1 - 0.0277)} + 0.04$$

$$= \frac{\text{Rs. } 5}{\text{Rs. } 106.953} + 0.04$$

$$= 0.0467 + 0.04 \qquad = 0.0867$$

$$= 8.67\%$$

(iii) Earnings-Price (E/P) ratio Approach:

This ratio represents the relationship between earnings and market price of shares. This approach is based on three conditions. They are:

(1) The firm does not employ debt.

(2) The dividend pay out is 100 per cent or in other words, the retained earning is equal to zero.

(3) The earnings of the firm are stable which implies a zero growth rate (g + 0).

$$K_e = \frac{E}{P}$$

The limitations of this approach are:

(a) All earnings are not distributed to shareholders in the form of dividends,

(b) Earning per share cannot be constant, and

(c) Share price also does not remain constant.

The earning-price ratio is a valid measure in case of a stable or expanding firm and growing firm. Adjustment regarding the flotation costs should be made when using E/P for measuring the cost of new issue of equity shares.

Example:

The Xerox Company is contemplating to raise Rs. 5 lakhs through the issue of equity shares. The company has already issued 10,000 equity shares of Rs. 100 each which is currently earning Rs. 1,50,000 per year. The new issue can be made at the current market price of Rs.112 on which the flotation costs would be Rs.3 per share. Compute the cost of new equity capital issue assuming that there is no debt and the earnings are stable.

If E/P ratio is used to measure the cost of equity, the formula is:

$$K_e = \frac{E}{P}$$

where E = Earning per share,

P = Net proceeds from sale of share.

$$\text{Earnings per share} = \frac{1,50,000}{10,000} = \text{Rs. } 15$$

$$= \text{Rs.}112 - 3 = \text{Rs.}109$$

Net proceeds per share =
Market price − Floatation Costs

$$K_e = \frac{\text{Rs. } 15}{\text{Rs. } 109} = 0.1376 \text{ or } 1376\%$$

(iv) Realised Yield Approach:

This approach is based on the rate of return actually realised for a period of time by investors in a particular company. If Rs.5 is paid as dividend on equity shares, it is the cost of equity

The cost of equity capital is normally greater than any other long-term financing cost. Since equity share dividends are paid from after-tax cash flows, no tax adjustment is required.

COST OF RETAINED EARNINGS

The cost of retained earnings is closely related to the cost of equity shares. If earnings are not retained, they would be paid out to the equity shareholders as dividends. Often the retained earnings are looked on as a fully subscribed issue of equity shares, since they increase the stockholders' equity in the same way that a new issue of equity shares would. The cost of retained earnings must therefore be viewed as the opportunity cost of the foregone dividends to the existing shareholders.

In determining the cost of retained earnings it is necessary to look at the alternative uses of funds. For what purposes other than reinvestment within the company could they be employed? On the one hand, they might he paid out to shareholders in the form of dividends. On the other hand, these funds might be employed to purchase a majority control in the assets of another company.

These approaches result in differences of treatment due to the fact that dividends paid to shareholders are taxable. For example, assume a company with a cost of equity capital of 10 per cent. The firm strives to achieve a minimum rate of at least 10 per cent. Under the dividend approach to cost of retained earnings, it is necessary to calculate the rate at which shareholders would have to invest any dividends received in order to equal the 10 per cent that the corporation could earn on the same funds.

This rate would depend upon the tax bracket of the shareholder. Thus, if his income were low and was in the 20 per cent bracket, he would first have to pay tax of 20 per cent on the dividends received from the corporation and then invest the net proceeds at a rate sufficiently high so that he would receive are turn of 10 percent after paying 20 per cent tax on the income received. On this basis, the cost of retained earnings would be 12.5 per cent.

$$\left[\frac{\text{Rs. } 10}{(\text{Rs. } 100 - 20)} \right]$$

But this cost would vary with the shareholder's tax bracket. If the shareholder is in the bracket of 50 per cent, the cost of retained earning would be 20 per cent.

$$\left[\frac{\text{Rs. } 10}{(\text{Rs. } 100 - 50)} \right]$$

Therefore, the cost of retained earnings

$$K_r = K_e \ (1 - T) \ (1 - B)$$

where K_r = Cost of retained earnings,

K_e = Cost of equity capital (either D/P or D/E approach)

T = Marginal tax rate of shareholder

B = Brokerage commission to acquire new shares

The value of K_r is thus fractionally less than the value of K_e. But it is difficult to get average tax rate and brokerage costs of a company's shareholders.

Example:

The cost of equity capital for Xerox Company is 8.67 per cent. The average tax rate of its shareholders is 25 per cent and the brokerage costs amount to 3 per cent. Fund out the cost of retained earnings.

Substituting the appropriate values into above equation, we get

$$K_r = 0.0867 \ (1 - 0.25) \ (1 - 0.03)$$
$$= 0.0867 \ (0.75) \ (0.97)$$
$$= 0.0631$$
$$= 6.31\%$$

In the above example, the shareholder should be able to earn at least 6.31% as dividend on his new investment which in turn equals 8.67% equity cost of his old company.

The second approach to estimate the cost of retained earnings is the 'external yield' on external investments. The company should invest these funds in an external investment which offers an income-risk package similar to its own existing assets. The line of approach suggests cost of retained earnings is equal to the cost of equity capital.

Instead of looking at these possible alternative uses of funds provided by retained earnings, we might shift to a consideration of the sources of funds. Assume that the corporation had paid out all its earnings as dividends. In order to obtain the amount of funds equal to the undistributed earnings that it would otherwise have had, the corporation would have to undertake new financing. If this were in the form of new equity capital, the cost would be our assumed 10 per cent cost of equity capital. Since virtually all companies rely primarily on equity sources of capital, we are inclined to favour this point of view, which means that the cost of retained earnings is the same as the cost of financing by means of equity shares.

One may question the reason for choosing the retained earnings to equity share issue by the companies when the cost of new equity issue is same as that of retained earnings. The reason being the matter of convenience and it does not involve any floating costs and at the same time gets all the advantages of equity. When retained earnings are use, the uncertainties of a public financing are eliminated, the delay caused by the preparation of registration statements and prospectuses is avoided and the difficulty of gauging market conditions bypassed. Management simply uses funds as part of its recovering activities. It does not draw fine lines of distinction between funds obtained from retained earnings. In effect, they both represent capital to be invested in the same type of projects at similar rates. Finally, as a matter of practice, if the overall rate of return is determined by using the market value of common stock, the need for determining a separate rate for retained earnings disappears, as both the weighing factor and rate of return are embodied in market values.

Some managements regard retained earnings as cost-free. Since internal sources of funds are much more important to corporations than equity funds raised externally. It becomes a matter of considerable importance whether or not to attach a cost of retained earnings.

Managements which assign no cost or a low cost, to internally generated funds may, as a result, undertake projects which they would not attempt if they had to raise equity capital externally, or if they had correctly calculated their combined cost of capital, attributing to retained earnings the same cost as to equity raised externally.

The weighted average cost of capital is found by weighing the cost of each specific type of capital by the historical or marginal proportions of each type of capital use. Historical weights are based on the firm's existing capital structure, whereas marginal weights consider the actual proportions of each type of financing expected to be used in financing a given project.

WEIGHTED AVERAGE COST OF CAPITAL BASED ON HISTORICAL COSTS

This is the most common method used for finding out the overall cost of capital. Under this method, the historical weights of capital used by the firm are used. The use of these weights is based on the assumption that the firm's existing mix of funds in the capital structure is optimal and there fore should be maintained in the future also. There are two types of historical weights that are used in the calculation of weighted average cost of capital. They are book value weights and market value weights.

Whatever method is used for calculating the average cost of capital, whether historical or marginal, the following steps are usually involved:

(1) Calculate the cost of specific source of funds such as cost of debt, cost of preference, cost of equity and cost of retained earnings.

(2) Multiply the cost of each source by its weight which is obtained in proportion to the total capital.

(3) Add these weighted costs from all the sources of fund to arrive at the weighted average cost of capital.

In the financial decision making, the cost of capital should be computed on an after-tax basis. Hence, the specific costs to be used to measure the weighted cost of capital must be the after- tax costs.

BOOK VALUE WEIGHTS

Here book values of different sources of capital already in use are considered for the purpose of obtaining the proportions in which they are used. These proportions of each source to the total funds are used as weights for calculating weighted average cost of capital. The use of book value weights in calculating the firm's weighted average cost of capital assumes that new financing will be raised using exactly assumes that new financing will be raised using exactly the same proportion of each type of financing as the firm currently has in its capital structure.

Example:

The specific costs and the book values of various types of capitals used by Xerox Company are as follows:

Source of Capital	Book Value	Cost
Debt	15,00,000	2.79%
Preference Shares	10,00,000	7.84%
Equity Shares	20,00,000	8.67%
Retained Earnings	5,00,000	8.21%
	50,00,000	

Calculate the weighted average cost of capital using book value weights.

The Weighted Average Cost of Capital for the Xerox Company based on Book Value Weights and the Values used in its Calculations

Source of Capital	Book Value Rs.	Percentage of each total	Cost %	Weighted Cost (2) x (3) %
	1	2	3	4
Debt Capital	15,00,000	30.00%	2.79%	0.837%
Preference Capital	10,00,000	30.00%	7.84%	1.568%
Equity Shares	20,00,000	40.00%	8.67%	3.468%
Retained Earnings	5,00,000	10.00%	8.31%	0.831%
	50,00,000	100.00%		6.704%

The weighted average cost of capital of Xerox Company, based on book values, is 6.704%.

MARKET VALUE WEIGHTS

Since the market values reflect the expectations of investors, market value weights may be preferred to book value weights while calculating weighted average cost of capital. Also the market values of the securities closely approximate the actual amount to be received from their sale. Moreover, since the costs of various types of capital are calculated using prevailing market prices, it seems only reasonable to use market weights too. However, it is more difficult to calculate the market values of a firm's sources of capital than to use book values. It is particularly difficult to allocate a market value to the firm's retained earnings. The weighted average cost of capital based on book value weights since most equity shares and preference shares have market values considerably greater than their book values. Since these sources of finds have higher specific costs, the overall cost of capital increases.

Example:

Using the costs given in the earlier example, calculate the weighted average cost of capital of Xerox Company when the market values of different capital are as follows:

	Rs.
Debt Capital	15,00,000
Preference Capital	15,00,000
Equity Share Capital	40,00,000
Retained Earnings	10,00,000
	80,00,000

Weighted Average Cost of Capital based on Market Values

Sources of Capital	Market Value Rs.	Percentage of each total	Cost %	Weighted Cost (2) x (3)
	1	2	3	4
Debt Capital	15,00,000	18.8%	2.79%	0.525%
Preference Capital	15,00,000	18.8%	7.84%	1.474%

Equity Share Capital	40,00,000	50.0%	8.67%	4.335%
Retained Earnings	10,00,000	12.4%	8.31%	4.335%
	80,00,000	100.0%		7.369%

The weighted average cost of capital based on market value weights of Xerox Company is 7.369% which is greater than book value weighted average cost of capital 6.704%. If the book value weighted average cost of capital is used, the firm may have to accept projects that would be unacceptable based on the market value approach. But the book value weights are used by many firms because of the following reasons. (1) Firms set their capital structure targets in terms of book value, (2) Book value information can be readily derived from the published information and (3) Investors normally use book value debt-equity rations to evaluate the riskness of the firms.

WEIGHTED AVERAGE COST OF CAPITAL BASED ON MARGINAL WEIGHTS

Cost of capital is significantly used in the selection of new projects by the firms and not with those under execution. Therefore, the relevant cost to be worked is the cost of raising new funds to finance the projects, not the historical cost which has been incurred in the past. Hence the weighted average cost of new capital issue for the project is more relevant than the one based on historical weights. The use of marginal weights involves weighing the specific cost of various types of financing by the percentage of the total financing expected to the raised. In using the marginal weights, we are concerned with the actual amounts of each type of financing used whereas in using historical costs our assumption is that the proportion of each type of financing in the firm's capital structure would remain the same. The use of marginal weights is more attuned to the actual process of financing projects. It recognises that funds are actually raised in discrete amounts using the various sources of funds.

This method does not consider the long-term implications of the firm's current financing. But all capital expenditures are long-term investments and therefore, the firm should have long-term financing strategy. If the firm uses cheaper debt capital for the current project the future project may have to be financed with equity share capital. For example, for the current project which is expected to give a return of 6 percent, the firm may obtain funds by selling debentures at 6 percent. But in the next year if the firm has to raise funds through equity, it has to offer, say 9 percent return in view of increased financial risk which makes the firm to reject a project offering 8 percent return. But if the weighted average cost of capital to the firm, say 7 percent return will be rejected and the 8 percent project will be accepted. Although the firm raises funds in lumps the use of historical weights to calculate the overall weighted average cost of a firm's capital structure is more consistent with the firm's long-term goal of maximising its owners' wealth. That is why most of the firms use historical market value weights to marginal weights while calculating their overall cost of capital.

Example :

The Xerox Company, whose specific costs of capitals are given in the previous example, is contemplating to raise Rs. 10 lakhs for plant expansion. They estimate that Rs. 2 lakhs will be available from retained earnings and intend to sell Rs. 6 lakhs worth of debentures and Rs. 2 lakhs worth of preference shares to raise the remaining Rs. 8 lakhs. Calculate the firm's weighted average cost of capital based on marginal weights.

Weighted Average Cost of Capital for the Xerox Company based on Marginal Weights.

Source of Capital	Amount Rs.	Percentage to total	Cost %	Weighted Cost (2) x (3)
Debt Capital	6,00,000	60%	2.79%	1.674%
Preference Capital	2,00,000	20%	7.84%	1.568%
Retained Earnings	2,00,000	20%	8.31%	1.662%
	10,00,000	100%		4.904%
Weighted average cost of capital = 4.904%				

The cost of capital obtained here is less than the average cost of capital calculated using either type of historical weights. This is because of the small amount of more expensive preference shares and retained earnings used. In view of the excessive use of debt financing, the firm may have to use expensive equity financing for future projects. as it has to maintain proper equity base.

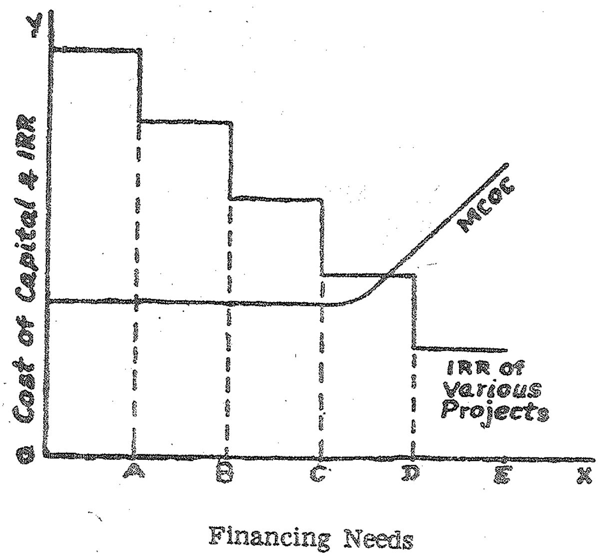
MARGINAL COST OF CAPITAL APPROACH TO CAPITAL EXPENDITURE DECISION

The marginal cost of capital to the firm can be measured by determining for various amounts of financing, the associated weighted average cost of capital for the desired capital structure. As the firm attempts to raise larger amounts of financing, the weighted average cost of the financing is expected to increase. The relationship between the amount of financing raised and the associated weighted average can be used to develop a marginal cost of capital function.

The marginal cost of capital function indicates the weighted average cost of funds, taking into account the fact that raising more than a certain amount of a given type of financing may cause the cost of that type of financing to rise, thereby raising the firm's weighted average cost of capital for all levels of financing.

Marginal cost of capital may be used for determining the group of acceptable capital expenditure projects.

The weighted average cost of capital is expected to increase as the firm raises funds. The increase in weighted average cost of capital due to increased financing is called marginal cost of capital. While considering the marginal cost of capital in capital budgeting decisions, one has to take the internal rate of return (IRR) of the various projects. If IRR is



greater than marginal cost of capital, the project can be accepted. In the above figure, it is found that projects A, B and C can be accepted as their IRR is greater than the respective marginal cost of capital. Project D cannot be accepted as its IRR is less than the cost of capital.

LEVERAGE AND COST OF CAPITAL

The financial manager can, within limits, cause his firm's cost of capital to rise or fall. If he starts with an all equity structure, adding debt to a limited extent will bring down the combined or average cost of capital. As debt is added the leverage ratio rises and the after-tax combined cost of capital falls. Of course the financial manager can achieve this effect in the range from pure equity to the optimal capital structure. Within this range, lower cost of debt increments are being averaged in with higher cost of equity, thus not only reducing the overall cost of capital but also maximising the wealth of the firm.

Test yourself

1. What is cost of capital? Explain the significance of cost of capital.
2. What is the relevance of cost of capital in capital budgeting?
3. What is weighted average cost of capital? Examine the rational behind the use of weighted average cost of capital.
4. A Company issues 10,000 10% preference Share, of rs 100 each redeemable after 10 years at a minimum of 5%. The cost of issue is Rs 2 per share. Calculate the cost of preference capital

5. A firm has the following capital stricture and after tax costs for the different sources of funds used:

Sources of Funds	Amount Rs.	Proportion %	After Tax Cost
Debt	15,00,000	25	5
Preference Shares	12,00,000	20	10
Equity Shares	18,00,000	30	12
Retained Earnings	15,00,000	25	11
Total	60,00,000	100	

Your are required to complete the weighted average cost of capital.

LESSON : 17

WORKING CAPITAL

Working Management is concerned with the problems that arise in attempting to manage the current assets. The current liabilities and the later-relationships that exist between them. The main objective of Working Capital management is therefore, of manage the firm's current assets and current liabilities in such a way that a satisfactory level of working capital is maintained. This is so because if the firm fails to maintain a satisfactory level of working capital, it is likely to become insolvent and may even be forced into bankruptcy. Hence each of the current assets must be managed efficiently in order to maintain liquidity of the firm while not keeping too high a level of any cne of them. This lesson would explain the basic ingredients of Working Capital Management.

Concept Working Capital

There are two concepts of Working Capital-gross and net. The gross working capital, 'refers to the firm's investments in total current assets. And the term 'net working capital' refers to the difference between current assets and current liabilities. Alternatively, the net working capital may be defined as that portion of a firm's current assets which is financed with long term funds.

The net working capital, being, the difference between current assets and current liabilities is a qualitative concept. It indicates (a) the liquidity position of the firm and (b) suggests the extent to which working capital needs may be financed by **permanent** sources of funds. It also covers the question of judicious mix of long term and short term funds for financing current assets. Efficient working capital management requires that firms should operate with some amount of net working capital. However, the exact amount of working capital required varies from business to business depending upon, among other things, the nature of business or industry.

The need for Working Capital

In its endeavour to attain the business objective any firm should earn sufficient return from its operations . For the purpose a successful sales activity is required. The firm has to invest enough funds in current assets for the success of sales activity. Current assets are needed because sales do not convert into cash instantaneously. There is always an operating cycle involved in the conversion of sales into cash.

Operating Cycle

The duration of time required to complete the following sequence of events, in case of a manufacturing firm, is called the operating cycle.

1. Conversion of cash into raw materials
2. Conversion of raw materials into work inn process
3. Conversion of work in process into finished goods.
4. Conversion of finished goods into debtors and bills receivables through sale.
5. Conversion of debtors and bills receivables into cash.

The cycle will repeat again and again.

A firm needs working capital (gross) because the production, sales and cash flows (payments and realisation) are not instantaneous. The firm needs cash to purchase raw materials and pay expenses as there may not be perfect matching between cash inflows. Cash may also be held to meet the future exigencies. The stocks of raw materials are kept in order to ensure smooth production and to protect against the risk of non-availability of raw materials. Similarly, stocks of finished goods have to be carried to meet the demands of the customers on continuous basis and sudden demands from customers. Goods are sold on credit for competitive reasons. Thus an adequate amount of funds has to be invested in current assets for a smooth and uninterrupted production and sales process. Because of the circulating nature of current assets, working capital is sometimes called circulating capital.

Permanent and Temporary Working Capital

The operating cycle is a continuous process and, therefore,' the need for current assets is felt constantly. But the magnitude of current assets needed is not always the same it increases and decreases over time. However, there is always a minimum level of current assets which is continuously required by the firm to carry on its business operations. This minimum level of current assets is referred to as permanent or fixed working capital. It is permanent in the same way is the firm's fixed assets are.

Depending upon the changes in production and sales, the need for working capital will fluctuate over and above the permanent working capital. Any amount over and above the permanent level of working capital is called as fluctuating, or variable or temporary working capital. This portion of the required working capital is needed to meet fluctuations in demand consequent upon changes in production and sales as a result of seasonal change.

Excess or Inadequate Working Capital

Both excessive as well as inadequate working capital positions are dangerous from the firm's point of view.

The dangers of excessive working capital are as follows:

1. It results in unnecessary accumulation of inventories. Thus, the changes of inventory mishandling waste, theft and losses increase

2. It is an indication of defective credit policy and slack collection period. Consequently, higher incidence of bad debts results, which adversely affects profits.

3 Excessive working capital makes management complacent which degenerates into management complacent which degenerates into managerial inefficiency.

4. Tendencies of accumulating inventories to make speculative profits grow. This may tend to make dividend policy liberal and difficult to cope with in future when the firm is unable to take speculative profits.

Inadequate working capital is also bad and has the following dangers.

1 It stagnates growth. It becomes difficult for the firm to under take profitable projects for nor availability of the working capital funds.

2. It becomes difficult to implement operating plans and achieve the firm's profit target.

3. Operating inefficiencies creep in when it becomes difficult even to meet day-to-day commitments.

4. Fixed assets are not efficiently utilised for the lack of working capital funds. Thus, the rate of return on investment slumps

5. Paucity of working capital funds renders the firm unable to avail attractive credit opportunities etc.

6. The firm loses its reputation when it is not in position to honour its short-term obligations. As a result, the firm faces tight credit terms.

Determinants of Working Capital

Number of factors exert their influences on the level and composition of working capital in an enterprise at different points of time. The following factors may be mentioned as significant among them.

Nature of Business

The nature of business has an important bearing on its working capital needs. For example a manufacturing concern will have more number of items in the list of current assets and current liabilities than a merchandising or a public utility undertaking and its requirements will therefore be large comparatively. Among manufacturing industries, seasonal industry's requirements may vary from non-seasonal industry's.

The Manufacturing Cycle

Obviously, the longer the period of manufacture the larger the working capital required. Most products have alternative processes of manufacture and, if working capital is considered a critical area, the choice has to fall on those processes that have shorter manufacturing cycle. This is a technological choice with an eye on working capital economy.

Credit Policy

Credit plays a considerable role in influencing the working capital needs of a concern. A concern, making purchases on credit and sales on cash will always require lower amount of working capital. On the contrary, a concern compelled to sell on credit and at the same time having no credit facilities may find itself in a tight corner. Prevailing trade practices and changing economic condition do generally exert greater influence on the credit policy of a concern.

Growth and Expansion Programmes

It is logical to expect that large amount of working capital will be required as a company grows. But is difficult to establish a link between the growth in the volume of a company's business and the growth of its working capital. an important point to note is that the need for increased working capital funds does not follow the growth in business activity but precedes it. Advance planning of working capital is thus a continuing necessity for a growing concern. The composition of working capital in a growing company also shifts with economic circumstances and corporate practices. Other things being equal, growth industries require more working capital those that are static.

Profit Levels

The enterprise with the high gross margins has the chance to restore to the working capital pool at the completion of the cash cycle a greater amount than would be the case if the gross margins were less in the first place. Profit levels in a case influence much directly on the working capital requirements. High profits can be regarded as the ultimate pointer to funds generation from within the enterprise.

Taxation

Taxation once assessed is a short-term liability payable in cash Advance tax may have to be remitted in installments, the computation being based on projected profits for the period. It is the first appropriation to be considered out of earned profits and the amount of tax is calculated as per tax regulations. It is natural that high taxation will put additional strains on working capital because it is an our flow of cash which brings no reproductive effects.

Reserve Policy

A prudent financial policy should strongly recommend for the fabrication of adequate reserves out of profit. Dividend policy should be formulated with this objects well in view. Reserves in the form of retained earning have provided a prolific base on which a concern's growth and expansion has been sustained. Reserve policy always gives priority to working capital or funds position of the concern.

Depreciation policy

Depreciation policy is simply considered as a part of financial policy which determines the amount to be provided as depreciation charge to make up the ultimate resources for replacement of worn out assets. Depreciation is only a book entry and will have no bearing on the working capital position at the time it is provided. However it has the effect to hold back distribution of dividends which may otherwise be made more liberally and help conserve cash. But at the time the active replacement take place there will be cash outflows. If current replacements fail short of depreciation provision the working capital position becomes strengthened. On the other hand excess current capital expenditure (replacements) causes much drain on the working capital pool. Only a well formulated depreciation policy can provide a sound working capital position to the business.

Price Level Changes

Rapidly rising prices create number of problems on the working capital front. For the same level of business activity comparatively more amount of working capital is needed. It depends to a large extent, how close to the price changes the pricing policy of the enterprise can run. It is true that not all prices move in pace, while some

Sources of short-term Working capital requirements

In the 1960s and '70s corporate sector depended heavily on bank credit for its working capital needs. More than 60% of its working capital needs were financed by banks. On the recommendations of the Tandon Committee and Chore Committee, the banks reduced gradually the working capital facilities to corporate sector. Now -a-days, corporate sector raises much of their short term resources through commercial papers, discounting of bills and non-bank financial institutions. This does not mean that the commercial bank have altogether stopped supplying the short-term working capital needs of the corporate sector. The commercial banks are still looked upon by the corporate sector to fund its working capital needs, but there are also other means through which they can raise their working capital funds requirement. The details relating to the features of these sources of working capital finance are described below.

1. **Trade Credit**

Credit extended by the suppliers of raw materials and finished goods is an important source for arranging short-term financial requirements. In the 1970, 6% of the total borrowings of corporate sector was in the form of trade credit. This is one of the cheapest, un secured and easily available source. The usual duration for this credit is thirty days. However, many big business enterprises have exploited their suppliers by not fulfilling the payment obligations and deferring payments beyond the due dates. Because of this the percentage of trade credit in the total borrowings have come down in the recent years. This position can be improved if the corporate sector adopts debt securitisation in meeting their short term demand. By choosing maturity dates of bills and trade debts in such a way that the receipts and payments on the bills and debts match each other evenly the firm can improve its repayment performance and increase its credibility Here each supplier's payment will be matched by equivalent debtors dues. Each bills payable will be matched with an equivalent amount of bills receivable. Thus each payment will be offset by equivalent receipt and this helps in maintaining a balance in the working capital.

2. **Commercial Bills**

Bills receivable representing the amount due from the customers is another short-term source of finance for business firms. Firms can encash these bills before due date by offering discounts to customers. Alternatively, these bills can be discounted with the commercial banks where the firms maintain its current account. However, banks will allow discounting of bills belonging to firms which have high credit worthiness. Normally, bills issued by institutional customers will be used by firms for discounting. In India, bill market has not grown to an appreciable level compared to developed countries.

3 Public Deposits

Reputed business firms may raise short term finance through public deposits. RBI and SEBI have issued certain guidelines relating to issue of public deposits. Business firms requiring funds will issue advertisements in newspapers calling for public deposits. They offer a maximum of 14% to 15% interest rate on these deposits. These deposits are unsecured. This is why only companies with good financial strength and reputation are allowed to take public deposits. Usually, firms should get credit rating from credit rating agencies and inform their rating to the public at the time of invitation of public deposits. There are also several disclosure obligations for inviting public deposits. As working capital finance from banks has been restricted, public deposits have gained popularity. But poor repayment records of many companies has led to several new regulations and restrictions on this source.

4 Commercial Papers

Commercial papers were introduced in India in 1990 Commercial paper is an unsecured promissory note negotiable by endorsement and sold at a discount rate to face value. Companies with a net worth of not less than Rs.10 crores and having fund based working capital facilities with banks of not less than Rs.5 crores can issue commercial papers. Shares of such companies should be listed in recognised stock exchanges. The company must have highest credit rating. Only those companies fulfilling the criteria prescribed by the RBI can issue commercial papers. Commercial papers are issued for a period of more than 3 months and less than 6 months. They are issued in multiples of Rs.5 lakhs minimum being Rs.25 lakhs. Since 1990,24 companies have issued commercial papers. Rs,50.7 crores worth of commercial papers were issued till 30th June 1991. The effective cost to the companies on the commercial papers is less than interest rate on bank loans. Hence reputed companies prefer to issue commercial papers for their short term financial requirements.

5. Factoring

Factoring is an asset based means of financing by which the factor buys up the book debts of a business firm on regular basis, paying cash down against receivables and then collects the amounts from customers to whom the firm has supplied goods. A factor usually performs the following services:

1. Purchasing accounts receivable for immediate cash.
2. Maintaining ledgers and performing other book keeping duties relating to accounts receivables.
3. Collecting accounts receivables.
4. Assessing losses which may arise from any customer's inability to pay.

In India, factoring service is rendered by financial institutions. The first factoring service was established by SBI in 1991 viz, SBI factoring and commercial services In the same year, Canbank Factors was also established. Fair growth factor in the private sector came in 1992. During 1991-92, SBI had a turnover of Rs.30 crores in factored debts and Canbank had Rs.25 crores in the same year. Factoring service is a boon to trade and industry as they are relieved from the burden of credit and collection and are able to convert their debts into immediate cash.

6. Bank Credit

Commercial banks provide short-term credit to business firms through several schemes. Overdraft, cash credit and term loans are the usual schemes through which working capital requirements are met. Besides, bill discounting, factoring are also provided to reputed firms. Pledging of inventory is the common method of providing working capital finance RBI regulates bank credit for working capital purposes. The detailed recommendations of the two Committees set by the RBI namely the Tandon and Chore Committees set by the RBI namely the Tandon and Chore Committee are given at the end of this lesson. After the implementation of the Tandon Committee recommendations by banks, dependence on banks for working capital finance has come down significantly. The rate of interest on these loans range from 16% to 20% per annum which is higher than the rates of public deposits and commercial papers. The aggregate bank credit for working capital purposes amounted to Rs.151030 crores at the end of March 1993. However, in the past, the banks have played a significant role in extending credit support to trade and industry. Between 1950-88 bank credit expanded by Rs.79,306 crores, of which 50% went to industrial sector 9about 2/3rd of this to medium and large units) compared to rs.12,400 crores extended by private capital markets to the entire corporate sector during the same period.

7. Credit from non-banking financial institutions

Private financial institutions, chit funds, money tenders also provide short-term credit facilities to business firms on the basis of the worth of the collateral/security. Usually the rates of interest on these loans are high compared to other sources of finance. However, the terms and conditions and procedure for obtaining and repaying these loans are simple and takes less time for availing them compared to formalities and procedures of availing commercial bank finance.

8. Sale of short term investments and securities

Many successful business firm invest their surplus funds in short term marketable securities which can be disposed of in times of need. However, there is a possibility of loss due to distress sale.

Thus, business firms have a variety of sources through which their short term financial requirements are met. However, the financial manager has to examine carefully the merits and weaknesses of each source of finance, understand the market conditions and make his choice judiciously. He should also attempt debt securitisation to overcome the problem of delayed payments.

Illustration : 1

Prepare an estimate of working capital requirements from the following information of a trading concern.

Projected Annual Sales	1,00,000 units
Selling Price	Rs. 8 Per Unit
% age of net profit on sales	25%

Average credit period allowed to customers	8 Weeks
Average credit period allowed by suppliers	4 Weeks
Average stock holding in terms of sales requirement	12 Weeks
Allow 10% for contingencies	

Solution :

Estimation of working capital Requirements

Current Assets :

Debtors (8 weeks) $\frac{6,00,000 \times 8}{52}$	92,308
Stock (12 weeks) $\frac{6,00,000 \times 12}{52}$	1,38,462
	2,30,770
LESS: Current Liabilities	
Creditors (4 weeks) $\frac{6,00,000 \times 4}{52}$	46,154
Net Working Capital	1,84,616
ADD: 10% for Contingencies	18,462
Working Capital Required	2,03,078

Working Notes:

Cost of Sales

1,00,000 units x 8 = 8,00,000 - 25% of = 6,00,000

Illustration: 2

A proforma cost sheet of a Company provides the following particulars:

Elements of Cost	Amount Per Unit Rs.
Raw Material	80
Direct Labour	30
Overheads	60
Total Cost	170
Profit	30
Selling Price	200

The following further particulars are available:

Raw materials are in stock on average for one month. Materials are in process, one average, half a month. Finished goods are in stock on average on months. credit allowed by suppliers is one month. credit allowed to supplies is two months. Lag in payment of ways is 1-1/2 weeks Lag in payments of overheads expenses in one month.

One -fourth of the output is sold against cash. Cash on hand and at bank is expected to be Rs 25,000.

You are required to prepare a statement showing the working capital needed to finance a level of activity of 1,04,000 units Production. you may assume that production is carried on evenly throughout the year, wages and overheads, accrue similarly and a time period of 4 weeks is equivalent to a months.

Solution :

Statement Showing the making Capital Requirement

Current Assets :

i.	Stocks of Raw Material (4 weeks)	1,60,000 x 4		6,40,000
ii.	Work-in-Process (2 weeks)			
	Raw Material	1,60,000 x 2	3,20,000	
	Direct Labour	60,000 x 2	1,20,000	
	Overheads	1,20,000 x 2	2,40,000	6,80,000
iii.	Stock of Finished Goods (4 weeks)			
	Raw Material	1,60,000 x 4	6,40,000	
	Direct Labour	60,000 x 4	2,40,000	
	Overheads	1,20,000 x 4	4,80,000	13,60,000
iv.	Sundry Debtors (8 weeks)			
	Raw Material	1,60,000 x 8 x 3/4	9,60,000	
	Direct Labour	60,000 x 8 "	3,60,000	
	Overheads	1,20,000 x 8 "	7,20,000	20,40,000
v.	Cash in hand and at bank			25,000
				47,45,000

LESS: Current Liabilities

i.	Sundry Creditors (4 weeks)	6,40,000	
ii.	Wages Outstanding (1-1/2 weeks)	90,000	
iii.	Overheads Outstanding (4 weeks)	4,80,000	12,10,000
	Net Working Capital Required		35,35,000

Notes :

(i) Weekly Raw material $\frac{1,04,000 \times 80}{52}$ = 1,60,000

(ii) Weekly Labour Cost $\frac{1,04,000 \times 30}{52}$ = 60,000

(iii) Weekly overheads $\frac{1,04,000 \times 60}{52}$ = 1,20,000

companies feel relieved of their working capital fully during price changes. others find them unmanageable.

Operating Efficiency

Operating efficiency has a significant bearing on the working capital position. Elimination of wastes and inefficiency are essential to improve the operating efficiency. The level of operating efficiency in these matters will influence the amount of working capital required. Efficiency of operations accelerates the pace of cash cycle and improves the working capital turnover. Working capital position is eased by improving the profitability and aiding with more internally generated funds.

Government regulations

Government through Reserve Bank of India exerts much controls on credit, bank rate, supply of money etc. For example a, Study Group, called tandon Committee constituted by the Reserve Bank of India has established norms for Inventory level and accounts receivables in 15 selected industries.

Management of Current Assets

Cash Management

Approximately 15% of the average industrial form's assets are held in the form of cash-which is defined as the total of Bank demand deposits and currency. Cash is the most important and at the same time the least productive asset that a business possesses. It is highly liquid and is essential to meet all current obligations since failure to meet such obligations denotes technical insolvency. Comparing with other current assets cash is the most unproductive or not directly productive, for example Inventories are essential to carry on production and fill customers' orders. Accounts receivables encourage customers place more orders and there by earn more profits. hence the goals of cash management are (a) to decide about the necessary minimum cash balance, (b) to synchronise cash outflows and inflows in such a way as to reduce both the average and minimum necessary cash balances and (c) to put the excess cash balances work in a productive way. The main factor that influences the cash management is the trade off between liquidity and profitability.

Motives for Holding Cash

Transactions, precautionary and speculative motives are the three traditional motives for holding cash of all firms, regardless of size, type or location. One more motive that may be added to the list is compensation to Banks for providing loans and advances. However a prudent financial manager will never hold cash for speculative purposes and will rely mostly on reserve borrowing power and marketable securities portfolios. Normally banks do not insist on compensating balances when the transactions of their clients are continuous and satisfactory. hence mostly the first two only will be considered for planning the Cash holdings of any business. Planning an optional cash balance at a particular point of time involves an estimate of cash flows from operations.

Advantage of Adequate Cash

A firm having sufficient cash, can take trade discounts. Adequate cash also improves the liquidity position of the business. Ample cash is always useful for taking advantage of favorable business opportunities that may come along from time to time. To meet emergencies such as strikes fires or marketing campaigns of competitors, adequate cash resources are necessary.

Methods to Improve Cash Flow Patterns

Synchronisation of Cash Flows

Cash inflows should be estimated carefully at periodic convenient intervals. Similarly outflows should be scheduled to synchronise with cash inflows.

Reduction of cash in transit

All efforts should be taken to minimise the delay in the realisation of customers, cheques, 'Lock Box' system can help realise the cheques of customers in for off places. Look box is postal box which can be established in the customers, places with necessary instructions to customers to send their payments to this postal box. Then the firm's banker at the customers place are instructed to pick up the cheques for local clearance and remit cash expeditiously to main bank account.

Utilisation of Liquid Resources

When a firm is under seasonal or cyclical fluctuations or expands sales or other current assets more cash is spent, More liquid resources may be required to meet contingencies or to stay in a highly competitive market or to meet development expenditures. These factors may cause for the investment of funds for a few weeks, a few months, a few year or more. The financial manager is to decide upon a suitable maturity pattern for his cash holding, considering the time required for holding such funds and the trade off between liquidity and profitability.

Cash Forecasting

Cash Budget

Generally, cash flows do no not mesh evenly in timing. Sometimes inflows exceed outflows due to large collections from debtors or seasonal sales and some other times the reverse will be the case due to dividend payments. tax payments or seasonal built up of inventories etc. A cash budget. prepared with short-time intervals such as weekly, monthly or quarterly basis, shall enable a firm to even out is cash flows. The most important tool in cash management is the cash budget, a statement showing the firm's projected cash inflows and outflows over some specified period of time. It is a short-term cash planning predicting all inflows and outflows of cash and expected excesses or shortages of cash at certain points of time. It is considered to be an effective tool of cash management since it aids to maintain liquidity by highlighting the firm's cash position. A firm can obviously know the cash deficit sometimes well in advance and is able to make up the deficit. In case there is surplus cash, the firm has enough time at its disposal to employ on profitable investments without impairing the liquidity position.

Investing Excess Idle Cash

Since the cash is an unproductive asset it should be minimised as far as possible without jeopardising the liquidity position of the business. The best way would be to invest excess of the required minimum cash balance in marketable securities or to reduce outstanding debt. The point to be considered here is to see that the interest earned over the expected holding period must, more than compensate for transaction and inconvenience costs. For the transfer between cash and marketable securities, to be performed efficiently the cash projections should be made more accurately.

Measurements of Efficiency of Cash Management
Ratio of Cash in Current assets

The liquid ratio i.e. the ratio of liquid assets to current liabilities indicates the firms ability to meet its current obligations without revealing anything about the efficient usage of cash resources. Hence the ratio of cash to current assets is used as an index of current operations which is computed thus:

$$= \frac{\text{Cash}}{\text{Current Assets}} \times 100$$

A high ratio indicates that the liquidity position of the firm is highly sound or the profitability of current assets is low. Similar ratios may be computed from a number of Balance Sheets and compared or this ratio may be compared externally with these applicable to similar companies within the same trade since this ratio, or itself, is not adequate to adumbrate the effectiveness of its control.

Cash Turnover Ratio

This ratio can be used as a measure of the velocity of cash. It is calculated as follows:

$$\text{Cash Turnover Ratio} = \frac{\text{Sales for a period}}{\text{Average cash balance}}$$

A high cash turnover ratio would point out that the profitability of the business is greater and this, itself is an evidence of efficient management. It would always be better to collate this ratio with those figures compiled from previous records or obtained externally from similar companies with in the same trade since laying down of any specific standard turnover ratio is generally not-feasible.

Marketable Securities Management

Meaning

Marketable securities are short term investments in which surplus cash is invested. The list of marketable securities subsumes securities like Treasury Bills, Commercial Papers, Fixed Deposits (short-term) and savings bank deposits in commercial banks etc. The essential features of these securities are ready marketability, safety and profitability.

Strategies regarding Marketable Securities

Seasonal firms may adopt any one of the three alternative strategies in order to meet their seasonal needs. Under Strategy-I, a firm would hold no marketable securities, relying completely on bark loans to meet seasonal peaks. Under Strategy-II, a firm may stockpile marketable securities during slack periods, then realise them to raise funds for peak periods. Strategy-III is a compromise, under this strategy, a firm may hold some securities, but not enough to meet all of its peak needs. There are advantages to each of these strategies. It is difficult to "Prove', that one strategy is better than another. However the basic policies regarding marketable securities holdings are generally formulated either on the basis of the judgement or by circumstances beyond the firm's control.

Accounts Receivables Management

Accounts receivables represent the uncollected portion of credit sales and normally subsume both trade debtors and bills receivable. The term ;trade credit" refers to accounts receivable, and it involves both the credit used to support and expand sales, and the resulting investment required of the firm. The above definition clearly glosses over the basic objective of creation of receivables as to improve or support sales Receivables constitute a major component of working capital and the investment in Accounts receivable is purely of a permanent nature, in the sense as some accounts (debts) are collected, others are created so that investment is always left in the receivables. Because of the size and permanence of the investment, good management of receivables is a highly significant aspect of the goal of maximising the value of the firm.

Objectives

Unless the firm's own finances are closely limited, the basic objective of receivables management, like that of inventory management, should be maximising return on investment. Bad debt losses may will be minimised by adopting strict credit policies and highly aggressive policing of collections. But such policies may have direct impact on sales and return on investment. Conversely, too liberal or sloppy credit policy management the investments in receivables and consequently bad debts losses without compensating increases in sales and profits. Hence, the obvious aim of receivables management is to maintain an optimum investment in receivables which helps to trade off between benefits and costs i.e., incremental expenditure on collection, bad debts and opportunity costs.

Factors Influencing the Credit Policy

The following factors are generally considered while formulating the credit policy, Credit terms, Credit risks and cash discounts.

Credit terms

A firm's cash needs and its industry's characteristic credit terms are the guidelines of its credit policy. Credit worthiness of the potential customer is the core of a credit policy which depends on six 'Cs' character, capacity, capital, collateral, conditions and capability of the customer. Usually the credit risk is judged on the basis of these six 'Cs'. The required information on these items are obtained either from internal sources such as sales representatives reports, direct visit etc., or external sources such as trade reference, credit agencies etc.

Evaluation of Credit Risks

Bad debts losses largely influence the selection of credit risks. The amount of risk is expressed as certain percentages on gross sales and depends on the profit margin of the firm, its competitive position, its collection policies and techniques and the susceptibility of its products to obsolescence. The seller's profit margin definitely influence the credit manager's decision concerning credit risks. However, a firm may accept a new or extend an existing credit risk provided the cost of granting such risk is less than the minimum expected rate of return.

Cash Discount Policy

Cash discount is allowed to the buyer provided he effects the payment within the period for taking such discounts. for example a trader allows 30 days time to pay the credit in full with a cash discount at 2% if paid within 10 days. It has the following merits.

a) Collection period is shortened and thereby the need for shorterm finance is reduced.

b) Bad debts and collection costs are reduced to the minimum and

c) This helps to take advantage of cash discounts from his suppliers As demerits the following points may be attributed

a) Even a small discount costs too much to the seller and reduces his firm's rate of return.

b) Large companies may pressurise for excessive cash discounts threatening to buy elsewhere and

c) Some customers may deduct cash discounts even if they pay after the discount period has elapsed.

Collection Policies

Collection policy refers to the procedures the firm follows to obtain payment of past due accounts. An effective collection policy is one which assures not only the collections of debts from customers who pay according to credit terms and from delinquent accounts but also promptness of payment. Am effective collection policy serves as the guide in developing procedures for (1) determining delinquent accounts, (2) developing collection correspondence, (3) dealing with discount 'chislers" (4) suits for collection, (5) 'adjustments" proceedings and (6) liquidation proceedings.

The collection process can be expensive in terms of both out of- pocket expenditures and lost goodwill, but atleast some firmness is needed to prevent an undue lengthening in the collection period and to minimise outright losses. Hence, an optional collection policy should be selected in which a trade off between costs and benefits of different collection policies can be established.

Control of Accounts Receivable

The implementation of credit an collection policies is the responsibility of the top management personnel. To exercise their unction in this regard they may fully rely on the following reports.

a) Receivables Turnover ration and Average collection period

b) Aging schedule

c) Percentage of collections

d) Bad debts and

e) Delinquent accounts.

Report on Receivable to turnover and average collection period.

This report enables the top management to apprehend the effectiveness of the credit and collection policies. This is computed as follows:

$$RTR = \frac{\text{Average Receivable}}{\text{Net Sales}} \times 100$$

Example: Sales for the quarter 1977 = Rs.420,000
Accounts receivable on 31.12.1977 = Rs.84,000

$$RTR = \frac{84{,}000}{420{,}000} \times 100 = 20\,\%$$

The ration can be collated with the ratio of the preceding year and any increase or decrease in the ratio may be viewed with caution. This may also be expressed in terms of average day's sales outstanding or average collection period and is calculated as follows:

$$\text{Average Collection Period} = \frac{\text{Account Receivables}}{\text{Net Credit Sales Per Day}}$$

$$\text{Net Credit Sales Per Day} = \frac{\text{Annual Credit Sales}}{365}$$

Average collection period indicates as to how many days credit sales remain uncollected. If the average collection period exceeds the credit period, it will adumbrate the laxity in the collection procedures and goad the collection department into action.

Aging Schedule

The Second useful control device is analyse the conditions of 'receivables is the aging schedule. This is nothing but a tabulation of receivables outstanding according to the length of time they have been outstanding. The average age of debtors may be calculated as follows.

$$\text{Average Age of Debtors} = \frac{\text{Debtors sat the end of period}}{\text{Credit sales during the period}} \times \text{No. of days in the accounting period}$$

If the average age of debtors rise over time, it denotes that increasing amounts of working capital are being tied up in credit. The aging schedule by revealing tendency for old accounts to accumulate, provides a useful supplement to the various receivables/sales ratios.

This report has gained an important place in receivables management for the following reasons.

a) It indicates the effectiveness of credit and collection policies

b) It reveals the general conditions of debtor's accounts and

c) It serves the basis for bad debts reserve

Percentage of Collection Reports

This is helpful to evaluate the efforts taken by the credit department Calculation is as follows:

Percentage of collections =

$$\frac{\text{Debts collecte for a month}}{\text{Receivables outstanding on the first day the month}} \times 100$$

A low percentage comparing with previous collection percentage will help to uncover the inefficiency of the collection personnel and the ineffectiveness of the collection policy adopted and may insist for a change in the collection approach.

Reports of Bad Debts

An acceptable bad debts ratio should be used for all comparative purposes. This is expressed as certain percentage credit sales and computed as follows:

$$\text{Bad debts ratio} = \frac{\text{Bad debts incurred}}{\text{Total credit sales}} \times 100$$

This may be compared with past ratios or with derived ratios or with those of comparable companies within the same industry. Any deviations from the acceptable norms should be viewed with caution.

Report on Delinquent Accounts

Yet another measure to evaluate credit and collection policies is the ratio of delinquent accounts be credit sales and is calculated as follows:

$$\text{Delinquent Account ratio} = \frac{\text{Delinquent accounts}}{\text{Total Credit Sales}} \times 1000$$

An examination of past experience or the experience of other companies in the same industry may serve as a basis for determining normal delinquency ratio, Any deviation of the actual ratic from the normal may be due to very strict or very lenient credit or collection policies.

TEST YOURSELF

1. Explain the concept of working Capital
2. Explain the concept of operating cycle. What is its significance/

3. Briefly explain the factors which determine the Working Capital needs of a firm.

4. What are the principal motives for holding cash?

5. Explain the techniques that can be used to accelerate the firm's collections.

6. Explain the objectives of Receivable Management.

7. Explain the techniques that can be used to control of Accounts Receivables.

8. Explain the objectives of Inventory Management?

9. Define the economic order quantity. How is it computed?

10. What is a selective control of inventory? Why is needed?

....

LESSON :·18

COMPUTER ACCOUNTING AND ALGORITHM

The notion of an algorithm is basic to all computer programming. Am early German mathematical dictionary, "Vollstandifes Mathematics Lexicon". "Under this designation are combined the notions of flow types of arithmetic calculations, namely addition, multiplication, subtraction and division. The Latin phrase 'algorithms infinitesimetes" was at that used to denote" ways of calculation with infinitely small quantities, as invented by Leibrity".

By 1950. the word algorithm was most frequently associated with 'Euchid's algorithm". a process for finding the greatest common division of two numbers which appear in Euchid's Elements. It will be instructive to exhibit Euchid's algorithm here:

Algorithm E [Euchid's algorithm]

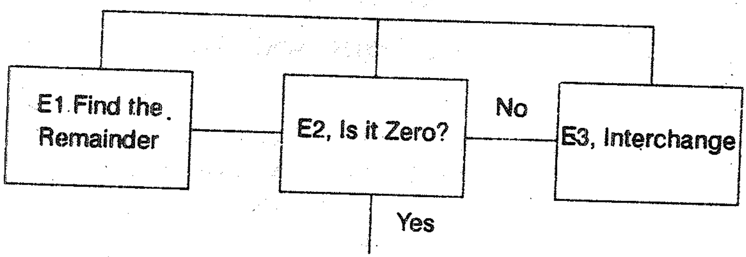
Given two positive integers m and n, find their greatest common division, i.e., the largest positive integer which divides both m and n.

E1. [Final remainder] : Divide m by n and let r be the remainder. (we will have $0 \leq r \leq n$)

E 2. [Is it Zero?] : If r=o, the algorithm terminates; n is the answer.

E 3. [Interchange] set $m \leftarrow n$, $n \leftarrow r$ and go back to step E1.

Flow Chart for Algorithm E



The modern meaning for algorithm is quite similar to that of receipe, process, method, technique, Procedure, róutine, except that the work "algorithm" Connotes something just a little different. Besides merely being a finite set of rules which gives a sequence of operations for solving a specific type of problem, an algorithm has five important features:

(1) Finiteness

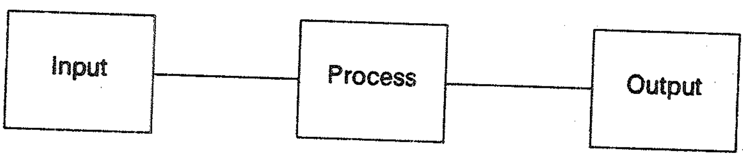
(2) Definiteness

(3) Input

(4) Output

(5) Effectiveness

An algorithm is nothing but a process which converts the input variables to desired output formats.



The sequence of instructions for solving a particular problem is called an algorithm. Actually whenever an algorithm is much more complicated, the algorithm is displayed though a flow chart (pictorial representation) rather than through numbered list of instructions.

Variables, Data-names, Programming Statements.

By a variable in a computer program, we mean a data items whose values may change during the execution of a program.

A data name or variable name is a programmer supplied label of a variable. Normally one chooses names so that they indicate the kind of data item they represent, e.g RECEIPTS, DISBURSE, NETFLOW, DEPREC, BONUS. Each programming language has certain reserved words such as STOP, END , DO, IF, that have specific operational meanings in the language. These words may not be chosen as variable names

Assignment statement

A Programming statement that assigns a value to a variable, by storing that value at the address of the variable. The general form of such a statement is

Name = arithmetic expression

Rate = 0.08

Similarly we would write the assignment statement

Bonus = 0.05 *Salary

Control Statement

The sequence of instructions is often controlled by conditional and unconditional branching statements.

Unconditional Statement

Go To 20

Here the sequence is branched to statement No.20. Sometimes logical decision must be taken before altering the sequence. The decision need not involve numeric data alone.

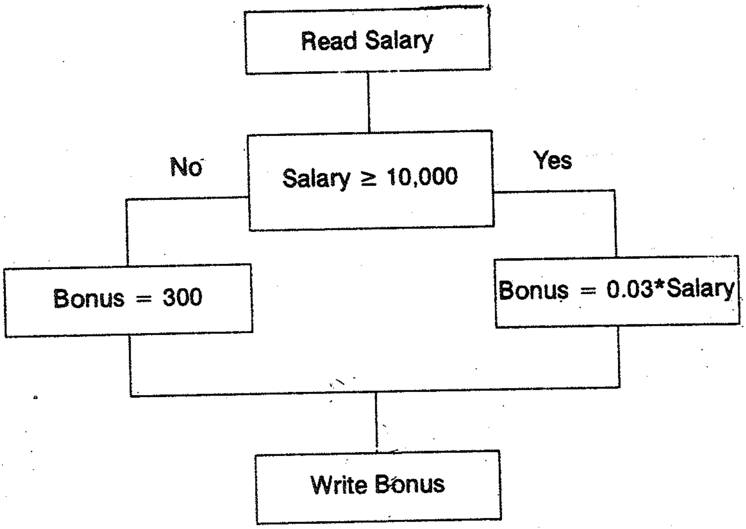
e.g. "Is student married"., 'Is student employed"

"Is student psysically handicapped. However if numeric data are involved then the decision normally involves one of the six mathematical relations or a combination of such relations.

$=$	Equal
$\neq$	Not Equal
$<$	Less Than
$\leq$	Less Than or Equal
$>$	Greater Than
$\geq$	Greater Than or Equal.

Example :

The square -Deal company plans to give a year-end 3% bonus to each of its employees earning Rs 10,000 or more per year and a fixed Rs.300 bonus for other employees.



If SALARY $\geq$ 10,000 then BONUS = 0.03 * SALARY
else BONUS = 300.

Loops and their control by a counter

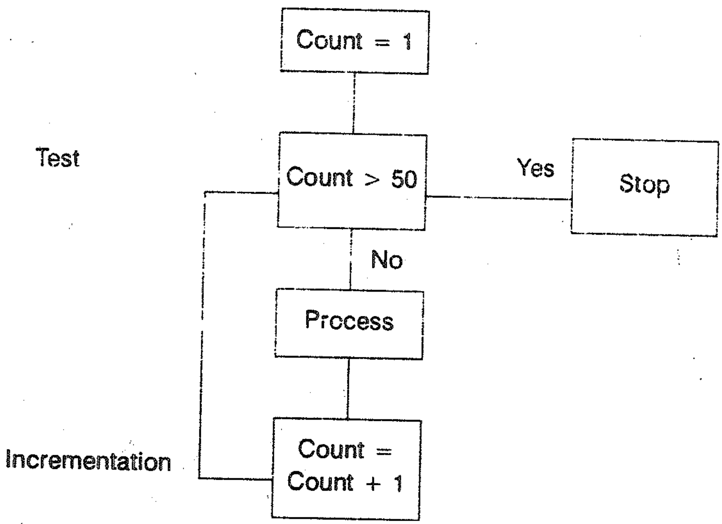
Frequently one wants to repeat a process a certain number of times. One way to accomplish this repetition is for the algorithm to transfer control of the computer from the end of the Process back to its beginning.

In the previous example, when square Deal company wants to calculate for every one of its employees the algorithm will be like this:

Step 1: Obtain employee's name and his salary
Step 2: If salary $\geq$ 10,000
Step 3: - Bonus = .03 *Salary (or) Bonus = 300

Step 4: Write employee's name and Bonus
Step 5: Go to step 1.

Clearly one wants to stop when all employees records have been read and all bonuses have been recorded. Give way to control a loop' i.e. to incorporate a stopping condition, is by using a counter, a variable whose value is increased by 1 for each repetition of the process. The following flowchart explains now an algorithm is repeated 50 times. The counter is denoted as count



observe that there are three essential steps in the use of a counter to control a loop:

Initialization	:	The counter is set equal one
Test	:	Before executing the process the counter is tested to see if it exceeds its maximum allowable value.

Incrementation : After the process has been executed, one counter is increased by one.

Accumulators

Frequently one wants to find the sum of certain numerical quantities that are read or calculated one at time.

PAGE – TOTAL = 0 ....(A)

PAGE – TOTAL = PAGE - TOTAL + AMOUNT................(B)

CUM – TOTAL = CUM - TOTAL + PAGE - TOTAL.............(C)

In step (A), a counter named PAGE - TOTAL is initialised to zero. In step (B) its value is incremented by the value of AMOUNT and the result is again stored in PAGE - TOTAL. For example assume that the entries of amount for a page are 5000, 2500, 7500 and 5000 and 2000.

(i) Page – Total = 0

(ii) Page – Total = 0 + 5000
now the present value of PAGE - TOTAL BEING 5000

(III) PAGE – TOTAL = 5000 + 2500
The Present value of page - Total is incremented by value of amount Now the result 7500 in being stored as PAGE -TOTAL.

(iv) PAGE -TOTAL = 7500 + 7500 = 15,000

(V) PAGE -TOTAL = 15000 + 5000 = 20,000

(VI) PAGE -TOTAL = 20,000 + 2,000 = 22,000

Now assume that in the three pages of the report we have calculated the page -Total as 22,000, 25000 and 20,000

The cumulative total counter is getting incremented as follows:

(i) CUM – TOTAL = O

(ii) CUM – TOTAL = CUM - TOTAL + PAGE - TOTAL
= 0 + 22,000
= 22,000

(iii) CUM – TOTAL = 22,000 + 25,000
= 47,000

(IV) CUM – TOTAL = 47,000 + 20,000
= 67,000

Simple Book-keeping System

The necessary accounting reports include

(1) a trial balance

(2) an income statement

(3) a balance sheet and

(4) a post closing trial balance.

A typical general -purpose journal should provide the following informations:

(1) Date

(2) Journal entry number

(3) Account to be debited

(4) Debit Amount

(5) Amount to be credited

(6) Credit amount.

The operation of this computer book -keeping system is similar to the operation of a manual system, with the exception of the assistance the computer offers in each of the steps. The sequence of actions listed below illustrates the operation of the system. BCREATE, B POST, BTRIAL, BINCOME, BSHEET, BJOURNAL, BFLIST, BPRINT and BCLOSE are the program names used here.

Time	Action	Program or Manual
Initialization at start of year	1. Determine required accounts	Manual
	2. Initialize files	B CREATE
Throughout accounting period	1. Gather transactions and Post Journal	Manual
	2. Post entries to file	B Post
End of accounting period	1. Prepare trial balance	B TRIAL
	2. Prepare Income statement	B INCOME
	3. Post net income / Loss Capital accounts	B Post
	4. Prepare Balance Sheet	B SHEET
	5. Enter Closing Journal Entries	B Post and Manual
	6 Close accounts	B CLOSE

As required	1. List account file contents	B FLIST
	2. Recreate journal entries	B JOURNAL
	3. Correct account information	B PRINT
	4. Display accounts	B PRINT

General Financial Programs

1. Break even Analysis(Basic)

The Program must accept cost and price information from the operation and Produce a table describing a products break even point and the cost breakdown for that level of production.

```
10   INPUT  "Enter FIXED COSTS" =F
20   INPUT  "ENTER VARIABLE COSTS PER UNIT" =V
30   INPUT  "ENTER UNIT PRICE" =P
40     Q    F/(P-v)
50     VI   V X Q
60     R    P X Q
70     C    P + (V X Q)
80     U    C/Q
85   PRINT  "BREAK EVEN POINT"
90   PRINT  "BREAK EVEN QUANTITY"; TAB (25) =Q
100  PRINT  "BREAK EVEN REVENUES" ; TAB(24);R
```

```
110    PRINT   "FIXED COSTS" ; TAB (15); F
120    PRINT   "VARIABLE COSTS" TAB (24) ;V
130    PRINT   "TOTAL COSTS"; TAB (15) ; C
140    PRINT   "UNIT COST" = TAB (15) :U
```

When we execute this program we will gel two following display.

ENTER FIXED COSTS ≥ 10,000 (operator entry)

ENTER VARIABLE COSTS PER UNIT ≥ .4 (operator entry)

ENTER UNIT PRICE ≥ .6 (operator entry)

The result will be as follows.

BREAKEVEN POINT		
BREAKEVEN QUANTITY		50,000
BREAKEVEN REVENUES		30,000
FIXED COSTS	10,000	
VARIABLE COSTS	20,000	
TOTAL COSTS	30,000	
Unit Cost	6	

Ratio Analysis

This Process calculates and prints a number of ratios that have been found to the useful in business, namely: the current ratio, the acid test ratio, the Net Profits on sales rates, the investment turnover ratio, the return on investment rates and the inventory turnover ratio.

Cash flow and Budget Analysis

Forecasting (i) Least square Regression Forecasting (ii) Moving average Forecasting (iii) Exponential smoothing forecasting.

Equipment Comparison

Depreciation

Amortisation Table

Return on Investment (ROI)

Mortgage Comparison etc.